TEAM INVICTUS

Systems Thinking & Design Phase 1 Submission

The Breakfast Paradox

IIIT Hyderabad Breakfast Mess System

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System Context Brief

IIIT Hyderabad Breakfast Mess System

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System Boundary and Context

System Boundary

The system under study is the breakfast meal as it is registered, billed, cooked and served in the four residential dining halls at IIIT Hyderabad: Kadamba, Palash, Bakul and Yuktāhār. Three layers sit inside the boundary.

The registration and billing layer is the dining portal (dining.iiit.ac.in), where students book, cancel and pay for meals, together with the QR scan at the counter that records who ate. The kitchen layer is the outside caterer that runs each hall: Prism at Kadamba, ABC Hospitality Services at Yuktāhār, and Vijayalakshmi Caterers at Bakul, which cooks at its Hafizpet kitchen (single-sourced). It covers quantity planning, cooking, food-safety checks before service, the counter, top-up during service, and what happens to leftovers. Food for Bakul and Palash comes from one approved off-site kitchen (confirmed), so for those two halls the layer also includes the transport link and the hot holding of food at the hall. Whether Vijayalakshmi Caterers also supplies Palash is a candidate. The governance layer is the set of bodies that fix the menu, the registration and billing rules and the operator contracts.

Several parties sit on the boundary, exchanging food, money or information with the system without being run by it: the operators' suppliers, the garbage contractor, the institution's site team (which relays complaints) and facility team (weekly quality inspection), and the tender through which each operator is hired. People outside the registration also eat from the same pots. At Kadamba these are 30 to 50 outsiders such as night-shift security, plus about half of the operator's own staff (single-sourced). The academic class schedule sits outside the boundary but shapes it, because the two run on overlapping, independently set timings. On-campus alternatives (Vindhya Canteen, the juice canteens, delivery apps) sit at the outer edge: a student who skips the mess eats from one of them.

Lunch and dinner at the same halls fall outside the boundary and are the comparison that justifies it. The same portal, rules and operator produce very different turnout at the two meals. At Yuktāhār in September 2026, lunch attendance ran from 56 to 89 percent of registrations (15 of 20 days at 70 percent or above) while breakfast on the same days ran from 18 to 40 percent (confirmed, from the operator's attendance register). The problem is specific to breakfast.

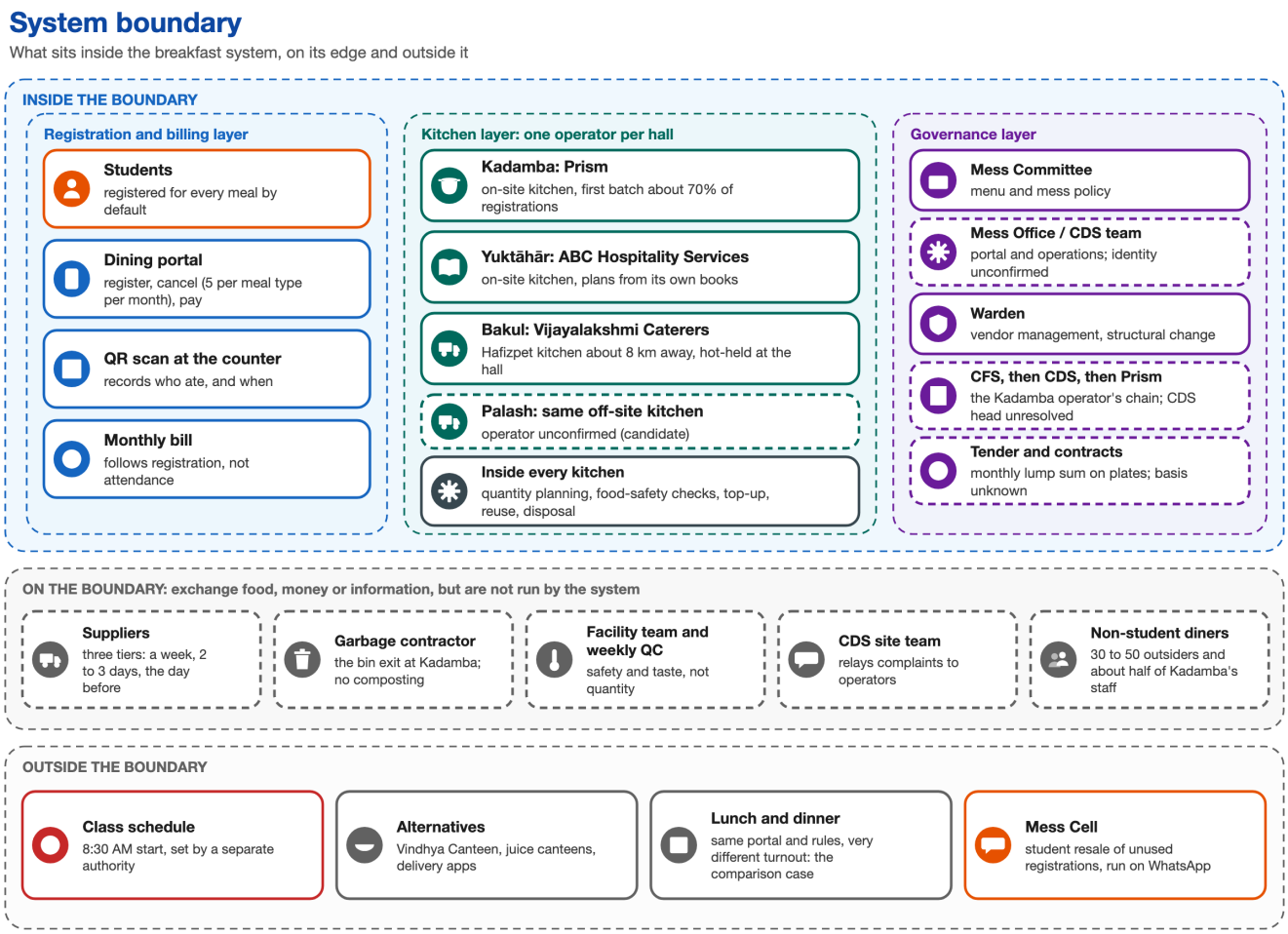


Figure 1. System boundary: three layers inside, boundary actors on the ring, class schedule and alternatives outside.

Context

The published breakfast window is 7:30 to 9:30 AM. At Kadamba in April 2026 the first scans came at about 7:28, service closed between 9:35 and 9:45, and the latest scan was at 10:07 (confirmed, scan records). Kitchen staff start between 5:00 and 5:30 AM. Batch items are ready by about 7:30, and at Kadamba live items such as dosa are made at the counter from 7:00 (single-sourced). Bakul's food is cooked off site, so labour-heavy items such as aloo paratha may be started around 3 AM to reach the hall by about 7 (single-sourced).

On one day in September the platform showed breakfast capacity of 2,755 registrations: Kadamba 700 vegetarian and 600 non-vegetarian, Bakul 350 and 350, Palash 400, Yuktāhār 340 regular and 15 Jain. Registered prices are Rs. 48 (vegetarian) and Rs. 66 (non-vegetarian) at Kadamba, Bakul and Palash, and Rs. 53 at Yuktāhār. Without prior registration the same breakfast costs Rs. 81, Rs. 111 and Rs. 92, about 70 percent more (confirmed, posted rates).

Registration is high and flat; turnout is low and variable. Kadamba took 32,162 breakfast registrations in April 2026, about 1,070 a day on every day of the week including Sunday, and scanned 11,941 breakfasts: 37.1 percent turnout, 62.9 percent no-shows (confirmed, scan records). Yuktāhār's breakfast registrations ran from 228 to 404 a day in September, with turnout averaging about 28 percent (confirmed, operator's register). These are the only two halls with turnout records. Bakul's manager estimates that at most about 60 percent of registered students come (single-sourced; nothing written behind it).

Registration is automatic. Every student is registered for every meal by default, on campus or not, holidays included. The only exit is a cancellation, allowed five times per meal type per month and at least four days ahead; beyond that a meal cannot be removed from the bill (confirmed). Billing is monthly and follows registration, so a student pays for every meal they are registered for, eaten or not. Operators describe counts reaching them in weekly blocks, while

students describe registering monthly. How the default is batched on the portal is unconfirmed (single-sourced).

The platform is shared but the kitchens are not. Kadamba serves South Indian food, vegetarian and non-vegetarian planned separately, from an on-site kitchen run by Prism. Bakul serves North Indian vegetarian and non-vegetarian food and Palash North Indian vegetarian food, both from the off-site kitchen until the Felicity Kitchen and Dining Facility opens, expected December 2026 (confirmed). Yuktāhār serves Satvik vegetarian food with a Jain variant, from an on-site kitchen run by ABC Hospitality Services (confirmed). Bakul's food travels about 8 km from Hafizpet, roughly 20 minutes early in the morning and 40 in normal traffic, in insulated boxes holding about 60 to 72 °C, and stands in a hot-water setup at the hall (single-sourced). Each operator buys its own ingredients on its own cadence. The four halls are unified only at the registration, billing and access layer. (Revised from the first submission, which had Bakul's vegetarian food cooked at Palash and carried over.)

Five adjacent systems act on the mess without being controlled by it.

- **Academic administration.** Classes run 8:30 AM to 6:40 PM with a 1:00 to 2:00 PM break, and most students have few classes on a given day. No channel links the mess to the academic calendar. Exams, events and holidays reach a kitchen only if someone raises them in its planning meeting: ”we don't come to know until we sit in the meeting,” as the Yuktāhār site head put it (confirmed).
- **Institutional governance.** A Mess Committee sets the menu and mess policy with student input about a month ahead; a Mess Office runs the portal, from which operators read their counts; the Warden holds vendor management and structural change. The Mess Committee is reported to be faculty-chaired, and a policy change tightening cancellation rules is reported to have been made to cut staff workload, both from a press summary we could not read in full (candidate). Operators speak of a ”CDS team.” The Kadamba operator describes its line of authority as CFS, then CDS, then Prism, then an operations manager, a manager, a supervisor and the service staff (single-sourced). Who heads CDS is open: the operator names one person and an earlier institutional interview names another. Whether CDS is the Mess Office, and how CFS relates to the Committee and the Warden, is unconfirmed.
- **The operators and their contracts.** Each firm is chosen by tender and paid monthly as a lump sum on plates, with deductions (single-sourced). Whether ”plates” means registered or served decides who bears the cost of a no-show, and the evidence disagrees. Kadamba's operator talks of being paid for what is registered, and Bakul's says its payment is on plates served (both single-sourced). The contracts may differ, or one remark may describe a duty rather than the payment formula. The basis is unconfirmed for every operator and Bakul's answer is not applied to the others. The institution sets the menu and the rules; the operator decides how much to cook.
- **Mess Cell.** A student WhatsApp group where people who hold a breakfast registration they will not use sell it at a negotiated price, transferring the personal QR credential. It runs outside the portal. Kadamba's manager describes students registering ”even if you are in the village” while a friend eats in their place, which corroborates informal transfer (single-sourced).
- **The wider campus food ecosystem.** Vindhya Canteen (Monday to Saturday from 10:00 AM), the juice canteens (mornings) and delivery apps (orders until 11:00 PM, riders kept outside the gate) all compete for the same students.

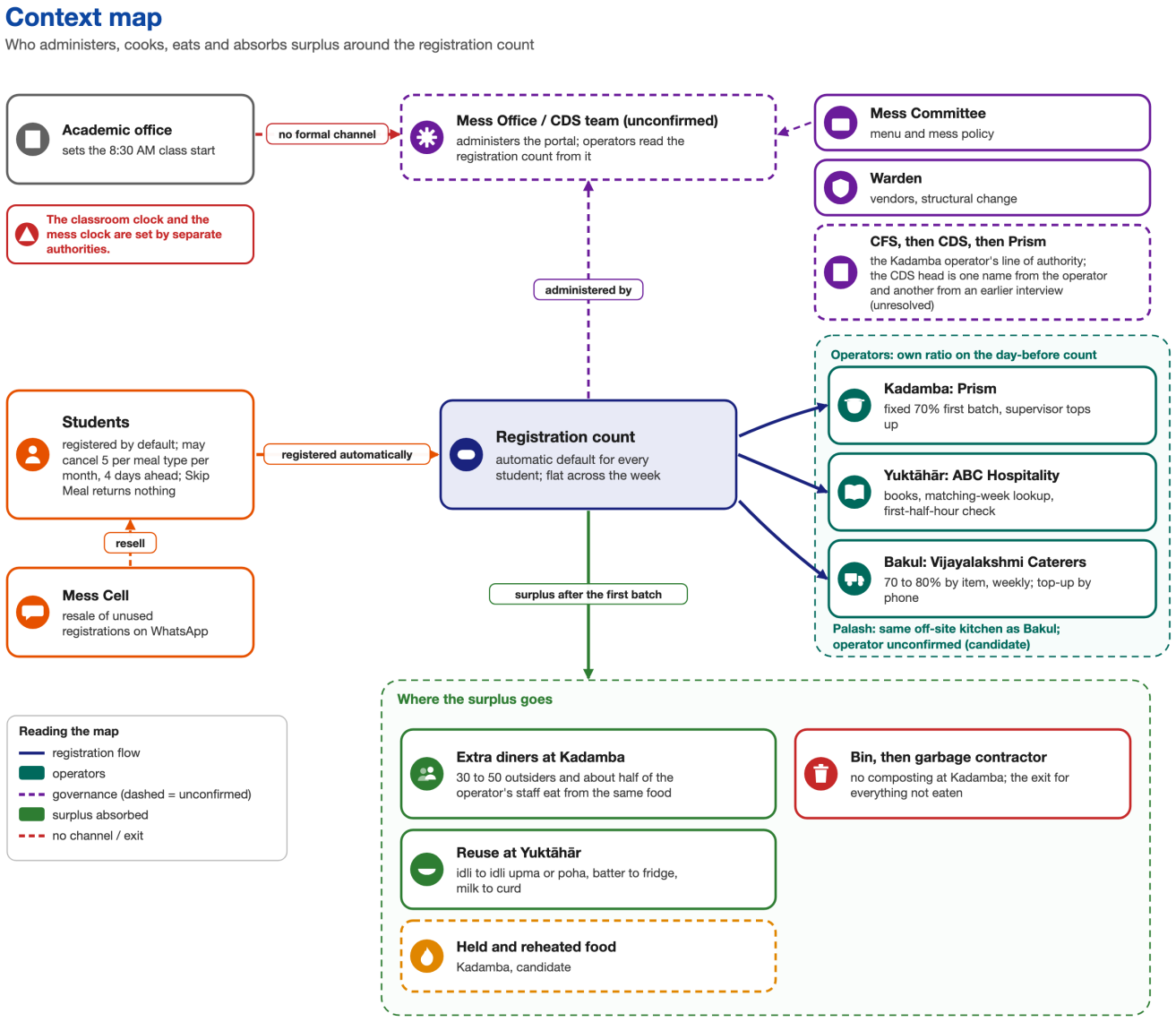


Figure 2. Context map: who administers, cooks, eats and absorbs surplus around the registration count.

Purpose and Intended Outcomes

Purpose

The system exists to give resident students reliable, adequate, good-quality food each morning at a predictable, affordable cost, covering vegetarian, non-vegetarian and Jain requirements, in support of health, energy and academic performance. The institution states its objective as to ”provide nutritious, hygienic, and affordable meals while ensuring efficient operations and minimizing food wastage through advance meal planning.” Advance registration is the mechanism it relies on to match food to demand.

Actual Mechanism

Registration is a default, not a choice. The only way out is one of five cancellations per meal type per month, made at least four days ahead (confirmed). The booking therefore records the absence of an opt-out. It stays flat across the week: Kadamba's roughly 1,070 daily registrations hold from Monday to Sunday while turnout runs from 42.1 percent on Tuesdays to 27.3 percent on Sundays (confirmed, scan records). Billing follows the booking, not the meal.

The kitchen does not cook to the booking. Each operator reads the closed count ahead of service, the day before,

and applies its own discount.

Yuktāhār plans in daily meetings of all staff, one for the next day's breakfast and lunch and one later for snacks and dinner. The inputs are portal registrations, the Wastage Book entry for the matching day of the menu cycle (the menu repeats fortnightly, so a week-3 Thursday is planned from the week-1 Thursday), an attendance register grouped by weekday, the month's turnout pattern, and any holidays or events someone mentions. The Wastage Book records, per item, what was issued, cooked and left, the percentage wasted and the time the item ran out, and it works as a rule: an item that runs out well before close is re-cooked, one that runs out near close is not (confirmed, operator and records). During service the first half hour is used as a sample to project the rest, which drives top-up cooking (single-sourced). That sample is taken before the closing crest, in which about a quarter of Kadamba's diners arrive; whether Yuktāhār has the same crest is unmeasured (candidate).

Kadamba cooks a first batch for about 70 percent of registrations, planned separately for vegetarian and non-vegetarian and portioned by a gram standard per dish (four idlis or 100 g of sambar a head, for example). The rest of the raw material stays ready, and a supervisor watching the counters tells the kitchen how much more to cook, ready in 10 to 15 minutes. No record carries one day's outcome into the next day's ratio, and the manager says of past attempts that ”it was wasted many times” (single-sourced). Biryani lunch, outside the boundary, is cooked at 100 percent of registrations (single-sourced).

Bakul cooks 70 to 80 percent of registrations, the share depending on the item. The manager sets it from past attendance, reviewed weekly: if Mondays have run near 70 percent he cooks for about 80 percent that day. Whether the history is written down or carried in his head is not known. His rule is that there should be no shortage even if some food is wasted, so on low weekends he still keeps food for about 70 percent of registrations. During service he checks every 30 minutes and phones the Hafizpet kitchen with a rough quantity, which is cooked there and driven over on a lead time nobody has stated (single-sourced). (Revised from the first submission, which treated Bakul's ratio as fixed like Kadamba's.)

Against the records, these rules sit well above breakfast turnout. A 70 percent first batch at Kadamba is about twice the 37.1 percent who arrive (candidate). On the Bakul manager's own figures, his first batch and his weekend floor both exceed the highest turnout he expects, about 60 percent, and the margin is larger if Bakul resembles the two recorded halls (candidate; no Bakul records exist). Yuktāhār's site head believes breakfast turnout is about 50 percent, while his own register averages about 28 (confirmed). Even the operator with the fullest books misremembers its own number.

Surplus has four exits. Yuktāhār reuses leftovers: idli becomes idli upma or idli poha for lunch, batter goes back to the fridge, surplus milk is set as curd (confirmed). Kadamba's surplus feeds the 30 to 50 outside diners, for whom about 30 extra portions are cooked, and about half the operator's staff (single-sourced); the team also saw food cooked ahead, reheated and moved between vessels several times before it is eaten, which suggests some surplus is held into later service (candidate). The rest goes to the bin. The earlier finding that composting absorbs the surplus is withdrawn: Kadamba's waste goes to a garbage contractor with no composting (single-sourced), and whether the contractor takes it to the compost facility the institution describes is unverified. The earlier figure of 20 to 30 staff eating leftovers is replaced by the outside-diner figure above.

Kadamba's operator puts daily waste at 5 to 10 percent, without saying of what (single-sourced). A 70 percent first batch against 37 percent turnout could leave that little waste only if the outside diners, reheating and small top-ups absorb most of the difference, or if the figure is measured on another base. Which holds is unknown, and the earlier kilogram figures for Kadamba waste are withdrawn.

The operators read the same count with different aims. Kadamba treats it as its obligation: it cooks to registrations whether or not people want to eat, ”because it is a responsibility and they are getting paid for it.” Yuktāhār names best quality, less waste, lower operating cost and higher profit together as its goals, so waste is a cost to it. Bakul states a third aim, no shortage at the price of some waste (single-sourced for all three).

Gap Between Purpose and Mechanism

Advance planning works only if advance registration expresses an intention to eat. Under automatic registration it does not, because the count records who has not opted out. For breakfast the distortion is large: registrations overstate turnout about 2.7 times at Kadamba and about 3.5 times at Yuktāhār (confirmed, records). Every operator

plans against a number it knows is inflated and discounts it by its own rule. The gap is already measured, in the institution's scan records and in the operators' registers, but nothing carries that measurement into the registration or billing rules, and inside the kitchens it has not produced a discount matched to breakfast turnout. At Bakul it is judged weekly by one manager and may exist only in his head.

Intended Outcomes

- **Every registered student gets a breakfast matching their dietary requirement inside the window.** Under strain at the close. Kadamba reopens the door for 5 to 10 minutes for latecomers, and students arriving after 9:30 also come in by the back door and are served because the mess cannot refuse them (single-sourced). Kadamba and Yuktāhār cook more during the final surge (single-sourced and confirmed). Bakul keeps food for about 50 people at 9:20 against a late tail its manager puts at 10 to 15 arrivals in the last ten minutes (single-sourced).
- **Quantity and quality stay consistent with actual demand, avoiding both shortage and waste.** Not met, and the best-measured failure. Kadamba's first batch is set well above turnout, yet it runs short in the last fifteen minutes when arrivals peak (turnout and arrival peak confirmed from records; ratios and shortages single-sourced). Quality is also at risk. At Kadamba the team judged that reheated, transferred food loses quality (candidate). At Bakul food arrives hot but steam in the boxes softens items such as aloo paratha: the transport keeps temperature and can cost texture (single-sourced).
- **Costs stay affordable and predictable across a semester.** Predictable, not efficient. In April 2026, 20,221 registered breakfasts at Kadamba were billed and not eaten (confirmed, scan records).
- **Access does not depend on navigating informal workarounds.** Not met. Students resell registrations outside the system, and operators describe friends eating on another student's registration.
- **The system responds to changing student needs through its own governance.** Met inside each operator, not between operator and institution. Operators adjust quantities and item mix from what they see, but the institution's exam, event and holiday calendar reaches them only informally, and they have little say over the menu.

Visible Symptoms and Underlying Issues

Visible Symptom	Underlying Issue
About two thirds of breakfast registrations go unused: 37.1 percent of 32,162 were eaten at Kadamba in April, about 28 percent at Yuktāhār in September. Yuktāhār lunch on the same days ran 56 to 89 percent (confirmed, records).	Registration is an automatic default, not a record of the morning's decision to eat (confirmed). The same platform and rules work at Yuktāhār lunch, the one lunch with records, so the cause is specific to breakfast. Timing against the 8:30 class start, late sleep and a four-day-ahead exit are the candidate causes. (Revises the first submission's working gap of about 30 percent, which fits lunch, not breakfast.)
Kadamba's registrations stay at about 1,070 a day on every weekday including Sunday while turnout runs from 42.1 percent (Tuesday) to 27.3 percent (Sunday). Yuktāhār's Sunday breakfasts run 18 to 24 percent, among its lowest. Bakul's manager calls weekend turnout very low (single-sourced).	The count reflects a default set days earlier, with an exit that closes four days ahead, so it cannot follow day-to-day variation in who eats (confirmed).
A student registers, does not attend, and is still charged the full price.	Billing follows registration, so the cost is incurred when the default is left standing, whether or not the meal is eaten.
The portal caps cancellations at five per meal type per month, each at least four days ahead. Beyond five there is no way off the bill (confirmed).	The cap limits exits from a default, not bookings. An explicit limit on leaving suggests unused registrations were frequent enough to need one.
”Skip Meal” lets a student flag a missed meal, but the charge stands.	The tool exists to inform the kitchen and returns nothing to the student. Whether it changes what Kadamba cooks is unresolved: the institution's account says declarations reach the kitchen, Kadamba's operator says it cooks on registrations alone, and no operator names Skip Meal as a planning input (single-sourced).
Students resell unused registrations in a WhatsApp group, ”Mess Cell,” at a negotiated price.	Resale returns value and Skip Meal returns none, so for a student who knows in advance, the informal market beats the official tool.
Unused registrations persist although cancellation and resale both exist.	Both need advance knowledge. A decision made on the morning (oversleeping, a change of plan) has no mechanism at all.

Visible Symptom	Underlying Issue
Walk-in breakfast costs about 70 percent more (Rs. 81 against Rs. 48). At Yuktāhār walk-ins are rare: unregistered, paying and faculty or staff diners together number 0 to 12 per breakfast (confirmed, records).	The price gap gives a standing reason to register even when attendance is uncertain, which inflates counts independent of intention.
Registration relative to capacity differed by cuisine on one September day: 51 to 79 percent at Kadamba's lines, 11 to 36.5 percent at the North Indian lines (Bakul and Palash).	These are registrations as a share of capacity from one portal snapshot, which disagrees with Yuktāhār's own register for the same day (116 registered on the portal, 276 in the register), so it was probably not a final count. Cuisine may influence where students register (candidate); it does not explain turnout. Kadamba's turnout (37.1 percent) is only modestly above Yuktāhār's (about 28), and within Kadamba the most-booked line, vegetarian, has the lowest turnout (32.4 percent against 48.5 for non-vegetarian). No records exist for Bakul or Palash. (Revises the first submission, which read these figures as attendance.)
Arrivals pile up at the end of the window. At Kadamba 26.5 percent of scans fall between 9:15 and 9:30 and 13.5 percent between 9:00 and 9:15, climbing to about nine a minute at 9:29 (confirmed, records). Yuktāhār's operator also describes a closing rush. Kadamba plans for about 150 arrivals in the last ten minutes and serves students after close. Bakul's manager describes an earlier peak, 8:00 to 8:30, with 10 to 15 in the last ten minutes (single-sourced).	Students who come arrive as late as the window allows; late waking is the candidate driver, and the one reason students most often give. Operators must keep food until the last minute, which drives both cooking ”just in case” and shortage at the close. It is the one symptom where shortage and waste occur together. The curve differs by hall: a closing crest is recorded at Kadamba and described at Yuktāhār, but not at Bakul. (Revises the first submission's claim that all three describe a closing rush.)
Kitchens cook well above turnout: Kadamba 70 percent of registrations against 37 measured; Bakul 70 to 80 percent, and about 70 on weekends, against its manager's own ceiling of 60; Yuktāhār's head believes 50 percent against a recorded 28.	Each operator applies its own factor to a count it knows is inflated: a fixed ratio with reactive top-up at Kadamba, a weekly judgement with a shortage floor at Bakul, a book system with first-half-hour projection at Yuktāhār. Only Yuktāhār feeds recorded outcomes back into planning. Bakul's 60 percent is a belief; if Bakul resembles the recorded halls at 28 to 37 percent, it is an upper bound (candidate).
At Kadamba, food is cooked ahead, reheated and moved between vessels, and the team judged its quality to fall fast (candidate).	A first batch far above turnout is held, not discarded, so surplus returns as reheated food. Whether this lowers attendance is not evidenced; it is an open chain, not a loop.
Some items are wasted every time they appear (at Kadamba uttapam, set dosa, upma) while others move beyond the per-head standard (puri, bonda, vada, idli). A student complained that dinner sambar was dropped while charges rose. Bakul's manager names pav bhaji as a draw, muesli and Chocos as items that go, and objects to some pairings students choose, such as Raj Kachori listed with chana where he would serve aloo curry (single-sourced; untested with Bakul residents).	Students and the committee choose the menu, and only operators see item-level waste and uptake. Nothing connects the two. Yuktāhār's records already work item by item on a fortnightly cycle (confirmed) but inform quantities, not the menu.
No student can name an instance of feedback changing registration, billing or hours.	Three channels exist (an in-app rating, a public email address, student-press coverage) and none has a confirmed instance of changing those rules. The one reported change driven by students, the fortnightly menu rotation, is single-sourced. Change does happen inside operators: Yuktāhār began weighing waste in August of last year, made it blame-free, and staff reported waste falling by December; Kadamba trimmed ragi idli and raised plain idli from what sold (single-sourced). None of it reaches institutional policy. The clash between breakfast hours and classes has never been formally raised because no path exists for a cross-authority issue.
Breakfast (7:30 to 9:30) overlaps the 8:30 class start, and separate authorities set the two. A late-waking student says they choose class over breakfast because of the attendance policy.	The mess bodies control operations, the academic office controls the timetable, neither controls the other's schedule, and nothing coordinates them. The conflict can persist because neither side can resolve it alone.
Students are registered for every meal whether or not they are on campus (confirmed). The portal also has a setting to notify a student of a randomly allocated meal.	The count records no opt-out. How random allocation works alongside the default is unconfirmed. (Revises the first submission, which treated only some registrations as system-generated.)
Some students skip for reasons unrelated to the mess: diet, sleep schedule, habit.	Non-attendance has more than one cause. Part of it would persist whatever changed in hours, pricing or process.
Students say attendance is more likely when a friend is going.	For some students breakfast is socially coordinated, so a registration alone does not predict who turns up.

Key Actors, Institutions, Processes, Resources and Constraints

Actors

- **Students.** Registrants and eaters; attendance and diet vary widely.
- **Mess Committee.** A policy body that sets the menu and mess policy with student input; reported to be faculty-chaired (candidate).
- **Mess Office.** The operational arm that runs the portal and issues operational policy. Operators say ”CDS team” and describe a site team that relays complaints; whether this is the Mess Office is unconfirmed.
- **CFS and CDS.** The institutional line above the vendor as the Kadamba operator describes it (single-sourced). Their relation to the Committee, Office and Warden is unconfirmed, and the identity of the CDS head is open.
- **Warden.** Holds vendor management and structural change.
- **Catering operators.** ABC Hospitality Services at Yuktāhār (confirmed); Prism, hired by tender, which runs all Kadamba operations (single-sourced); Vijayalakshmi Caterers at Bakul, from Hafizpet (single-sourced). Whether it also serves Palash is a candidate. (Revises two first-submission readings: Prism as the name of Kadamba's floor team, and Vijayalakshmi's hall as unnamed.)
- **Bakul staff.** A manager, a month in the post but with about 20 years of North Indian food experience, who sets the first batch, checks the counter every 30 minutes and phones Hafizpet for more (single-sourced).
- **Kadamba staff.** A chain under Prism of operations manager, manager, supervisor and service staff, about ten chefs under a main vegetarian chef; the supervisor watches the counters and calls for more cooking (single-sourced).
- **Yuktāhār staff.** Sixteen people, no hierarchy, rotating across every job, in tasting, cutting and cleaning teams who cross-check one another's records. The mess in-charge keeps the accounts and the Wastage Book, and a colleague cross-verifies the grocery register. Two project managers, with some site staff sharing the work, run records and hygiene. There is no leave policy: absence is raised in the daily meeting and colleagues agree to cover (single-sourced). (Revises an earlier five-person project team.)
- **Non-student diners at Kadamba.** 30 to 50 outsiders such as night-shift security, and about half the operator's staff (single-sourced). (Revises an earlier 20 to 30 staff.)
- **Suppliers and the waste contractor.** Grocery, vegetable, milk and paneer suppliers for each operator, and the contractor that removes Kadamba's food waste.
- **Quality control.** A weekly QC inspection; at Kadamba a facility team tastes each meal with two Prism staff before service (single-sourced; the tasting was observed). Faculty visitors are also described as inspecting operator stores (single-sourced).
- **Academic administration.** Sets the timetable; no authority over the mess.
- **Mess Cell sellers.** A subset of registered students.
- **Guests and walk-ins.** Parents, relatives, faculty, staff and other campus affiliates paying the spot rate.

Institutions

- Three governance actors hold different parts of mess governance: the Mess Committee (policy), the Mess Office (operations) and the Warden (vendors and structural change). The Kadamba operator names a different line, CFS then CDS (single-sourced). How the two descriptions map onto each other is unconfirmed.
- The academic administration and timetable authority, independent of all of them.
- The tender and monthly per-plate payment through which each operator is contracted (single-sourced; the plate basis is unknown).
- The operators' record systems. Yuktāhār keeps a set of paper books, all photographed (confirmed): a Wastage Book (per dish: issued, cooked, left, percentage, run-out time, re-cook quantity, registered and eaten counts, student plate waste), a weekday attendance register, a wastage board kept in view, a grocery issue book headed with the day's registrations, a store and ingredient book, a stock register, a vegetable book and indent with spoilage remarks, a milk, fruit, gas, electricity and water register, and fridge temperature and cleaning logs. The issue book must reconcile with what was cooked, and a mismatch is raised in that day's meeting (single-sourced); on the day checked, the issue book and stock register agree (confirmed). The site head describes cross-checking as ”not to find fault,” and weighing was made blame-free on purpose (single-sourced). Together these hold the system's most detailed demand data.
- The food-safety regime: an external ”Eat Right Campus” five-star certification and laboratory testing of suspect

food. At Kadamba every meal passes a fixed sequence: cooking temperature checked, a sample kept for 72 hours so any illness can be traced, a wrapped display plate set at the entrance, serving temperature taken per item, then a tasting by the facility team with two Prism staff and corrections. QC inspects weekly, and water TDS is read daily with a failure triggering a full check (single-sourced; display plate and tasting confirmed by observation).

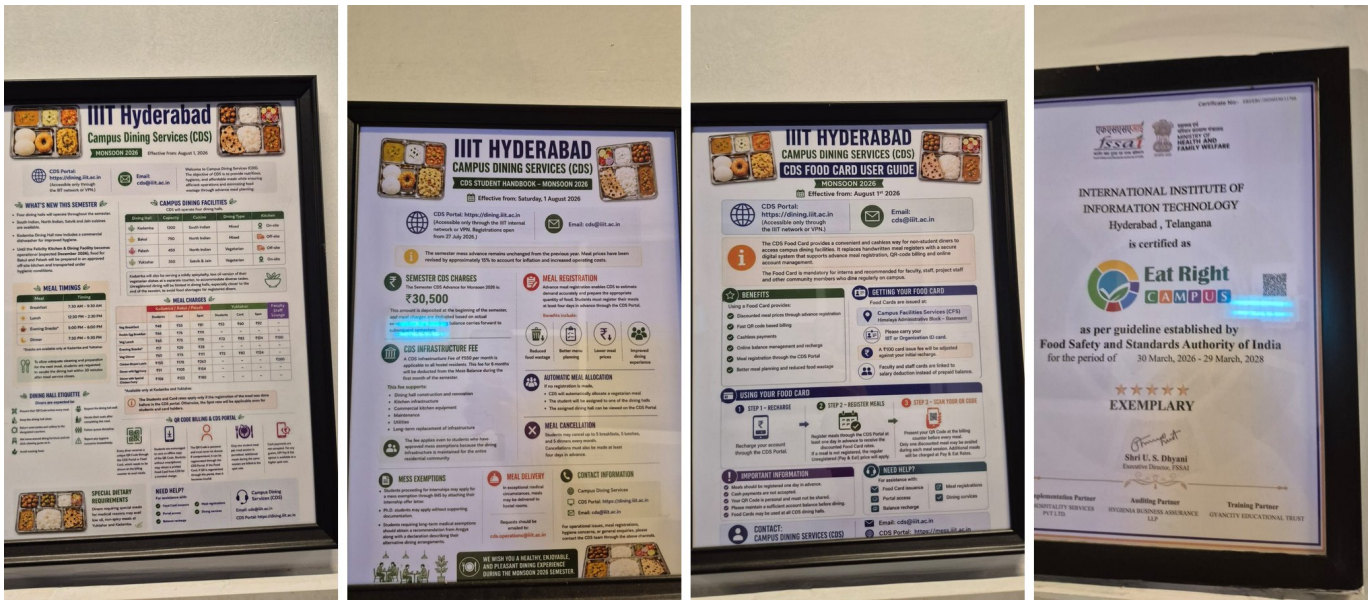


Figure 3. What the dining halls post: the main overview poster, the student handbook, the food-card guide and the food-safety certificate.

Processes

- **Registration.** Students are registered automatically for their hall and diet category, on campus or not. The student's action is to opt out (confirmed).
- **Billing.** Monthly, calculated from registrations, independent of attendance.
- **Cancellation.** At least four days ahead, five per meal type per month, removing the charge. No other route removes a meal from the bill (confirmed).
- **Skip Meal.** A flag to the kitchen that does not affect billing; its effect on Kadamba's cooking is unresolved (see symptoms).
- **Menu setting.** The Mess Committee sets the menu with student input; operators describe a student committee choosing from options they offer. Bakul's manager says students choose combinations from a balanced-diet, protein and preference view (he mentions a target of about 30 to 40 g of protein per meal), while he would pair dishes by traditional and regional pattern and by what people eat; his account of the students' reasoning is untested (single-sourced). The menu runs on a two-week cycle, weeks 1 and 3 repeating and weeks 2 and 4 repeating (confirmed, Yuktāhār records).
- **Procurement.** Each operator buys on its own cadence. Yuktāhār buys groceries fortnightly, vegetables two or three times a week, milk daily (confirmed). Kadamba buys dry goods a week ahead, some vegetables two or three days ahead, and milk, fruit, meat, eggs and most vegetables the day before (single-sourced). Yuktāhār buys from many local shops and godowns, and its store cleans and sorts raw material before issue against an expected impurity norm per item, above which the supplier is challenged or changed (cleaning confirmed, norm single-sourced); the operator visits mills to check how inputs are made (single-sourced).
- **Quantity planning.** The operator reads the closed count the day before and applies its own conversion: Yuktāhār through the meeting and books, producing a written per-meal issue list (confirmed for books and list); Kadamba a fixed 70 percent with gram standards; Bakul 70 to 80 percent by item revised weekly (single-sourced). (Revises the first submission, which had the Mess Office placing food orders four days ahead. The four-day lead applies to registration; the operator sets quantities the day before.)
- **Cooking and top-up.** The first batch is cooked from 5:30 (from about 3 AM for labour-heavy items bound for

Bakul) and at Kadamba passes the food-safety checks before service. At Yuktāhār the first half hour of service projects the rest of the meal; from about 45 minutes before close staff check the counter every 30 minutes, open a second counter during surges, and at peak run two counters of handmade rotis cooked without oil (confirmed). At Kadamba the supervisor calls for more, ready in 10 to 15 minutes on site. At Bakul the manager phones Hafizpet, which cooks and sends it by road on an unknown lead time (single-sourced).

- **Access.** A personal QR code is scanned at the counter to receive a plate matching the registered diet. The code can be reset at any time and staff can reassign it if a student eats at a different hall. When the scanner fails, staff write down ID and name by hand (single-sourced).
- **Late entry.** At Kadamba the door reopens for 5 to 10 minutes after close, and students arriving after 9:30 also enter by the back door and are served; on one biryani day 25 came after close (single-sourced).
- **Walk-ins and extra portions.** A student without a registration may pay the spot rate from a separate, smaller allocation. Kadamba cooks about 30 extra portions beyond registrations, described by the operator as for outside diners such as night security and by an earlier recording as 30 to 40 portions for paying walk-ins. Whether these are one allocation or two is unconfirmed (single-sourced).
- **Leftover reuse and disposal.** Yuktāhār reuses leftovers and weighs student plate waste and kitchen waste (confirmed). At Kadamba held food is reheated for later service (candidate) and waste goes in a dustbin emptied by a garbage contractor, with no composting (single-sourced).
- **Resale.** Outside the portal, a student transfers a registration through Mess Cell for a negotiated payment.
- **Feedback.** Student input arrives through an in-app per-meal rating, a public email address and student-press coverage. Operator complaints pass from the institution's site team or email to the operator's manager and then the kitchen, at Kadamba through a WhatsApp group (single-sourced).

Resources

- The portal and the scan-level record it produces of who ate and when. The institution holds this record, which measures the registration-to-turnout gap for every meal it scans (confirmed for Kadamba breakfast; assumed for other halls and meals).
- The operators' records. Yuktāhār's books are listed above; utility readings feed decisions too, for example a Wednesday electricity peak from grinding idli batter informs how long machines run (single-sourced). Yuktāhār tried a commercial ERP to simplify purchasing and predict the rush and dropped it, judging its books enough for the cost and its staff unsuited to digital tools; office staff digitise the books later (single-sourced). Kadamba holds per-dish gram standards, a live count watched during service, a waste figure, retained 72-hour samples and a daily water-quality log (single-sourced). No written record of attendance, consumption or waste is known at Bakul. The registration-to-turnout gap is therefore already measured locally, by the people who cook.
- Four halls of differing capacity and origin: Kadamba (purpose-built, largest), Bakul (a converted warehouse), Palash and Yuktāhār.
- Production capacity split by cuisine: on-site kitchens at Kadamba and Yuktāhār, and one off-site kitchen with a transport link serving Bakul and Palash, which for Bakul means the Hafizpet kitchen, insulated boxes on the road and a hot-water holding setup at the hall (single-sourced).
- Ingredients chosen on a quality-and-cost stance at Yuktāhār, which has used rock salt and cold-pressed oil for about six months (single-sourced).
- Kitchen equipment and storage, including a cold room on first-in, first-out labels at Yuktāhār (confirmed). Kadamba's toaster is broken and not yet replaced, and an LPG shortage was handled by cutting quantities and buying some items outside (single-sourced).
- The personal QR credential issued to each student.



Figure 4. *Yuktāhār's kitchen on the day of the visit: the registers laid out, the cold room, and staff sorting vegetables at the weighing station.*

Constraints

- Registration is automatic, billing follows registration, and cancellations are capped at five per meal type per month, each at least four days ahead (confirmed).
- The exit closes four days ahead, but Kadamba buys dry goods a week ahead and all other perishables one to three days ahead, so only non-perishable stock is committed before the exit closes and the kitchen's own supply chain does not need the four-day lead (candidate).
- At Kadamba each item passes a cooking-temperature check, a retained sample, a display plate, a serving-temperature check and a tasting before service (single-sourced), so the first batch must be cooked and approved before any sign of the day's turnout exists (candidate). At Bakul the first batch is fixed earlier still, when it leaves Hafizpet to arrive at about 7 AM (single-sourced).
- Cooked food is usable for only three to four hours (single-sourced). Raw material held back can be cooked later or on another day; cooked surplus cannot be recovered beyond a few reuses. The binding constraint on waste is therefore the cooked first batch, set as a share of registrations (fixed at Kadamba, adjusted weekly at Bakul), not raw procurement (candidate).
- Cooked quantity can rise during service through top-up; nothing lowers it once the first batch is cooked. (Revises the first submission, which said nothing lets same-day activity change what is cooked.)
- A Bakul top-up needs a phone call, cooking at Hafizpet and a 20 to 40 minute drive, against 10 to 15 minutes on site at Kadamba (single-sourced). Cooking time is unstated, so the true lead time is unknown. Late in the window a top-up cannot arrive before close, so food already held is the only cover for late arrivals; the manager's reserve of about 50 portions at 9:20, several times the 10 to 15 late arrivals he describes, reads as that cover (candidate).
- Bakul keeps food for about 70 percent of registrations even on low weekends (single-sourced), so any change to its quantities meets that floor first.
- Food cooked off site must be held hot from dispatch to eating, at about 60 to 72 °C in insulated boxes. This keeps temperature and can cost texture (single-sourced for aloo paratha; untested for other items).
- Kadamba cannot hold excess stock (single-sourced). Since dry goods are bought a week ahead, the limit reads as applying to perishable and cooked stock (candidate).
- Cutting first-batch surplus at Kadamba also cuts food that 30 to 50 outside diners and about half the operator's staff rely on (candidate).
- Walk-in pricing is set about 70 percent above registered pricing.
- Vegetarian and non-vegetarian food is segregated by registration type at service and planned separately at Kadamba; Palash and Yuktāhār serve vegetarian food only.
- At Yuktāhār records are on paper, staff are not digitally inclined, and waste measurement took hold only once it was made blame-free (single-sourced).
- Operators control quantities, but not registration rules, billing rules or the menu (consistent across three operators).

- Exams, events and holidays reach a kitchen only informally, through whoever raises them in the planning meeting (confirmed).
- Mess hours and the class schedule are set independently by separate authorities with no coordination mechanism.
- Authority over mess operations is split across the Mess Committee, Mess Office and Warden, with CFS and CDS named by the Kadamba operator as its line; how these relate and who heads CDS is unconfirmed. The academic administration has no formal role.
- Record-level turnout exists only for Kadamba breakfast in April 2026 and Yuktāhār in September 2026. Anything said about Bakul, Palash, or lunch and dinner outside Yuktāhār is an assumption until records are obtained.

Initial Problem Framing

Most breakfast registrations go unused. At Kadamba in April 2026 about 63 percent of registered breakfasts were not eaten; at Yuktāhār in September about 72 percent. At Yuktāhār lunch the same system is mostly used, with 56 to 89 percent of registrations eaten. Students are charged for every registered breakfast whether they eat it or not.

The kitchens know registration overstates breakfast, so none cooks for every registration. Each cooks to its own discounted guess and tops up during service. Kadamba's first batch is still far above what arrives, while arrivals pile into the last fifteen minutes and cause shortages at the close. Surplus is partly reused, partly eaten by outsiders and the operators' staff, at Kadamba possibly held and reheated into later service, and the rest is thrown away. Kadamba's operator puts waste at 5 to 10 percent a day, without saying of what, which is hard to square with a first batch near twice the turnout. Bakul shows the same pattern on its manager's own account. He estimates at most about 60 percent come, cooks for 70 to 80 percent, and keeps food for about 70 percent even on low weekends, because his rule is no shortage even if some food is wasted (single-sourced; no Bakul records). He attributes low turnout partly to students not waking and partly to taste and dish pairing. The first agrees with what students interviewed elsewhere most often say. The second, like any effect of food softened in transport, is a candidate influence untested with Bakul residents.

Students do not choose to book breakfast; they are booked, on campus or not. To avoid paying for a breakfast they will not eat, they can cancel it at most five times a month per meal and at least four days ahead, or mark it skipped, which returns nothing. A student who decides on the morning itself has no option. Students built their own workaround, reselling registrations, because it pays back more than the system's tools do.

The problem: breakfast registration does not reflect who is going to eat, by a margin of roughly two thirds, while the same system works for lunch at Yuktāhār, the one hall with lunch records. Why not, and what in the system causes that gap? And why does each kitchen have to guess its own conversion from registrations to plates, when the data needed to calibrate it is already being recorded, and when one operator learns from its own books, another applies a fixed ratio, and a third adjusts its ratio by judgement?

Stakeholder Map

IIIT Hyderabad Breakfast Mess System

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Use Case

The breakfast system at IIIT Hyderabad registers, bills and serves breakfast to resident students in four messes (Kadamba, Palash, Bakul and Yuktāhār) through one digital platform. Registration counts, not attendance, drive billing and the quantity each operator cooks. Three caterers run the four messes. Two cook on site, and Bakul's, Vijayalakshmi Caterers, cooks at the Hafizpet kitchen and trucks the food in. This map identifies every actor, institution and informal structure that takes part in the system, shapes it, benefits from it or is affected by it, and the relationships, dependencies and conflicts between them.

Operator visits, the Yuktāhār registers, Kadamba's scan-level April records and a later account of the Bakul manager overturned several first-version findings about the vendors, the four-day lead time and who places food orders. Each change is marked as revised where it appears. There are no Bakul records, so every Bakul figure here is the manager's own statement or estimate.

Stakeholders

Primary, Secondary, and Indirect Stakeholders

Thirty-two candidate actors were identified through seventeen lenses (student, social, food production, governance, money, technology, academic, sleep and time use, substitutes, campus infrastructure, waste, health, policy, informal systems, absent or sleeping actors, temporal, intervention). Seven became visible only once the operators' own practice was observed: operator site management, the CDS site team, suppliers, the garbage contractor, event and exam organisers, quality-check visitors and the food-safety regulator. One more, the Student Parliament's mess seats, came from the Parliament's published records. Twenty-four are in the final map.

Five were treated as minor. Cleaning staff and the juice canteens add no leverage beyond serving staff and Vindhya. Guest diners are counted by the operators themselves and stay too small for a full stakeholder: Yuktāhār's register shows none to twelve unregistered, paying or faculty and staff diners per breakfast on the legible September days (confirmed, records). Quality-check visitors, described by Kadamba's manager as IIIT faculty who inspect the store

and by the operator as a weekly QC inspection, are read as a role of the Mess Committee's faculty, which is a candidate identity. The food-safety regulator certifies the campus but has no part in attendance. Three were excluded: a campus health office (its only link to the messes is food safety), senior-to-junior norm transmission, and a parental financial relationship (mess fees are bundled into hostel fees).

Revised: Kadamba's extra portions were first described as 30 to 40 for paying walk-ins, with security staff sharing a leftover meal. The operator's own account is that about 30 extra portions are cooked in advance for 30 to 50 people outside the student registration, such as night-shift security, and that about half of Kadamba's staff also eat from the same food (single-sourced). Together they are a large, uncounted sink for first-batch surplus (candidate), so they now count as a stakeholder: non-student diners fed at Kadamba.

Primary stakeholders: direct, daily interaction with the system.

Stakeholder	Type
Students	Human group: registrant, billed party, consumer
Operator: Prism, Kadamba (hired by tender; South Indian, veg and non-veg, on-site kitchen)	Commercial
Operator: Vijayalakshmi Caterers, Bakul (North Indian; cooks at Hafizpet; Palash served from the same off-site kitchen, candidate)	Commercial
Operator: ABC Hospitality Services, Yuktāhār (Satvik vegetarian and Jain, on-site kitchen)	Commercial
Operator site management (Kadamba: operations manager, manager, supervisor; Yuktāhār: project managers, in-charge, coordinator, store function; Bakul: the manager who phones Hafizpet)	Human group: decides tomorrow's quantities and today's top-ups
Mess serving and counter staff	Human group
Kitchen and cooking staff	Human group

Secondary stakeholders: govern, execute or manage the system without daily direct interaction.

Stakeholder	Type
Mess Committee	Institution: policy
Mess Office (candidate: CDS, Campus Dining Services)	Institution: operations, registration rules
CDS site team	Institution: day-to-day liaison with operators
Warden	Individual role: vendor management, structural change
Student Parliament Mess Secretary, and a Mess Deputy Secretary per mess (Kadamba, Palash, Yuktāhār)	Elected roles: student representation

The institution's posters and the operators name a Campus Dining Services (CDS) body under Campus Facilities Services (CFS), with a faculty CDS Committee, a CDS Student Council and one admin per mess. Kadamba's operator describes its contract chain as CFS, then CDS, then Prism (single-sourced). Five identities are unconfirmed and kept apart here: whether CDS is the Mess Office, whether the CDS Committee is the Mess Committee, whether the CDS Student Council is the Parliament's mess seat or part of it, whether the Warden's vendor and structural role belongs to CFS, and who leads CDS. The Kadamba operator names one person as head of CDS and the CFS Chair earlier named another as CDS Chair, which may be two roles or one wrong name. The first identity is now the stronger candidate, because the CDS team supplied the scan-level records and its site team is the operators' day-to-day counterpart.

Indirect stakeholders: affected by or connected to the system without direct engagement, including sleeping actors that are inert now but hold potential leverage.

Stakeholder	Type
Academic administration and timetable authority	Institution: latent
Event and exam organisers (clubs, hackathons, other campus food events)	Institutional and informal: latent demand shifters
Facilities and Estate team	Institution: latent
Portal and IT system owner	Institution or individual: latent
Suppliers (local shops and godowns, vegetable vendors, milk, paneer, oil and flour mills, LPG)	Commercial: upstream supply

Stakeholder	Type
Garbage contractor	Commercial: waste exit
Non-student diners fed at Kadamba (night-shift security and other outside workers; about half of Kadamba's staff)	Human group: surplus absorber
Mess Cell WhatsApp community	Informal group
Unofficial mess-tool developers	Individuals: latent
Ping (student publication)	Institution: latent
Vindhya Canteen and the juice canteens	Commercial: substitute
Delivery platforms and riders	Commercial: substitute

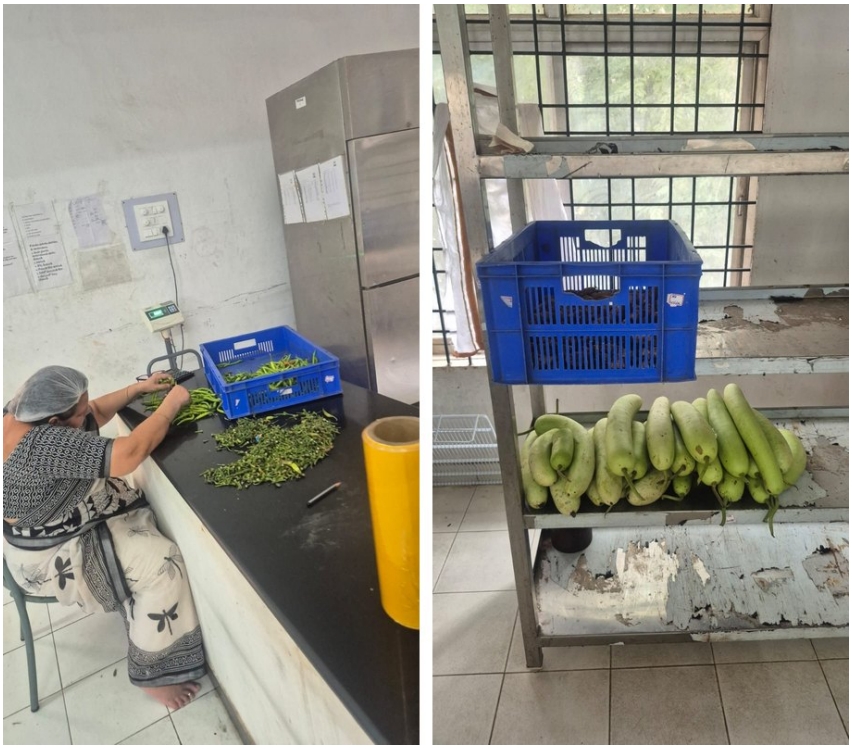


Figure 1. Staff at Yuktāhār's weighing station and the vegetable store, the people behind the books.

Stakeholder Interests, Needs, and Expectations

Stakeholder	Interest	Need	Expectation
Students	Convenient, affordable, timely food; not paying twice for a missed meal; keeping menu items they like	A working registration and billing system, predictable hours, food matching their diet, and entry when they arrive in the last minutes before close	Registering reliably means being fed if they attend; being charged regardless is a tolerated rule, not an agreed one. Registration is automatic and mandatory, even on holidays or off campus, and the only exit is five cancellations per meal type per month, at least four days ahead (confirmed). It is a default, not a daily decision: April breakfast registrations sat near 1,070 every day while turnout ran from 42.1% on Tuesdays to 27.3% on Sundays (confirmed, records). A Kadamba student objected in person that dinner sambar was dropped ”while they are taking more money from us”

Stakeholder	Interest	Need	Expectation
Operator: Prism (Kadamba)	Renewal and margin under a tendered per-plate rate; whether payment is per registered or served plate is unconfirmed	Registration counts split veg and non-veg, released about four days ahead; dry goods a week ahead, most perishables the day before; a raw reserve cookable in 10 to 15 minutes; live counter counts	Revised: the expectation that the four-day count approximates demand fails for every operator. Kadamba cooks about 70% of registrations first against 37.1% turnout (confirmed, records) and treats the count as an obligation it is paid for. Its manager says there is no forecasting method (”it was wasted many times”). It puts waste at 5 to 10% a day on an unstated base, which fits a 70% batch against 37% turnout only if extra diners, reheating or small top-ups absorb the rest (single-sourced)
Operator: Vijayalakshmi Caterers (Bakul; Palash candidate)	Renewal and margin, paid monthly as a lump sum with deductions; says payment is on plates served (single-sourced, flagged under Dependencies). Its manager's principle is ”there should not be a shortage of food, even if some wastage occurs”, and he wants a say in the menu, above all in pairings	Registration counts; its own supply (paneer moved from Amul to Heritage to Milky Mist after quality complaints); a phone line to Hafizpet and a 20 to 40 minute transport leg for every top-up, with total lead time unstated	Revised: the reading that Bakul cooks a fixed 70%, lately raised to 80%, is withdrawn. The first batch is about 70 to 80% by item, set from weekly review of attendance history, and about 70% even on very low weekends. The manager checks stock about every 30 minutes and estimates at most about 60% of registered students come; no records exist to test it. He is a month into the post with about 20 years in North Indian food (single-sourced)
Operator: ABC Hospitality Services (Yuktāhār)	Renewal and margin, with an explicit goal of best quality, less waste, an optimised workflow, lower cost and higher profit, in which waste cutting is a cost lever; keeping prediction in its own hands (”I want to remain close to the data”)	Registrations read as regular plus Jain plus cash meals; its Wastage Book for the matching week; the weekday attendance register; the month's pattern; and news of holidays and events, which arrives only if someone raises it in the planning meeting. The first half hour of service is a sample to project the rest (single-sourced)	Plans from its own records yet believes breakfast turnout is at most about 50%; its September register shows 17.7 to 40.2%, about 28% across the thirteen legible days (confirmed, records)
Operator site management	Getting tomorrow's quantities right; staff buy-in to waste tracking	Yuktāhār: a twice-daily meeting with the books on the table. Kadamba: the supervisor's reading of the counters. Bakul: the manager's weekly review, 30-minute checks and a reserve for about 50 people at 9:20. Whether Bakul's review is written down is unknown	Waste measurement is for improvement, not fault-finding. Yuktāhār's cross-checkers are paid less than the person checked, weighing was made blame-free and discussed daily, and staff who first said ”they are catching us” reported waste falling by December (single-sourced)
Serving and kitchen staff	A manageable load and predictable headcounts	Live counts, a second counter during the closing surge (Yuktāhār opens two roti counters at peak, confirmed by observation), reserve raw material, and a working QR scanner (on failure, ID and name are written by hand). Kadamba's chef starts at 5:30 and dosa is cooked live from 7:00. At Hafizpet, labour-heavy items such as aloo paratha may start about 3 AM for a 7 AM arrival, and steam in the boxes softens them (single-sourced)	Revised: the figure of 20 to 30 cooking and security staff eating what students leave is replaced by about half of Kadamba's staff eating from the same food. Whether that is an entitlement or a norm is unconfirmed (single-sourced)
Non-student diners fed at Kadamba	A meal on shift without leaving campus	Cooked food available alongside student service, planned through the extra portions	Food is simply there. They absorb first-batch surplus that no waste or attendance record counts (candidate)

Stakeholder	Interest	Need	Expectation
Mess Committee	Balancing student satisfaction against feasibility; menu authority, exercised by choosing from operator-offered options	A working way to gather and read student input	The menu-input process is enough for mess matters generally, including hours and billing it was never built for. No source confirms that operators' item-level waste figures reach it
Mess Office (candidate: CDS)	Lower operational workload, the stated motive behind its one policy change on record (candidate, from a secondhand press summary); ownership of the four-day registration rule	Revised: the earlier ”need” for a same-day picture of intent does not hold, since the CDS team already holds a scan for every registration (11,941 of 32,162 Kadamba breakfasts availed in April). It lacks a route from those records into the rules	Counts are a workable proxy for attendance. Its own records contradict this, and no operator acts on it
CDS site team	Keeping operations and complaints moving	A direct line to each operator: a WhatsApp group relaying complaints and instructions (single-sourced)	Not established
Warden	Stable vendor relationships, orderly infrastructure	A clear split of responsibility from the Mess Office	Vendor and structural authority stay in the Warden's own lane
Student Parliament Mess Secretary and per-mess Deputy Secretaries	Representing students on mess matters	A working channel into the Mess Committee and Mess Office	Not established. Whether the seat carries a vote in the institute-level Committee or only in a Parliament-internal one is unconfirmed
Academic administration	Class scheduling, run without outside interference	No stated need connects it to the mess	No expectation of any relationship to mess governance; confirmed to hold no formal mess role
Event and exam organisers	Running their own events	None connects them to the messes	Not established. Exams and hackathons cut turnout and a food event elsewhere means a kitchen should cook less, but nothing formal tells the kitchen (single-sourced, Yuktāhār operator)
Facilities and Estate team	Infrastructure upkeep	Budget and approval for mess-facility work (the Kadamba renovation occurred; its toaster and grill are broken with no replacement, single-sourced)	Not established; the actor's identity is itself unconfirmed
Portal and IT owner	Uptime and reliability of registration	Working scanners; the system already gives operators their counts, the counter live scans and CDS a timestamp per scan	Not established; no source names this actor
Suppliers	Steady orders and repeat business	Orders at each operator's cadence. Yuktāhār buys groceries fortnightly from many local shops and godowns (confirmed, stock register), vegetables two or three times a week and milk daily; Kadamba buys dry goods a week ahead, most perishables the day before and some vegetables two or three days ahead (single-sourced)	Operators check and return poor stock. Yuktāhār holds an expected grams-of-waste norm per raw item (50 g of stones in 2 kg of grain was weighed), complains to or changes the supplier when it is exceeded, visits the mills that make its oil, and has used rock salt and cold-pressed oil for about six months. Vijayalakshmi Caterers changed paneer brand twice (single-sourced)
Garbage contractor	A collection business	Daily access to each operator's bins	Not established. At Kadamba all uneaten cooked surplus goes to the bin and then to this contractor, with no composting (single-sourced)
Mess Cell community	Recovering value from a registration that will go unused	A way to transfer a registration, since Skip Meal returns no money	Buyers and sellers find each other and trade on trust. Kadamba's manager holds a related belief that students register while away and a friend eats in their place (single-sourced; his belief, not an observation)

Stakeholder	Interest	Need	Expectation
Unofficial mess-tool developers	Not established beyond building the tool	Not established	The tool's own documentation says it is built ”for students of IIIT Hyderabad”
Ping	Reporting on mess issues as student interest	Material worth publishing	Not established whether coverage produces any institutional response
Vindhya Canteen and juice canteens	Capturing demand when mess breakfast is skipped	Steady walk-in flow	Not established
Delivery platforms and riders	Capturing the same demand	Gate access, currently restricted	Not established

Formal and Informal Roles

Every institutional and commercial actor holds a formal, rule-defined or contract-defined position: students as registrants, the Mess Committee, the Mess Office, the CDS site team, the Warden, the three operators, serving and kitchen staff, suppliers, the garbage contractor, the canteens and the delivery platforms.

Formal roles inside the operators. Each operator runs its own organisation, which the institution does not see. The two on-site ones are opposites, and Bakul splits across two sites.

Kadamba sits at the end of a contract chain from CFS to CDS to Prism, and inside Prism from an operations manager to a manager, a supervisor and the service staff (single-sourced). A separate staff recording adds a director above the operations manager; reading that as Prism's ladder is a candidate. The supervisor is the in-service quantity decider: they watch the counters, track how much of each item is left at what time and tell the kitchen how much more to cook. The kitchen has about ten chefs with separate veg and non-veg cooks, a main vegetarian chef, and a head chef who joins on biryani days. Up to fifty staff work at the mess (single-sourced). Every meal passes a fixed safety sequence before service: cooking temperature checked, a sample kept 72 hours, a wrapped display plate set at the entrance (confirmed, observed), serving temperature per item, then a tasting by a facility team with two Prism staff (confirmed, observed) and corrections. Water TDS is read daily and QC inspects weekly. Who the facility team is (CDS site team, CFS, Facilities and Estate, or faculty visitors) is not confirmed and is kept separate from each.

Yuktāhār has no hierarchy: everyone does every job and peers cross-check. ABC runs the mess through two project managers, with about three site staff sharing the work (single-sourced; revised from a five-person team). The mess in-charge keeps the accounts and the Wastage Register, and colleagues check those records. A cross-verifier writes and checks the grocery register, and their signature is on the photographed fridge cleaning log (confirmed). A co-ordinator keeps the registrations, the weekday attendance register and a wastage board with a falling waste-share target. The store function pulls, sorts and cleans raw material but is nobody's dedicated job. A tasting pair checks every breakfast, with cutting and cleaning teams alongside, and sixteen kitchen staff start cooking at 5:30 to be ready by 7:30. A site head has run the mess since the takeover and writes in the registers daily; whether they are one of the two project managers is unconfirmed. Each role attaches to a book (Wastage Book, weekday attendance register and board, grocery issue book, store ingredient book, stock register, vegetable book and indent, milk, fruit, gas, electricity and water register, fridge logs), all photographed (confirmed), and each book is checked by someone other than its keeper. What was issued must match what was cooked, and a mismatch goes to the daily meeting; on 18 September the grocery book and stock register agreed on toor dal (confirmed). An office digitises the paper books later.

Bakul (Vijayalakshmi Caterers) splits two sites: Hafizpet cooks, and the mess holds, serves and decides. The operator describes a line from staff to the manager to a managing director, and a complaint route from the CDS team to the manager, the kitchen and the managing director (single-sourced). The manager is the single quantity decider at the mess. This puts Bakul between the other two, with a weekly judgement rather than Kadamba's fixed share or Yuktāhār's books and no book of its own that any source has seen (candidate).

Informal roles belong to the Mess Cell community, Ping and the tool developers. None has official standing and each does a real job: a resale market, a public-pressure channel and an alternative registration interface. The operators add informal structures of their own. Yuktāhār's twice-daily meeting is the system's real forecasting institution, and no institutional document names it. It also carries Yuktāhār's informal organisation. There is no leave policy, so absences are raised there and colleagues agree to cover. One project manager took all staff to his farmhouse for training on what good quality looks like, and staff describe a family-like culture (single-sourced). That informal layer

holds the formal books together. The WhatsApp group between the CDS site team and each operator carries formal business informally. At Kadamba, meals for outside diners and half the staff, the door reopened for 5 to 10 minutes after close (25 students came after close on one biryani day) and the back door that, in the operator's words, the mess ”can't say no” to, are accommodations that absorb surplus and lateness with no rule requiring them (single-sourced).

Roles that are not separate stakeholders. Attending regularly, skipping regularly, having an early class, selling on Mess Cell and using an unofficial tool are states one student moves through in a week. Peer influence is a relationship inside the Students node, not a separate ”Friends” stakeholder. Guest diners have a different formal relationship (no registration, a higher spot rate) but too little volume for their own node. Students also act as menu selectors. In the Bakul manager's account, students choose combinations from balanced diet, protein, carbohydrate and preference, while he reasons from traditional pairings, regional habit and what is eaten (single-sourced; residents have not been asked). Whether residents differ in regional familiarity, diet and breakfast habit in ways that matter for menus is untested, so menu research at Bakul should sample along those lines (research implication, candidate).

Create the Map

Stakeholder Map

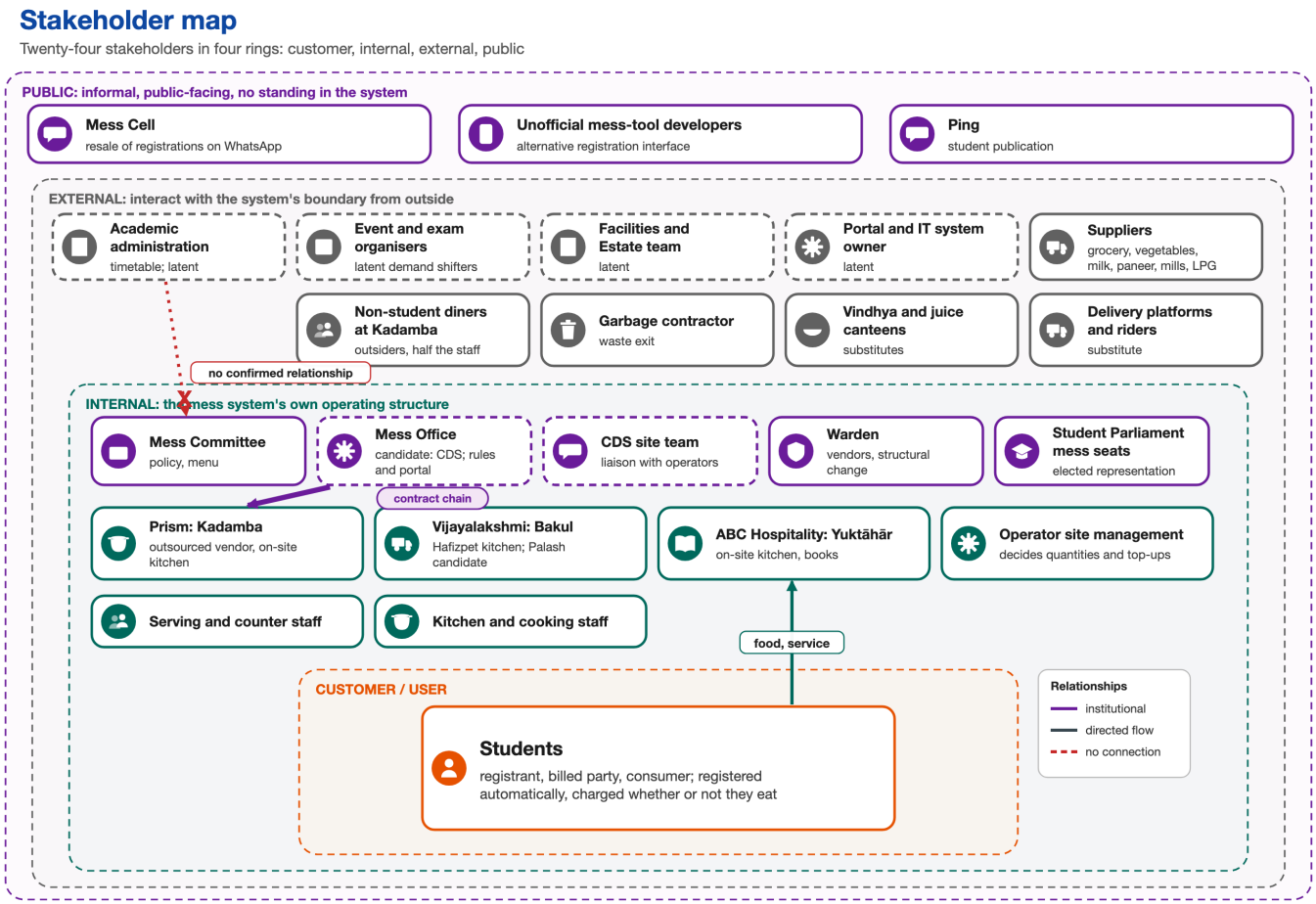


Figure 2. Stakeholder map in four rings: customer, internal, external and public.

Legend: relationship types

Style	Relationship
Solid line, plain label	Relation
Dotted line	Unclear, informal relationship

Style	Relationship
Thick line	Institutional relationship
Solid arrow, labeled	Directed flow of information
Red line	Relationship with conflict potential
Dotted line ending in an X	Interrupted relationship (no confirmed connection)

Placement follows the four-ring model. The customer or user is Students. Internal stakeholders are the Mess Committee, Mess Office, CDS site team, Warden, the Parliament's mess seats, the three operators with their site management, and serving and kitchen staff. External stakeholders are Academic administration, event and exam organisers, Facilities and Estate, the Portal and IT owner, suppliers, the garbage contractor, the non-student diners (placed next to the operators), the canteens and delivery platforms. Public stakeholders are Mess Cell, the tool developers and Ping: informal, public-facing actors with no standing in the system.

Relationship Network

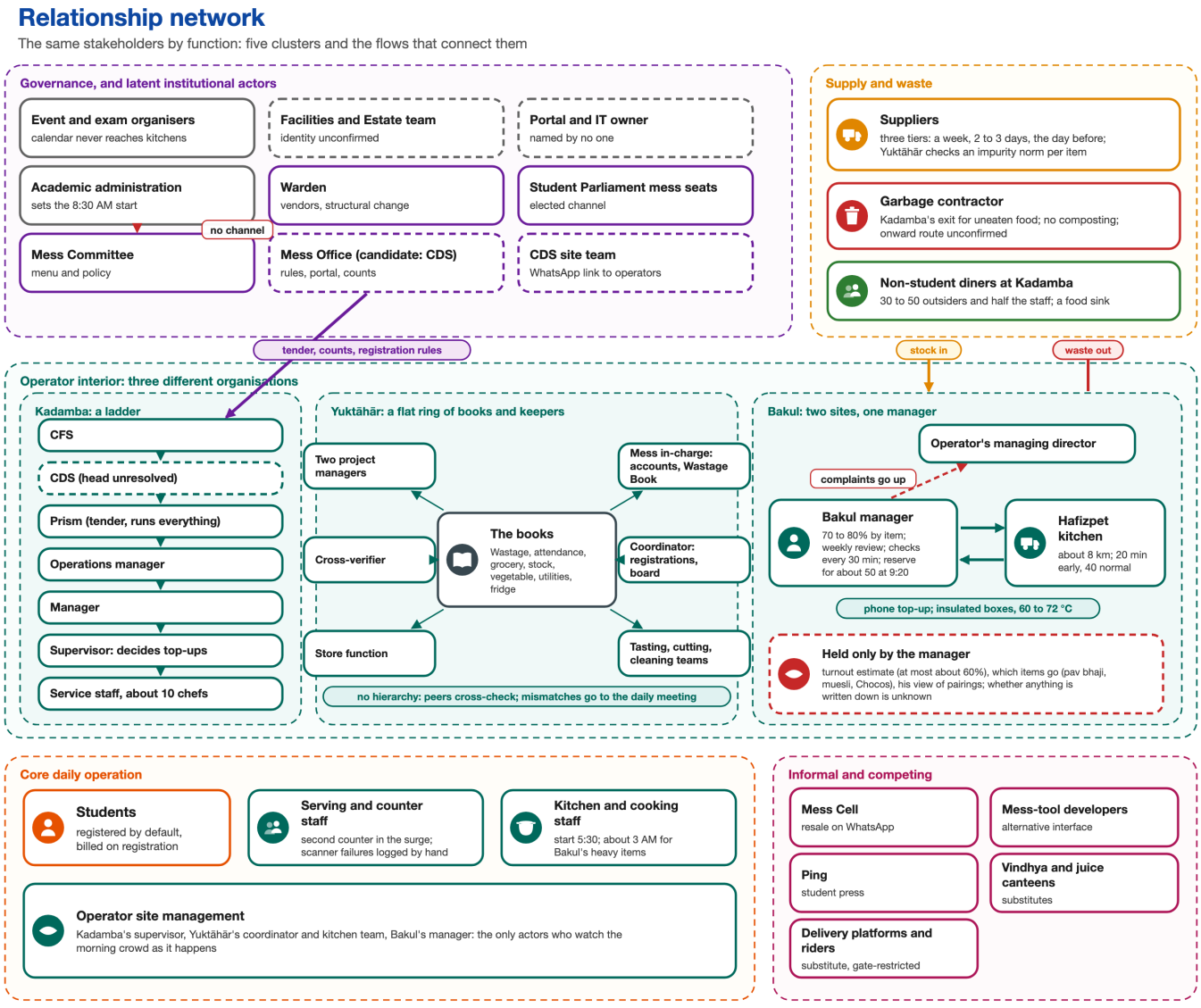


Figure 3. *Relationship network: five clusters, with the flows of authority, money, food, information and complaints.*

The same twenty-four stakeholders fall into five clusters: core daily operation, operator interior, governance, supply and waste, and informal or competing actors. The operator interior holds two opposite organisations, Kadamba's ladder from CFS through CDS to Prism and the supervisor, and Yuktāhār's flat, peer-checked ring of books and keepers. Bakul adds a third arrangement: one manager at the mess, linked by phone and road to a kitchen 8 km away. The manager takes registration counts from the portal and sets the first batch from a weekly review of attendance history, 70 to 80% by item and about 70% on low weekends. He checks stock about every 30 minutes, phones Hafizpet for top-ups on a lead time nobody has stated, and holds a reserve for about 50 at 9:20. Complaints run from the CDS team to the manager, the kitchen and the operator's managing director. Money arrives as a monthly lump sum on plates ”served”, with deductions (basis flagged). The manager's turnout estimate, his view of which items go and his view of good pairings are held only by him, routed nowhere known (single-sourced; no Bakul records exist).

Proximity to the Student's Lived Experience

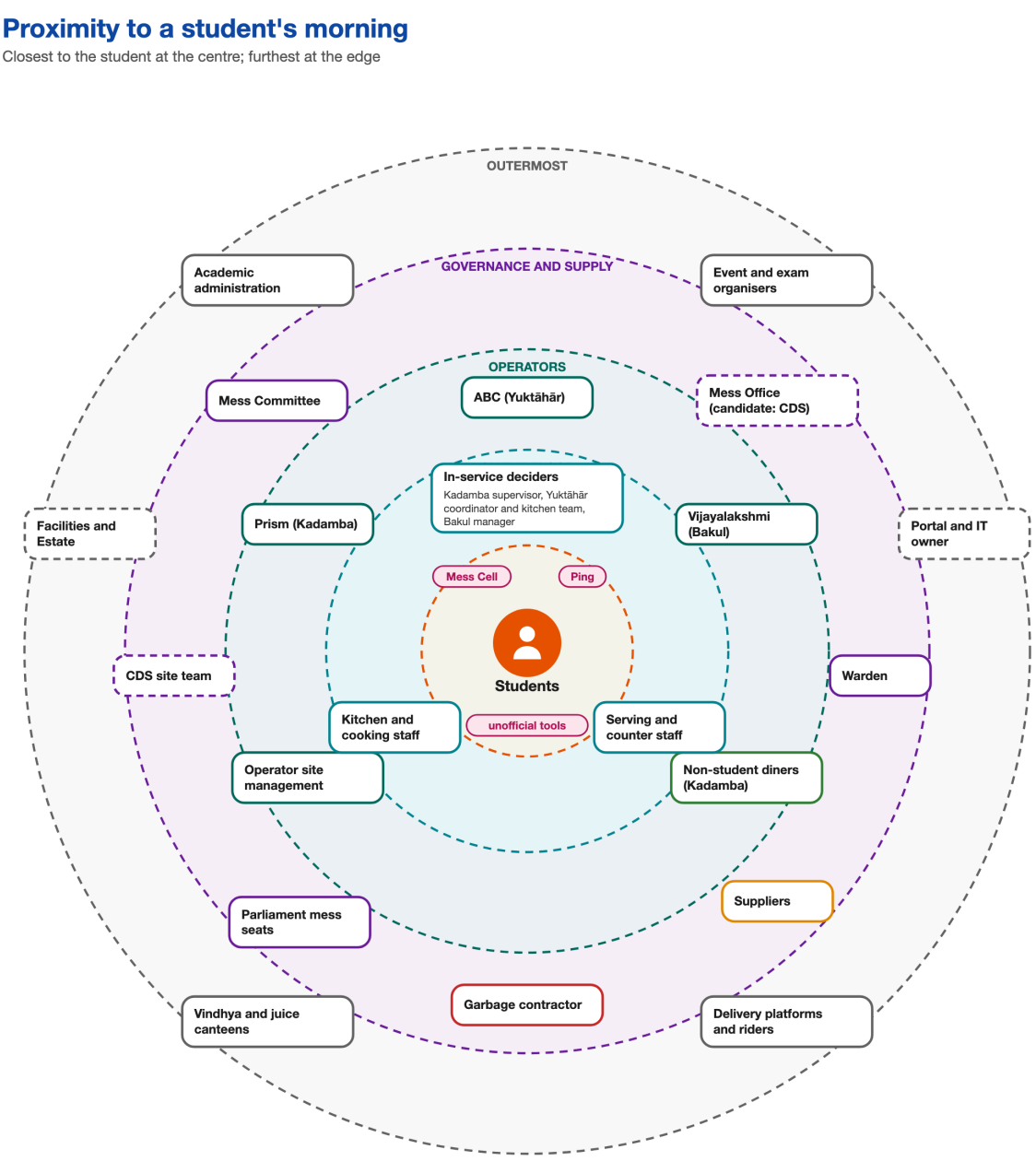


Figure 4. Proximity map: how close each stakeholder is to a student's actual morning.

Clustering by closeness to a student's actual morning gives a different picture from the two maps above. Academic administration and the event and exam organisers sit in the outermost ring with no adjacent relationship at any layer, while Mess Cell, the unofficial tools and Ping, none of them part of formal governance, sit in the innermost ring beside the students. The counter staff and the in-service deciders (Kadamba's supervisor, Yuktāhār's coordinator and kitchen team, Bakul's manager) also sit close. They are the only actors who see the morning crowd as it happens.

Relationships, Dependencies, and Conflicts

Relationship Matrix

Cells marked *hypothesis* are not independently confirmed. Blank cells mean no source describes a relationship.

	Students	Mess Cttee	Mess Office (CDS)	Warden	Academi admin.	Operators	Staff	Mess Cell	Ping
Students	n/a	Rating system exists; whether read is un-confirmed	Register and pay; automatic, 5 cancels per meal type per month		Bound to its timetable	Counter service, complaints, late entry	Service	Pay, serve	
Mess Committee	Authority (menu, feedback)	n/a	Sets policy	Hypothes overlap	None	Menu from operator-offered options; faculty store visits (candidate)			Hypothesis: subject of cov-erage
Mess Office (CDS)	Billing, rules (four-day exit)	Hypothesis: reports to	n/a	Hypothes overlap	None	Contract chain CFS, CDS, Prism; counts at four days; WhatsApp complaint relay; monthly pay with deductions; facility tasting; scan records not shared back	Indirect		Hypothesis: subject of cov-erage
Warden		Hypothesis	Hypothesis	n/a	None	Contract authority			
Operators	Food, plate-waste weighing, late entry	Menu options; item-level waste data with no route to the Com-mittee	Depend on it for counts and pay	Under con-tract	Event news only by chance	Compete for the same diner	Kadamba: Prism ladder. Yuktāhār: flat, peer checks. Bakul: staff, manager, MD, phone to Hafizpet		
Staff	Service; half of Kadamba's staff eat the same food					Report to (Kadamba); cross-check and taste in pairs (Yuktāhār)	n/a		
Non-student diners (Kadamba)	Eat the same food					Cooked for in advance, about 30 extra portions; outside registration unconfirmed	Share the food		
Suppliers						Supply, impurity norm, returns, brand switching, mill visits			
Garbage contractor						Daily bin collection (Kadamba); onward route unconfirmed			
Mess Cell	Pay, serve, informal influence							n/a	

The empty cells matter as much as the full ones. Academic administration has no confirmed relationship of any kind to any of the three governance actors. That is a structural gap, not a modelling omission, and the most consequential finding in this map. The operator side shows the same gap from below, since the academic and event calendar reaches a kitchen only if someone mentions it in a meeting. The Mess Office row shows a second gap: the institution and the operators each measure turnout, and nothing connects the two measurements.

Dependencies

Students depend on operators for food quality and timing. Operators depend on no individual student, an asymmetry consistent with a walk-in price that needs no negotiation. Operators depend on the Mess Office for counts released four days ahead and on the contract for payment. Revised: operators do not depend on that count to procure food. Procurement is theirs and runs on its own cadence (Yuktāhār groceries fortnightly, vegetables two or three times a week, milk daily, confirmed from its registers; Kadamba dry goods a week ahead, most perishables the day before). What the count sets is how much cooked food is prepared, since every operator converts registrations into a cooked quantity and cooked food lasts only three to four hours. Kadamba uses a fixed first-batch share. Bakul uses 70 to 80% by item, adjusted weekly and not below about 70%. Yuktāhār uses its records, its meeting and a first-half-hour projection. The Mess Office depends on students to register accurately, but nothing in billing rewards accuracy: a student pays the same whether or not they attend and is registered automatically unless one of the capped cancellations is used. The weakest link in the chain has no incentive attached. The Mess Committee depends on the Mess Office to execute policy. The Office's one confirmed policy change, motivated by workload (candidate), did not visibly come from the Committee, which suggests operational pressure may drive policy from below as often as governance drives it from above.

Operators depend on suppliers for clean raw stock, on the garbage contractor to remove what is not eaten and, at Bakul, on the road. Revised: that leg was first given as 40 minutes from a kitchen near the airport, with top-ups arriving within 30 minutes. The kitchen is about 8 km away, a drive of about 20 minutes early in the morning and 40 in normal traffic, and a top-up needs a phone call, cooking and that drive. Total lead time is unstated, so the 30-minute figure survives only as a noisy-audio reading that transit alone makes doubtful (single-sourced). A top-up called late in the window would miss the close, which is a candidate reason the manager holds food for about 50 at 9:20, several times the 10 to 15 he says arrive in the last ten minutes. The same boxes keep food hot and soften it, so Bakul's quality depends on the transport link as well as the kitchen (aloo paratha only; other items untested). Kitchen staff depend on the QR scanner and portal for live counts. Operators also depend on one another without planning to: Yuktāhār's site head blames weak Wednesday paneer turnout on another mess serving something more attractive that day, so the messes substitute for each other as well as for Vindhya, the juice canteens and delivery.

A dependency runs the other way at Kadamba. The outside diners and about half the staff eat what the fixed first batch produces (single-sourced), so any cut to that batch takes food from people nobody has consulted unless their portions are planned explicitly. They are a latent opponent of waste reduction (candidate).

Revised: the first version said vendors and students have no direct relationship and everything routes through the Mess Office or portal. They meet daily at the counter. A student complained in person to Kadamba's manager about the dropped sambar and got a direct answer, operators absorb latecomers through a door and a back door they say they cannot refuse, and Yuktāhār weighs student plate waste. Only formal complaints route through institutional channels, running from a director's email or the CDS team to the manager and the kitchen, and at Bakul on to the managing director (single-sourced). No source describes the Portal and IT owner having any direct relationship with the Mess Committee, although the Committee sets policy that the system presumably enforces.

One contract dependency matters more than any other cell. Vijayalakshmi Caterers says it won a tender and is paid monthly on plates with deductions, with tender figures unclear on the recording, and that payment is on plates served (single-sourced). If plates mean registrations, an operator cooking for well under all of them keeps a margin on every no-show and has no financial interest in closing the gap. If plates mean meals served, the reverse holds. Bakul's ”served” and Kadamba's ”we cook on registrations because they are getting paid for it” pull opposite ways. The contracts may differ, ”served” may be a normalised word for billed plates, or Kadamba's remark may describe a duty rather than a formula. One word in one account does not settle Bakul, still less the others, so the question stays open for every operator until the contract wording or CDS confirms it. If Bakul is paid on plates served, it bears the cost of over-cooking and accepts it to avoid a shortage (candidate).

Influence

Influence splits by what each actor controls. Operators hold high local influence over how much is cooked, when a batch is topped up and whether a second counter opens, and they respond within 10 to 30 minutes. At Kadamba that minute-to-minute decision sits with the supervisor, at Yuktāhār with the kitchen team reading the first half hour. At Bakul the manager decides, but every top-up adds a 20 to 40 minute drive to the cooking time. Latecomers are absorbed, not decided on, since students at the back door cannot be refused. Operators hold no influence over the registration cut-off, the billing rule or the service window, and their influence over the menu is contested. Bakul's manager is the clearest case: he holds pairing knowledge and sees what is eaten, yet in his account students choose the combinations, and which menu decisions he can override is unknown (single-sourced). The Mess Office holds the registration and billing rules and the only scan-level attendance data, and has no confirmed influence over how much any operator cooks. The Mess Committee holds menu authority and the final say on policy. Academic administration holds the class timetable, the variable most in conflict with breakfast hours, and exerts no influence on the mess at all. Students supply the input everyone else plans around and have no influence over any rule, except informally through Mess Cell and direct complaint at the counter.

Conflicts

Conflict	Side A	Side B
Billing certainty vs. attendance uncertainty	The Mess Office locks registration four days ahead and bills on it; operators turn the count into cooked food at a share each sets: fixed at Kadamba, weekly at Bakul, from books at Yuktāhār	Students' attendance is uncertain day to day and nothing in billing rewards reporting it. Revised: vendors do not need the four-day number to buy food, since perishables are ordered one day ahead and groceries in bulk
Flexibility vs. workload	Students and the Mess Cell market want easier cancellation and resale; they face automatic registration, five cancellations per meal type per month at least four days ahead, and no way off the charge beyond that (confirmed)	The Mess Office's one confirmed policy change tightened rules to reduce staff workload (candidate)
Convenience vs. waste	Students want flexible registration	The Mess Office wants predictable, low-workload ordering
Operator stance on registrations	Kadamba treats the count as an obligation and cooks against it whoever is expected to eat	Yuktāhār treats waste as a cost it can cut and ties reduction to profit. Bakul's manager holds a third stance: shortage is the failure to avoid and waste is its price. A run-out is visible and draws complaints up to the managing director while a surplus is silent, so the stance may be the kitchen's answer to which failure gets noticed (candidate). All three work under one registration-based system, so the difference is operator culture (candidate; single-sourced)
Waste reduction vs. who eats the surplus	Any cut to Kadamba's first batch reduces surplus that becomes waste	30 to 50 outside diners and about half the staff eat from that food and would lose it unless planned for (candidate)
Over-preparation vs. food quality at Kadamba	The fixed first batch and live top-ups keep the counters full	By the team's own judgement, food is cooked ahead, reheated and moved between vessels, and degrades fast; students complain about quality. The link from held food to complaints is a candidate, not evidenced

Conflict	Side A	Side B
Menu ownership vs. waste visibility	Students, the Committee and the Parliament own menu choice and feel the loss of liked items, as with the dropped dinner sambar	Operators see which items go to waste (uttapam, set dosa, upma at Kadamba; puri, bonda, vada and idli move) but hold no menu seat, and Yuktāhār's head says turnout data is not used for the menu. Kadamba's manager said both that students choose the menu and that the operator does, so authority there is unresolved. Bakul's manager says some pairings are wrong rather than the dishes (kachori belongs with chole, Raj Kachori with aloo curry), that students plan for nutrition where he plans for familiarity, that pav bhaji draws diners and muesli and Chocos go readily, and that a protein target of 30 to 40 g a meal exists, set by someone unknown. He blames low turnout partly on taste and partly on students not waking. Not waking matches what interviewed students say most; taste is not a leading reason in those interviews, none with a known Bakul resident (single-sourced; untested)
Late arrival vs. service close	26.5% of Kadamba's April breakfast scans fall between 9:15 and 9:30, building to about nine a minute by 9:29 (confirmed, records). The operator's rule of thumb matches: about 100 early, 100 in the middle, 150 in the last ten minutes	Operators reopen the door and top up with food near the end of its safe life, and students use the back door after 9:30; the same surge drives cooking ”just in case”. The surge differs by hall: Bakul's manager puts his main rush at 8:00 to 8:30 with 10 to 15 in the last ten minutes, and still keeps food for about 50 at 9:20 (single-sourced; no Bakul scans seen)
Fixed cuisine allocation vs. uneven turnout	Turnout differs by line and item even inside one mess: Kadamba's April breakfast turnout was 48.5% for non-veg and 32.4% for veg	Each hall's cuisine and kitchen are fixed by designation and contract and do not adjust to demand. Revised: procurement itself is not fixed
Temperature vs. texture in off-site transport (Bakul)	Early dispatch, insulated boxes at about 60 to 72 °C and hot-water holding keep food hot through the drive and service	Hot parathas steam and soften, so food arrives hot but less pleasant. The food held longest, including the late reserve, may suffer most; other items are untested (single-sourced; candidate)
Operator vs. supplier on quality	Operators need clean, consistent stock	Suppliers deliver grain with stones and paneer that draws complaints, which operators answer with an impurity norm, returns and brand or supplier switches
Low-bid tender vs. quality	The institution picks the lowest or closest bid against a published base rate (single-sourced)	Operators must hold quality and portions within that rate; margin pressure is a candidate tension
Staff vs. management over waste tracking	Yuktāhār's management introduced item-level waste recording	Staff first read it as surveillance (”they are catching us”). Managers made it blame-free, discussed it daily and paired it with cross-checking ”not to find fault”, which settled the conflict by December
Operator autonomy vs. centralised prediction	Yuktāhār keeps prediction on paper and in its own meeting, and dropped a commercial ERP because paying lakhs of rupees for prediction made no sense with the data already in its books for a mess of 300 to 350 diners, and its staff are not digitally inclined (single-sourced)	An institution-designed forecasting tool would take that judgement out of the operator's hands
Academic schedule vs. meal schedule	Academic administration owns class timing and has no stake in mess outcomes	Students bear the cost of the 8:30 AM overlap with breakfast, and operators absorb it as the closing surge

Findings

Thirty-two candidate stakeholders were narrowed to twenty-four on direct evidence. Three lenses that might plausibly have produced a stakeholder (student health, senior-to-junior norms, a parental financial relationship) produced none.

The Mess Office is distinct from the Mess Committee, separating who decides policy from who executes it. Every earlier account, including this team's first drafts, had collapsed the two. The Office owns the four-day registration rule, passes registration counts to operators, runs the platform and issued the one confirmed policy change on

record, which was motivated by workload (candidate). Revised: it does not place food orders. Operators buy their own stock on their own cadence, and the four-day count sets only how much cooked food they prepare.

The operators are the largest part the first map left out, and each runs a different planning regime. Yuktāhār plans at a twice-daily meeting from the Wastage Book for the matching week (the menu repeats in weeks one and three, and two and four), the weekday register, the month's pattern, holidays and events, and a run-out rule (”if it runs out at 1:30 we re-cook; if at 2:20 we don't”). The books are confirmed; the rule is single-sourced. During service it samples the first half hour to project the rest. Kadamba cooks about 70% of registrations first against a gram standard for every dish, and its supervisor calls top-ups from a raw reserve. Its manager says there is no forecasting method, and the gram standard is its only standing record. Revised: Bakul was read as a fixed 70% raised to 80%, with top-ups inside 30 minutes. Its manager cooks 70 to 80% by item, adjusted weekly from attendance history, keeps about 70% even on low weekends because there should be no shortage even if food is wasted, and tops up by phone on an unstated lead time (single-sourced). Yuktāhār's matching-week learning against Kadamba's fixed ratio is the cleanest comparison in the map, and Bakul's weekly judgement with a floor sits between them (candidate). Every recorded planning figure sits well above actual turnout: 37.1% at Kadamba in April and about 28% at Yuktāhār in September (confirmed, records), against Yuktāhār's belief of at most about 50% and Bakul's manager's estimate of at most about 60%. His first batch and weekend floor both exceed even that, and if Bakul resembles the recorded halls the gap is larger and its size unknown (candidate). At the two recorded halls, then, the operators' conversion factors are miscalibrated upward.

The registration-to-attendance gap is already measured twice and reaches nobody who sets the rules. The CDS team holds a timestamp for every Kadamba breakfast scan (11,941 of 32,162 registrations availed in April, confirmed). Yuktāhār records registered against eaten by weekday and waste by item, in books each checked by someone other than its keeper. Kadamba keeps a gram standard per dish and watches its counters but records nothing of what was eaten. At Bakul the measurement is the manager's weekly review, and no source says whether any of it is written down, so it may live in one head (candidate). None of these measurements is confirmed to reach the owner of registration and billing policy, and they do not reach each other: no source shows CDS sharing scans with operators, or operators' registers reaching CDS or the Mess Committee.

Academic administration holds the lever most likely to affect the core problem, the overlap between breakfast hours and the 8:30 AM class start, and has no relationship of any kind to the three mess-governance actors. It is the clearest sleeping stakeholder in the map: total authority over the one variable in conflict, and zero engagement with the system it conflicts with. The operator evidence extends the gap downward. Exams, hackathons and campus events cut turnout, yet reach a kitchen only if someone raises them in a meeting.

Three feedback channels exist on the student side (an in-app per-meal rating, a public email address and Ping) and none has a confirmed instance of changing a policy. The operator side adds channels the first map missed: complaints from a director's email or the CDS team to the manager and kitchen (at Bakul on to the managing director), the WhatsApp group between the CDS site team and each operator, the facility team's tasting with Prism before service, and complaints made in person at the counter. It also adds the one working internal feedback loop anywhere in the system, Yuktāhār's Wastage Book, run-out times and weekday register feeding the next meeting, which took hold only once measurement was made blame-free (single-sourced). The fortnightly menu rotation that makes that loop possible is a confirmed policy change, and it came through the governance route from CDS to the Student Council to the Student Parliament (single-sourced, CFS Chair), not through any student feedback channel.

Revised: the first version found registered attendance markedly lower, against capacity, at every North Indian mess, read as three independently run messes pointing to cuisine. That no longer stands. Bakul is served by Vijayalakshmi Caterers from Hafizpet (single-sourced) and Palash from the same off-site kitchen (candidate), so the pattern does not come from three independent messes, and an off-site kitchen where food is held hot for hours and softens in transit is a competing explanation (candidate). The Bakul manager's view that North and South Indian eating patterns differ is a hypothesis from experience, not evidence for cuisine. Turnout records show cuisine and item driving attendance inside a single mess too: at Kadamba non-veg breakfast registrations were availed at 48.5% and veg at 32.4%. Cuisine is one factor among several.

The Mess Committee's chair, reported as a faculty member by a secondhand source we could not verify, is a second, subtler candidate sleeping stakeholder if that report holds: a person with cross-institutional standing that no

student channel has and a plausible bridge to Academic administration, with no evidence the bridge has been used. Kadamba's manager describes IIIT faculty inspecting the store, and the operator describes a weekly QC inspection and a facility team that tastes every meal with Prism. These are the only signs of institutional contact with the kitchens, and whether any belongs to the Committee is a candidate only.

The Student Parliament has an elected Mess Secretary and a Mess Deputy Secretary for each mess (Kadamba, Palash, Yuktāhār), known from the Parliament's published records (single-sourced). It is a real, structured channel of student representation, and it sits in the internal ring beside the Committee, Office and Warden.

Surplus leaves the system through people and a bin, not a compost heap, at least at Kadamba. Revised: staff eating leftovers is replaced. Kadamba cooks about 30 extra portions for 30 to 50 outside diners, about half its staff eat from the same food (single-sourced), latecomers are fed through the reopened door and the back door, Yuktāhār turns leftover idli into idli upma or idli poha for lunch and sets surplus milk as curd, and the rest of Kadamba's cooked surplus goes to a dustbin the garbage contractor empties. Kadamba's 5 to 10% waste figure, on an unstated base, sits uneasily with a 70% batch against 37% turnout: either the extra diners, reheating and small top-ups absorb most of the surplus, or the percentage uses another base. The evidence does not say which. The first version's assumption that composting absorbs the physical surplus is revised for Kadamba, whose manager says there is no composting (single-sourced). The CFS Chair earlier said a collector takes waste to an outside compost facility, and the manager may mean only that nothing is composted on site, so the onward route stays open and the garbage contractor is the single point where all of Kadamba's waste leaves.

Revised: the first version named the four-day procurement lead time as a hidden actor capping the system's response to any same-day signal. Kitchens respond within 10 to 30 minutes and buy perishables a day ahead, so that finding no longer holds. Three other hidden actors carry outsized influence without being visible to any one respondent. The Portal and IT owner is used constantly and named by no one. The four-day registration lock is a Mess Office rule that fixes the student's commitment long before they know whether they will eat. The first-batch share is a rule of thumb held by each operator, fixed at Kadamba, a weekly judgement at Bakul that can fall only as far as its floor of about 70% (single-sourced), and it turns the four-day commitment into cooked food that lasts three to four hours.

Several questions remain open after the operator visits. The most consequential is whether operators are paid per registered or per served plate, and what the deductions are, since that decides who gains from no-shows. Bakul's ”served” needs confirming from the contract or CDS, along with whether Kadamba's basis differs. Whether Palash is served by the same operator and kitchen as Bakul needs confirming. At Bakul it is open whether anything behind the weekly adjustment is written down, how long a top-up takes, which menu decisions students control and the manager can override, and why Bakul residents skip breakfast. Who heads CDS is open, as is whether Skip Meal changes the count Kadamba cooks from, which needs a direct answer. At Yuktāhār, whether the mess in-charge is the person the recording calls ”Adit Amma”, and whether the site head is one of the two project managers, are unconfirmed. Also open: who the facility team is, what the outside registration for night security is and who pays for those portions, what base Kadamba's waste figure uses, who decides the menu at Kadamba, whether any operator's item-level waste figures reach the Committee, whether CDS ever shares scans back, the onward route of Kadamba's waste, the identity of faculty quality-check visitors, whether staff meals are an entitlement, and whether any channel could carry the academic and event calendar to the kitchens. The identities between CDS, CFS and the bodies named earlier need a direct answer from CDS leadership, which has not been interviewed.

Stakeholder Power-Interest/Leverage Map and Power Analysis Brief

IIIT Hyderabad Breakfast Mess System

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Executive Summary

Power in this system is split across several governance bodies whose relationship to one another is only partly established, and some of it is exercised by actors with no official standing. Its two most consequential features are absences. Nothing connects the one actor who controls the class schedule to the actors who control everything else about breakfast, and nothing carries the kitchens' own records of the gap to anyone who sets the rules that produce it.

The first central finding is that students control an input nothing requires to be accurate. Registration is an automatic default, not a daily decision to eat. At Kadamba in April 2026, 32,162 breakfast registrations produced 11,941 meals eaten, a turnout of 37.1%. Registrations stayed near 1,070 a day on every weekday while turnout ran from 42.1% on Tuesdays to 27.3% on Sundays (confirmed, scan records). The operators, the only actors who plan food against this input, each correct it privately and in different spirits. Yuktāhār distrusts the count and replaces it with its own

records. Kadamba treats it as a paid obligation, cooks a fixed 70% and tops up from the counter (single-sourced). Bakul (Vijayalakshmi Caterers) sits between: its manager cooks 70 to 80% by item, adjusts the share weekly from past attendance, and keeps food for about 70% of registrations even on very low weekends (single-sourced). The institution's scan records measure the inaccuracy, but nothing routes that measurement into a rule, and the institution never sees any operator's correction. (Revised: the first version said every operator distrusts registration. That holds for Yuktāhār only.)

The second central finding is that the operator is the data-rich, authority-poor actor. It sets the parameter with the largest effect on breakfast waste, how much of each registration is cooked. Yuktāhār keeps the system's only per-item, per-weekday record of registered, eaten and wasted food, in books it reconciles every day (confirmed). Kadamba holds per-dish gram standards and live counter knowledge but no per-item waste record was seen. Bakul's weekly adjustment rests on its manager's reading of past attendance, and whether any of it is written down is unknown. None of the three operators has authority over the registration cut-off, the cancellation rules, billing or the menu, or a route by which its records reach someone who does.

Several first-version findings were revised on the September 2026 operations evidence. Vendors were described as passive producers working to a forecast fixed four days ahead. No operator cooks the full registration count for breakfast, and the four-day lead is the students' registration cut-off, not a procurement lock. The Mess Office was presumed to hold the system's evidence base, but that sits with the operators and, separately, with the CDS team's scan records. Students were said to have no channel to vendors, yet a weak one runs through menu selection and the counter. A first-hand account of Kadamba and Yuktāhār then added the Kadamba command chain from the institution to the service staff, the opposite stances of the two operators under one billing basis, and the finding that institutional oversight of the kitchen covers safety and taste, never quantity.

The most consequential open question is whether operators are paid per registered or per served plate, because that decides whether the registration gap costs an operator money or earns it money. Kadamba's remark that it cooks on registrations because ”they are getting paid for it” leans to the registered-plate reading without being contract evidence. Bakul's manager says payment is on plates served (single-sourced). The two sit in tension and stay unresolved for every operator.

Power is treated as issue-specific. An actor may hold high authority over one decision and none over the next, so each claim names its domain and separates formal authority from observed influence.

Assess Decision-Making Authority

Formal governance structure

Three institutional actors hold mess governance. The first drafts collapsed ”who decides” and ”who executes” into one.

- **Mess Committee.** A policy body that sets the menu about a month ahead with student input. That it is separate from the Mess Office is confirmed. That a faculty member chairs it rests on a secondhand press summary we could not read in full (candidate). Its authority beyond menu content is unconfirmed.
- **Mess Office.** The operational arm. It runs the registration platform, which closes about four days before service, and issues operational policy. The one policy action on record, a December 2024 tightening of cancellation rules, was reported as issued by this office alone and motivated by staff workload, again from a secondhand press summary (candidate). The current cap, five cancellations per meal type per month at least four days ahead, is confirmed from the published rule and from students' experience. (Revised: the first version said the Mess Office places food orders on a four-day lead. Operators buy their own food, on cycles far shorter than four days.)
- **Warden.** Holds vendor management and structural change. No source describes the Warden taking a specific action, so the authority is stated, not observed.

Academic administration sets the class schedule, including the 8:30 AM start that overlaps the breakfast window, with total authority in that domain and no formal relationship to any mess-governance actor. Nobody has raised the overlap with it, and in the one direct account on the point, nobody has tried. The evidence favours inattention over refusal: the bridge has never been attempted, not tried and blocked. The same gap reaches the kitchen. Exams,

events and holidays reach Yuktāhār only if someone raises them in the planning meeting, ”we don't come to know until we sit in the meeting” (confirmed), and no formal calendar feed reaches any operator.

The Student Parliament's mess roles are a further actor. The Parliament, elected by single transferable vote at roughly one representative per 50 students, maintains a Mess Secretary and a Mess Deputy Secretary for each mess, known for Yuktāhār, Kadamba and Palash from its published records (single-sourced). Student representation therefore sits in the Parliament, not in a seat the Committee appoints. Parliament representatives also hold a standing meeting with a Mess Warden, a title not yet confirmed as the same as the Warden above.

The CDS team is the institutional counterpart operators actually name. At Bakul, complaints arrive from the CDS team and pass from the manager to the kitchen and on to the operator's managing director. At Kadamba they arrive by email from a director and the manager relays them into a WhatsApp group that a site or facility team also reports into (both single-sourced). The April 2026 Kadamba scan records came from the CDS team (confirmed). Whether CDS is the Mess Office is unconfirmed, and the two are kept separate. Oversight at Kadamba is routine but narrow. Every meal passes a fixed sequence: cooking temperature checked, a sample kept 72 hours, a wrapped display plate at the entrance, serving temperature per item, then a facility team tastes with two of the operator's staff before corrections and service. The display plate and joint tasting are confirmed by observation, the rest single-sourced. Quality control inspects weekly, including the store, and water TDS is read daily. Whether QC visits can penalise or only inspect, and whether the facility team is the CDS site team, is unknown.

The Kadamba command chain runs from CFS to CDS, then the contractor Prism, then inside Prism from an operations manager to a manager, a supervisor and the service staff (single-sourced). Prism is an outsourced vendor that wins by tender and runs all of Kadamba's operations. The head of CDS is named in the team's site account, while an earlier interview names a different CDS Chair. Whether these are two roles or one misnamed is open and they are not merged. A separate staff account of an operator's reporting chain (staff, supervisor, manager, operations manager, director) matches the lower part, but which mess it describes is unconfirmed, so reading it as Prism's is candidate.

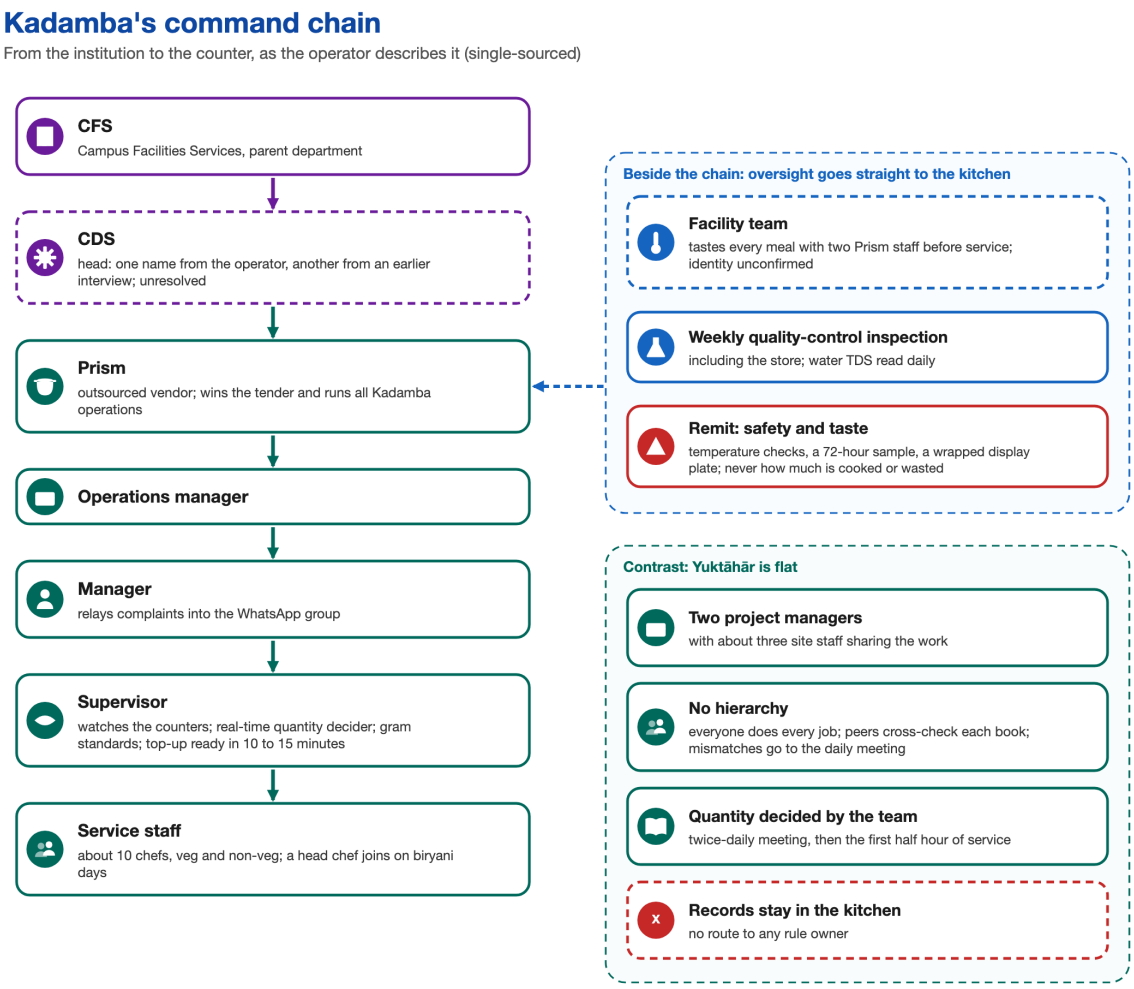


Figure 1. *Kadamba's command chain, with the facility team's side channel and Yuktāhār's flat contrast.*

A separately named ”Mess Council,” raised earlier as the recipient of a public complaint email, is most likely a naming artefact: a Mess Monitoring Council exists at a different institute, and IIIT Hyderabad's own site never mentions one. A sharper ambiguity replaces it. Secondhand sources describe what may be two bodies both called ”Mess Committee,” a Parliament-internal committee of elected representatives that negotiates with the administration, and the institute-level, faculty-led policy body above. Operators add a third name: the Kadamba and Bakul operators both say a ”student committee” chooses the menu from options they supply (single-sourced). Whether that is either Committee or the CDS Committee is unconfirmed, and none is assumed to be another.

For several decisions, including breakfast timing, the registration cut-off and contract terms beyond the tender, no source says who proposes, approves or can veto a change. That absence, across every source, is itself a finding.

Operator-side authority

”Vendors” was one box in the first version. The operators' accounts show a layered actor whose internal structure decides who holds which power.

- **Contract and price terms** sit with the operator's owners or managing director. Kadamba's manager does not know the plate rate: ”our owners are directly dealing with them” (single-sourced).
- **Quantity rules** sit with the operator, and no institutional approval of them is evidenced. At Kadamba the standing rule is Prism's: a first batch of 70% of registrations, veg and non-veg planned separately, a gram standard per dish (four idlis or 100 g of sambar a head, with separate standards for poha, pulao and the rest) and the remaining raw material ready to cook in 10 to 15 minutes. Biryani lunch is the exception, cooked at 100% at two pieces of

150 to 200 g with unlimited rice, with a head chef brought in. At Bakul the manager sets 70 to 80% by item from attendance history reviewed weekly, and keeps about 70% even on very low weekends. It is a judgement-based weekly calibration with a floor, not a fixed ratio like Kadamba's (all single-sourced).

- **The in-service top-up decision** sits with a different person in each mess. At Kadamba the supervisor watches the counters and tells the kitchen how much more to cook, staff watch each dish's level against its gram standard, and items such as bonda, puri, vada and idli ”float” above the standard. At Yuktāhār the decision is collective: a twice-daily team meeting discounts portal registrations using the matching week's Wastage Book, the weekday register, the month's turnout and any events (confirmed), and the first half hour of service is sampled to project the rest (single-sourced). At Bakul the manager alone compares food left against registrations and expected demand about every 30 minutes and phones the Hafizpet kitchen, which cooks and drives the extra over. The drive is about 20 minutes early and 40 normally, the total lead time is unstated, and the earlier ”within 30 minutes” came from noisy audio (single-sourced).
- **Re-cook, portioning, reuse and supplier choice** also sit with the operator. Yuktāhār's Wastage Book works as a rule (”if it runs out at 1:30 we re-cook; if at 2:20 we don't”), and leftover idli, dosa batter and milk are reused (confirmed). It holds an impurity norm for each raw item, changes supplier when it is exceeded, and visits the mills that supply its oil. Bakul's operator switched paneer brands twice after complaints. Kadamba's operator cut quantities and bought items outside during an LPG shortage on its own initiative (single-sourced). Who may replace broken equipment is unknown: Kadamba's toaster was broken with no replacement.
- **Menu content** is the one kitchen domain where knowledge exceeds authority. The Bakul manager says students select the combinations and some do not work as pairings: kachori belongs with chole, and Raj Kachori, listed with chana, would go better with aloo curry. He reasons from traditional and regional combinations, taste and what people eat, with about 20 years in North Indian food, and describes students as reasoning from balanced diet, protein, carbohydrate and preference. He mentioned a protein target of about 30 to 40 g a meal, origin unknown. He sees which items move: pav bhaji draws students, and muesli and Chocos go readily. He holds consumption knowledge the menu process does not use and no formal power over it. Whether his view matches what Bakul residents prefer is untested, so it is not evidence of what students want (single-sourced).
- **Internal structure** is the sharpest contrast. Yuktāhār is flat: two project managers, no hierarchy, no storekeeper-only role, peers checking one another's records. The mess in-charge keeps the accounts and Wastage Register while a colleague cross-verifies the grocery register (the cross-verifier's signature on a photographed fridge log is confirmed). Leave is shared by agreement in the daily meeting. The site head describes a proposal-and-review practice: ”you follow this; if you find any problems come back to me” (confirmed). Kadamba runs the layered chain, where the supervisor decides and no staff voice was described. Bakul's structure is known only through its complaint route and a manager who decides both first batch and every top-up for a kitchen 8 km away.

The operator therefore holds high local authority over the quantity, timing and reuse of food, and none over the rules that shape how many students register.

Authority at a glance

Domain	Proposes	Approves	Confirmed?
Breakfast timing	Unknown; no evidence anyone has proposed a change	Unknown	No. Service is meant to close at 9:30, yet late diners come in by the back door and the operator says it cannot refuse them (single-sourced); 1,100 of 11,941 April scans (9.2%, about 37 a day) fall at or after 9:30 (confirmed, records)
Class timing	Academic administration	Academic administration	Confirmed in its own domain, disconnected from mess governance
Registration cut-off and cancellation rules	Mess Office	Unclear whether Committee sign-off is required	One case on record (December 2024, candidate). Registration is automatic, applied even off campus; the only exit is five cancellations per meal type per month at least four days ahead, and beyond that the charge cannot be removed (confirmed)

Domain	Proposes	Approves	Confirmed?
Menu	Operator supplies options (single-sourced)	A student committee selects (single-sourced, identity unconfirmed); formal Committee approval is reported	Student selection confirmed in outline, the body's identity not. The Bakul manager disputes some pairings but has no approval power
Quantity and first-batch ratio	Yuktāhār's daily meeting; Prism's standing ratio and gram standards with the supervisor deciding top-ups; Bakul's manager's weekly adjustment with a floor and phoned top-ups	Operator; no institutional approval evidenced	Confirmed at Yuktāhār; single-sourced at Kadamba and Bakul
Leftover reuse and supplier choice	Operator	Operator	Confirmed at Yuktāhār; single-sourced elsewhere
Vendor contracts	The institute publishes a tender; operators bid; Prism holds Kadamba this way (single-sourced)	The closest or lowest bid; the Warden per the Stakeholder Map	Tender route single-sourced; the award authority unconfirmed
Complaint routing	CDS team or a director's email, to the manager, to the kitchen	Operator's manager acts locally	Single-sourced
Food safety and quality inspection	Per meal, a facility team tastes with two Prism staff after temperature checks and a 72-hour sample; QC weekly; water TDS daily	Corrections before service; whether QC can penalise is unknown	Display plate and tasting confirmed by observation, the rest single-sourced. No check covers quantity cooked or wasted
Portal and scan data	CDS team can export registration-level scan records	Formal owner unconfirmed	Export capability confirmed
Mess Cell resale	Students, self-organised	Not applicable	The market's existence is the only ”approval”
Waste handling	Operator	Operator; who pays the garbage contractor is unknown	At Kadamba waste goes to a dustbin collected by a contractor with no composting (single-sourced); whether it reaches an outside compost facility, as the CFS Chair described, is unverified. Yuktāhār's route was not stated

Several domains now have a nameable proposer and approver: the menu, the one operational policy change on record, and every quantity and reuse decision inside the kitchen. The domains that still have none, breakfast timing and the registration cut-off above all, are the ones that set how many students register in the first place.

Identify Formal and Informal Power

Formal power sits with the governance actors and Academic administration, as above. Four informal actors exercise real influence with no formal standing.

- **Roommates and friends.** Social prompting changes individual attendance in a minor way, a secondary effect that six student respondents support. Friends also eat in place of registrants who do not come, which Kadamba's manager describes as routine and which corroborates informal redistribution of registrations (single-sourced).
- **The Mess Cell community.** A self-organised resale market showing real peer-to-peer coordination, mostly applied to individual trades, with one exception below.
- **Ping.** The student publication has covered mess issues at least four times, according to a secondhand search summary: a November 2024 piece on a unilateral non-veg cancellation, a December 2024 opinion survey, a January 2025 follow-up, and a 2023 piece on an unrelated water-quality issue (candidate). It gives problems public visibility, and no source shows that visibility alone producing a policy response.
- **Unofficial registration tools.** At least one public open-source tool lets a student manage registration through an AI agent, described in its documentation as built ”for students of IIIT Hyderabad.” Its existence is confirmed, its adoption and effect not.

Late diners hold a small, routine informal power over the operator. Kadamba closes at 9:30, but when the front door closes students come in by the back door and demand food, and the operator says it cannot refuse (single-sourced; 25 came after close on one biryani day). The records show this is not occasional: 9.2% of April scans, about 37

a day, fall at or after 9:30 (confirmed). (Revised: the first version had the operator reopening the door at its own discretion.)

The operator also exercises informal power beyond its remit. Kadamba cooks about 30 portions for 30 to 50 people outside the student registration, such as night-shift security, and about half its own staff eat from the same food (single-sourced). Nobody grants or forbids these accommodations. Each absorbs part of the gap between registration and turnout without being reported, and the people fed have no voice in how much is cooked. (Revised from 20 to 30 staff eating leftovers.)

Analyse Stakeholder Incentives and Motivations

Stated interest vs. what the system rewards vs. observed behaviour

Stakeholder	Stated interest	What the system rewards	Observed behaviour	Gap
Students	Convenient, affordable, timely food	Nothing rewards registering accurately; billing is identical either way	Register as a standing booking and attend far less: 37.1% at Kadamba (April), about 28% at Yuktāhār (September), against 56 to 89% for Yuktāhār lunch (confirmed, records)	Large, and breakfast-specific
Mess Office	Not stated by any source	Reduced operational workload, its reported motive for its one policy change (candidate)	Tightened cancellation rules	Small; an unusually clean alignment
Mess Committee	Balance student satisfaction against feasibility	Unclear	No independent action distinct from the Office's observed	The least-evidenced pairing here
Kadamba operator (Prism)	Meet the registration as an obligation, ”a responsibility and they are getting paid for it” (the operator's attitude, not contract text)	A contract won by tender; the manager does not know the plate rate; whether pay follows registered or served plates is unconfirmed	Cook a fixed 70%, veg and non-veg separately, then top up on the supervisor's call; 100% on biryani days	Cannot be resolved until the payment basis is confirmed
Yuktāhār operator (ABC)	Best quality, less waste, optimised workflow, lower cost, higher profit, an explicit management goal	A plate-rate structure assumed to match Kadamba's; waste reduction is a cost lever	Discount registration with its own books, re-estimate from the first half hour, reuse leftovers, weigh waste daily	Small; goal and practice match
Bakul operator (Vijayalakshmi)	”There should not be a shortage of food, even if some wastage occurs”; more say over pairings	A tendered plate rate paid monthly as a lump sum with deductions; the manager says payment is on plates served (figures unverified, in tension with Kadamba's remark)	Cook 70 to 80% by item, adjust weekly, keep about 70% on low weekends, top up by phone, hold food for about 50 at 9:20	Small between goal and practice; whether the waste it accepts costs the operator depends on the payment basis
Academic administration	Its own scheduling goals	No incentive tied to mess outcomes	Total non-engagement	None, which is why the actor is dormant
Mess Cell participants	Recoup the cost of an unused registration	Direct, immediate financial return	An active, working market	None; the cleanest match in the system

(Revised: the first version said vendors ”continue producing to the forecast.” None of the three cooks the full breakfast count, and all three watch live counts and top up. Kadamba cooks 100% only for biryani lunch.) The two operators with the fullest accounts are assumed to share one billing structure yet take opposite stances. Yuktāhār pursues lower cost by learning from its records, and Kadamba applies a fixed ratio as an obligation. The difference tracks management culture rather than the payment rule, which makes operator culture a candidate lever. Bakul adds a third stance, shortage as the failure to avoid and waste as its price. If its manager's report of payment on plates served is confirmed, the operators do not share one billing structure and this comparison would need restating.

Incentive chains

1. **Billing follows registration, not attendance.** That removes any reason for a student to register only when intending to attend. The cause is structural, not a personal failing. Registrations at Kadamba stay flat across the week, Sunday included, while turnout falls to 27.3% on Sundays, and veg registrants attend at 32.4% against 48.5% for non-veg (confirmed, records).
2. **Skip Meal returns nothing, and resale returns value.** A student with advance notice has a rational reason to prefer resale over the system's own tool. Six respondents: none used Skip Meal for a return of value, and four used Mess Cell in some form (confirmed).
3. **Walk-in pricing about 70% above the registered rate** (Rs. 81 against Rs. 48 for vegetarian) gives a standing reason to register under uncertainty, inflating counts independent of any no-show. All six respondents register in advance, none day by day. Walk-in demand is real but small: Yuktāhār's breakfast register shows none to twelve unregistered, paying, faculty and staff diners a day, usually four or fewer, twelve on the peak day with nine paying (confirmed).
4. **The Mess Office's workload incentive produced tighter cancellation rules,** which cut student flexibility and plausibly strengthened the pattern they targeted. A candidate case of a fix that fails, unconfirmed by any governance-side account of its effect.
5. **Academic administration feels none of the consequences of the overlap,** so the issue never reaches its agenda and persists with no actor able to resolve it alone (confirmed).
6. **The payment basis decides which side of the gap an operator sits on.** If operators are paid per registered plate, they are paid for the whole registration and cook only part of it (70% at Kadamba, 70 to 80% at Bakul), so the gap is margin and accurate registration has no financial appeal. If paid per served plate, the operator bears every portion cooked and not eaten and wants accuracy. No operator has stated a position on the cut-off. Kadamba's remark leans to the first reading. The Bakul manager says payment is on plates served (single-sourced, from a team member's normalised summary, not a transcript). The statements pull apart and are left unresolved: contracts may differ, ”served” may mean plates billed, or Kadamba's remark may describe a duty. If Bakul pays for plates eaten, it bears the cost of food held against a shortage and accepts it, which would mean it judges shortage costlier than waste, and a change letting students leave the booking would not cut its revenue (candidate). Under per-registered-plate billing, Yuktāhār's profit goal and Kadamba's fixed ratio both gain from cooking less than is registered, and differ only in pursuing the saving through records or through a rule. Both readings stay candidate, and this is the most important unresolved incentive question in the analysis.

Bargaining power and dependency

Students depend on operators for food. Operators depend on no individual student, a strongly asymmetric relationship consistent with a walk-in price that needs no negotiation. Operators depend on registration only as a ceiling. Their real planning input is their own books at Yuktāhār, a fixed ratio at Kadamba, and at Bakul a weekly-adjusted ratio kept at about 70% or more. (Revised: the first version treated Bakul's ratio as fixed.) Neither Kadamba's ratio nor Bakul's floor is calibrated to recorded breakfast turnout. Bakul's manager expects at most about 60% at breakfast, below both his first batch and his floor, so on his own numbers the batch exceeds turnout by design (candidate; no Bakul records).

At Kadamba a 70% first batch against 37.1% turnout implies about 1.9 times over-preparation before any top-up (candidate, combining records with a single-sourced ratio). Because the ratio applies to veg and non-veg separately, the gap differs by stream: 70% of about 757 daily veg registrations against about 245 eaten is about 2.2 times, and 70% of about 315 non-veg registrations against about 153 eaten is about 1.4 times (candidate). The operator reports only 5 to 10% daily waste on an unstated base (single-sourced). That fits only if outside diners and staff eat most of the surplus, reheating carries it into later service, top-ups are smaller than the batch implies, or the base differs. Whichever absorber takes the surplus has a stake in the ratio staying where it is.

(Revised: vendors do not depend on the Mess Office for a four-day forecast. Operators procure on their own cycles and adjust cooking on the day.) The Mess Office depends on students to register accurately, yet nothing in billing requires it. Academic administration and the mess depend on nothing in each other, which is a total absence of mutual dependency, not an imbalance. The Mess Cell market and the formal system are similarly independent, though the market may quietly absorb part of the formal system's waste problem without either side acknowledging

it.

Operators show clear bargaining power over suppliers: Bakul switched paneer brands after complaints, and Yuktāhār changes supplier when its impurity norm is exceeded and has used rock salt and cold-pressed oil for about six months (single-sourced). No instance of an operator bargaining with the institution is evidenced, no operator has objected to the four-day cut-off, and none is recorded raising it with a rule owner.

Historical precedent

The precedents are thin, and several are secondhand.

- **Bakul's creation,** triggered by Kadamba's non-veg capacity reaching its ceiling, is a confirmed structural response to a limit. Who raised it and how is unknown.
- **”Let Them Eat Frogs” (November 2024).** A frog was reportedly found in Kadamba's chicken biryani and the Mess Office cancelled non-veg meals at Kadamba without stating the real reason, prompting debate across mail threads. This is secondhand and not verified in full text (candidate), but a clear example of unilateral, opaque decision-making.
- **The December 2024 cancellation tightening,** reportedly triggered by Mess Office workload and a Kadamba renovation, came with no evidenced Committee involvement or student consultation. Pushback reportedly occurred: Ping ran a student opinion form that returned majority disappointment, and Parliament's mess representatives reportedly met the Mess Office twice before the policy took effect and then told Ping they were pushing back. The policy went ahead, a partial answer to whether organised student action worked (candidate). Whether it is the ”organised email effort” the project has otherwise only heard about is unconfirmed.
- **Non-compulsory registration during shortages and holidays (date unknown).** During an LPG shortage, and during holidays and fests, registration was made non-compulsory and the cancellation cap raised from five to ten, then returned (single-sourced). It is the one evidenced loosening of the rules, and it shows the rule owner can change them quickly when a supply shock or predictable low attendance forces it.
- **The Yuktāhār waste-tracking change (August to December of the operator's first year)** is the one evidenced case of an actor deliberately changing its own practice to cut waste. Management introduced per-item recording, staff first resisted for fear of blame (”they are catching us”), and by December staff themselves reported waste falling (confirmed). The project managers made the weighing blame-free and discussed it daily in the team meeting (single-sourced), which is the condition under which measurement changed behaviour. The operator also tried and dropped a commercial ERP: ”prediction will be based on our data only; I want to remain close to the data” (confirmed), because paying lakhs of rupees for prediction made no sense with the data already in its books and its staff are not digitally inclined (single-sourced). The change stayed inside the kitchen and nothing shows it reaching any rule owner.
- **”Water Mess” (2023).** In a different domain at the same institution, complaints about drinking-water quality went unaddressed for months, reportedly until 38 or more typhoid cases forced a reactive response (secondhand; candidate). The complaint, ignore, crisis, reactive-fix shape recurs across two domains.

For breakfast timing and class-timing coordination no precedent was found. That is recorded as unknown, not zero.

Identify Who Can Influence Decisions and Who Cannot

Power asymmetries

- **The Mess Office over students on registration and cancellation rules.** It holds unilateral operational authority and students hold none, so rules can be tightened without a confirmed consultation step.
- **Operators over students on food quality and quantity on a given morning.** Operators decide how much is cooked, when to re-cook and what to reuse. Students reach them only through a weak informal channel: menu selection through a student committee, and complaints at the counter.
- **Operators adjust on the day; students are locked in four days ahead.** Kadamba keeps raw material ready to cook in 10 to 15 minutes, Bakul phones a kitchen 20 to 40 minutes away with an unmeasured lead time, and Kadamba orders perishables a day ahead and dry goods a week ahead (single-sourced). The flexibility runs against

the party who knows least about the morning. (Revised: the first version said the four-day lead capped every actor's responsiveness equally.)

- **Academic administration over every mess-governance actor combined on class start time,** the variable most likely to resolve the core overlap, because it holds total, independent authority and mess-side actors have no channel to it.
- **Students over every formal actor on the accuracy of registration data,** because they are its only source and nothing requires it to be accurate. Yuktāhār replaces it with its records, Kadamba discounts it by a fixed ratio and Bakul by a weekly ratio with a floor. The correction stays private, so the inaccuracy shows up in the kitchen and nowhere else.
- **The institution over the kitchen, every meal, but only on safety and taste.** Temperature checks, the 72-hour sample, the display plate, the tasting, weekly QC and daily water readings all ask whether food is safe and good, and nothing asks how much is cooked or wasted. Formal power is present in the kitchen daily and pointed at a different variable from the one producing the gap.

Consequence and control

High consequence, low power. Students bear the most visible cost of the central problem (wasted food, lost money, energy crashes) while holding no confirmed authority over billing structure, cancellation rules or class timing. Registration is mandatory and automatic, even off campus or on holiday, and beyond five cancellations a month made four days ahead the charge cannot be switched off (confirmed). Skip Meal tells the kitchen but does not remove the charge, so the booking is a default the student cannot opt out of. Their influence over the menu is real but runs through a committee of unconfirmed identity.

Consequence absorbed downstream, no power. Operators absorb over-preparation and closing-rush shortages. Kadamba's scan records show one continuously building crowd, with 26.5% of scans between 9:15 and 9:30 and the rate still rising at 9:29, and the Kadamba and Yuktāhār operators describe the closing rush as the moment they re-cook or hold reserve (confirmed, records). Kadamba's rule of thumb, about 150 diners in the last ten minutes of any meal, matches the crest, and the records put 78.8 scans a day, 19.8% of the total, between 9:20 and 9:29. The Bakul manager places his main rush earlier, 8:00 to 8:30, with 10 to 15 arriving in the last ten minutes, yet aims to have food for about 50 at 9:20 (single-sourced; no Bakul scans). (Revised: the first version said all three operators describe a closing rush.) A top-up from Hafizpet could not arrive before close late in the window, so held food is Bakul's only hedge, and the reserve is several times the late tail he describes (candidate).

Whether over-preparation costs the operator depends on the unconfirmed payment basis. At Kadamba, 30 to 50 outside diners and about half the staff eat from the same food, an uncounted absorber with no voice in how much is cooked, and the garbage contractor removes the rest with no voice at all. Cooked food lasts only three to four hours, so cooked surplus cannot move to another day, while Yuktāhār reuses what it can as a new dish, curd or batter (confirmed). Within a meal, Kadamba's surplus is not simply discarded. In the team's own observation, food is cooked ahead, reheated and moved between vessels and degrades fast, so diners bear part of the cost of over-preparation as lower quality (candidate; no link to skipping is evidenced). Bakul's diners bear a different cost. Labour-heavy items such as aloo paratha may start around 3 AM at Hafizpet for a 7 AM arrival, and insulated boxes holding about 60 to 72 °C keep food hot but, in the manager's account, let steam soften the parathas. The food held longest for late diners is likely the softest (single-sourced; candidate).

Low consequence, high power. Academic administration holds total authority over the variable most likely to resolve the overlap and experiences none of the consequences of leaving it, and nothing in its mandate connects it to the mess. That is the structural reason the overlap can persist indefinitely.

Missing voices and missing relationships

Question	Answer
Affected but not represented	Students, on the registration cut-off, cancellation rules and billing
Consulted but cannot decide	Students, on cut-off, cancellation and billing: a named structure exists (Mess Secretary and per-mess Deputy Secretaries) but whether it carries a vote on these rules is unconfirmed. Operators, on the menu: they supply options and follow the selection

Question	Answer
Decides but does not experience consequences	Academic administration, on class timing
Oversees the kitchen but not the quantity	The facility team and QC, whose tasting and weekly inspection cover safety and taste only
Has the best data but no authority	The operator, most fully Yuktāhār. Its books record registered, eaten and wasted food per item, meal and weekday, with run-out times, re-cook quantities, plate waste and leftovers, alongside grocery, stock, vegetable, milk, gas, electricity and water; the store book must match what was cooked and mismatches go to the daily meeting (confirmed). Kadamba holds gram standards and reports 5 to 10% daily waste on an unstated base; no per-item waste record was seen. Bakul's manager adjusts weekly from attendance history and knows which items move, but whether any of it is written down is unknown (single-sourced). (Revised: the first version named the Mess Office here, and a later one said Kadamba records waste by weight, a claim whose figures were withdrawn)
Has authority but no data	The student committee and the Mess Committee on the menu, which decide without the operator's per-item waste record: uttapam, set dosa and upma consistently waste while puri, bonda, vada and idli move at Kadamba (single-sourced), non-veg registrants attended at 1.5 times the veg rate in April (confirmed), and at Bakul pav bhaji draws students and muesli and Chocos go readily. Also whoever owns the four-day cut-off and cancellation cap
Observes waste but cannot change the menu	The operator. Yuktāhār's site head says waste and attendance records are not used to set the menu (confirmed); Bakul's manager asks for more menu input and disputes the pairings chosen (single-sourced; untested with residents)
Holds information the kitchen needs, with no channel	Academic administration and the institute calendar: exams, events and holidays reach the kitchen only if mentioned in a planning meeting (confirmed)
Implements decisions it did not help design	Operators, on the cut-off and cancellation rules that shape the numbers they plan against
Absorbs consequences created upstream	The 30 to 50 outside diners and about half Kadamba's staff (single-sourced); diners who receive held and reheated food (candidate); the garbage contractor
No formal escalation channel	Any cross-authority issue, specifically the class and mess overlap and the mismatch between the four-day cut-off and one-day perishable ordering

Between operators and students the relationship is thin, not absent. (Revised: the first version said students have no direct channel to vendors.) Students reach operators through menu selection, since operators say they hand a student committee a menu and follow its choice, and at the counter. During the Kadamba visit a student complained directly that dinner sambar had been dropped ”while they are taking more money from us,” and the operator pointed back to the students' own menu choice (single-sourced). No channel carries the operator's waste records back to students or to the committee that picks the menu. No source describes a relationship between the portal owner and the Mess Committee, although the Committee depends on that system to enforce its decisions.

Coalitions and conflicts

Students and operators may share an interest in a shorter registration cut-off. At Kadamba, registration closes four days ahead while perishables are ordered the day before and dry goods a week ahead (single-sourced). The perishable tier falls after the cut-off and the dry-goods tier before it, so the four-day lock serves no evidenced procurement need, while students, for whom it is the source of most unwanted registrations, stand to gain from a shorter one. No operator has stated a position and neither side has a channel to the rule owner, so this is a potential coalition (candidate). If operators are paid per registered plate their interest in accurate registration may be weak, a counter-reading that is also candidate. Bakul may be the exception: if it is paid on plates eaten, fewer unwanted registrations would not cut its revenue and its view on a shorter cut-off would likely be neutral or favourable (candidate, resting on the unresolved payment basis).

A second conflict runs over menu criteria. In the Bakul manager's account students choose for balanced diet, protein and preference, while he would choose for traditional pairing, regional familiarity and what is eaten. Only his side has been heard, and whether Bakul residents share his view is untested. It is recorded as a conflict between the holder of formal selection power and the holder of consumption knowledge, not as evidence of what students prefer.

Convenience (students) against reduced workload (the Mess Office's reported incentive) is not a symmetric negotiation: the Office has shown unilateral authority to settle it in its own favour. Billing certainty against attendance uncertainty is structurally misaligned, not negotiable. No rule change removes the underlying uncertainty, only shifts who bears its cost.

Examine Unequal Access to Resources or Opportunities

Resource control

Resource	Controls it	Advantage created
Money (billing)	Mess Office	Sets terms students cannot individually negotiate
Food and kitchen infrastructure	Operators, per mess	Controls what every other actor ultimately needs
Waste and consumption records	Operators: Yuktāhār's book system (confirmed); Kadamba's gram standards and waste estimate (single-sourced); Bakul's weekly reading of attendance history, held by its manager, no written record known (single-sourced)	The only per-item, per-weekday record of the gap. It never leaves the kitchen, and no route upward is evidenced
Food-safety oversight	The facility team with the operator's staff, per meal; QC weekly	A daily institutional presence in the kitchen, pointed at safety and taste, not quantity
Registration and scan records	CDS team and the portal side (confirmed by the April 2026 export); formal owner unconfirmed	Exportable per registration, with no evidence it is reviewed against billing or the cut-off
Complaint data	Received by the CDS team or a director, relayed to the operator's manager and a WhatsApp group (single-sourced)	Reaches the operator for local fixes; no visibility beyond that
Formal complaint channels (in-app rating, email)	Institute side, as recipient	One-directional; no visibility into what follows submission
Informal escalation (Ping, Mess Cell)	Students and Ping, collectively	Two tiers of access, depending on whether a student knows these channels exist
The four-day registration cut-off	Mess Office or portal owner (unconfirmed)	Locks students in about four days ahead while operators adjust on the day, itself a power finding. (Revised: the first version called it a procurement lead time binding every actor equally)
Institutional legitimacy	The Mess Committee	Not acquirable by students individually, only collectively and only if organised

Students experience the consequences of every resource here and control direct access to none. The evidence that would show the pattern exists in two disconnected places, the operators' hand-kept records and the CDS team's scan records, and neither reaches the actors who set the rules.

Information flows: who holds what, and where it stops

Information breaks in four places, and each sits at the operator.

- **Upward, from kitchen to policy.** Yuktāhār keeps the richest record set in the system, and every book terminates inside the kitchen. No source describes a route from these books to CDS, the Mess Office, any Committee, CFS or billing. The institution's one routine channel into the kitchen, its safety checks, carries nothing about quantity back out. The CFS Chair's own awareness of low breakfast turnout never reached a registration or billing discussion either (confirmed).
- **Downward, from institute to kitchen.** Exams, events and holidays reach Yuktāhār only when someone raises them in the meeting (confirmed), and no calendar feed exists.
- **From student to kitchen.** Whether a Skip Meal declaration changes the number Kadamba cooks from is in conflict. Earlier findings say declarations reach the kitchen, while Kadamba's operator says it cooks on registrations only (single-sourced). Skip Meal may adjust the count the operator sees while the operator treats whatever count it sees as the obligation, but this is not established.
- **Laterally, between operators.** Operators do not share figures. The Bakul operator believes Kadamba sees 90 to 95% of registrants (single-sourced, unclear in the recording, possibly about lunch or dinner), while the April scans show 37.1% for Kadamba breakfast (confirmed).

Holding the data is also not the same as seeing it at breakfast level. Yuktāhār's site head believes breakfast turnout is about 50% (confirmed as his belief), while his September register shows 17.7 to 40.2%, a mean of about 28% (confirmed, records). His lunch figure, above 70% on most days, matches the register. The Bakul manager estimates

at most about 60% (single-sourced; no records), and if Bakul resembles the recorded halls at 28 to 37%, that is an upper bound. He explains the shortfall partly by taste and menu and partly by students not waking or being reluctant, a view untested with Bakul residents, of which only the second matches what interviewed students most often say.

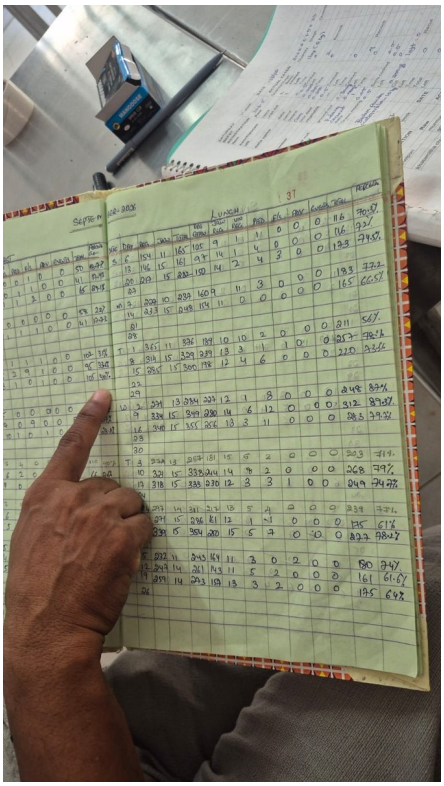


Figure 2. *The operator's register: Yuktāhār's September attendance book, lunch on the right and the edge of the breakfast page on the left.*

Leverage analysis

Power is the capacity to influence a decision. Leverage is the capacity for a small action to produce a large system effect. They are separate.

(Revised: the first version said the Mess Office holds both.) The Mess Office holds power and leverage over student flexibility rules, shown by a small operational change producing a system-wide behavioural effect in December 2024 (candidate), but that is a different lever from waste. The operator holds high leverage and high local power over waste and no rule-making power, and two kinds of leverage need keeping apart. Parameter leverage, the first-batch ratio, the top-up threshold and the batching of each item, belongs to the operator. At Kadamba the ratio is a standing Prism rule and the fastest lever, the in-service top-up, belongs to the supervisor; at Yuktāhār both sit with the daily meeting; at Bakul both sit with the manager, bounded below by his no-shortage floor. This leverage is fast, local and needs no rule change. But tightening it alone makes the gap even less visible upstream, and cutting the first batch would take food from the outside diners and staff who eat from it. Information-flow leverage, routing the operator's records and the CDS scan records to whoever owns the cut-off, cancellation rules and billing, belongs to nobody. The data exists and the flow does not. The institution's per-meal tasting and weekly QC already put it in the kitchen every day, on a route that carries safety and taste only. Operator culture is a further candidate lever: under one billing structure a management that ties waste to profit built a record system and a blame-free measurement habit, and one that treats registration as an obligation did not.

Academic administration holds power without leverage over this problem. Students individually hold neither, but the Mess Cell market shows that students collectively have real organising capacity, which has been aimed at governance at least once. That is better described as exercised-but-unmeasured leverage than as sleeping leverage.

Sleeping stakeholders

Academic administration remains the clearest case: total authority over class timing, no relationship to any mess-governance actor, and no incentive pulling it toward engagement.

The Mess Committee's chair, if the report of a faculty chair holds, is subtler: a person with plausible informal access to academic-side actors that no student channel has, and no evidence the access has been used to raise the overlap.

The operators' records are a sleeping resource, not a sleeping stakeholder. At Yuktāhār they are kept per day and per item and reconciled book against book, though the books do not always agree (the Wastage Book showed 77 breakfast diners and the attendance register 110 on 3 September), and nobody outside the kitchen has asked for them.

Students, collectively, are ambiguous. Mess Cell proves peer-level organising capacity, and an organised effort has been aimed at governance at least once. What it targeted and whether it changed anything are unknown, so the open question has narrowed from whether this capacity has ever been aimed at governance to what happened the one time it was.

Stakeholder Power-Interest/Leverage Map

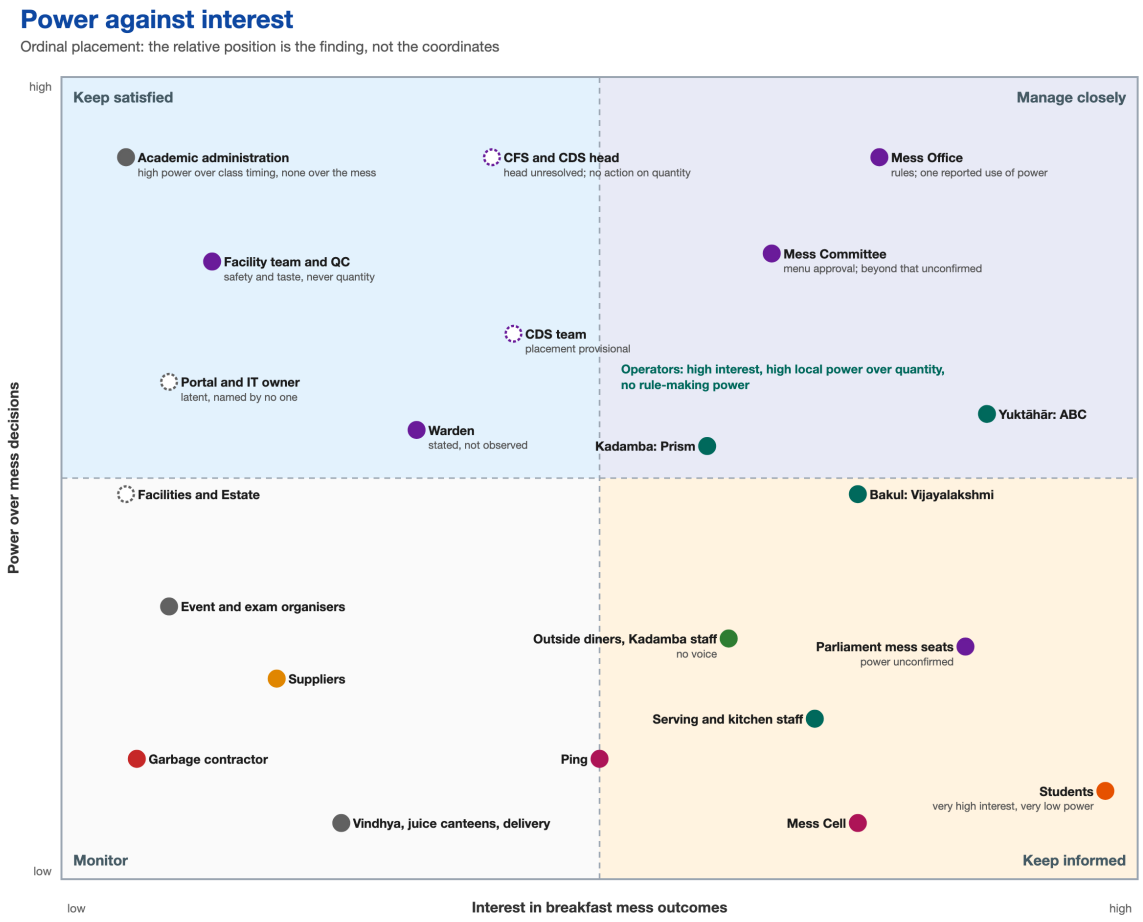


Figure 3. Power against interest. Placement is ordinal.

Coordinates on these charts are ordinal placements, not measured values. The relative position of each actor across quadrants is the finding. The **Mess Committee** has high power over menu approval and unconfirmed power beyond it. The **Mess Office** is the highest-confidence high-power placement over rules, with one reported instance of

exercising it. **Academic administration** has high power over class timing and essentially none over, or interest in, mess outcomes. **Mess operators** have high interest, shown through waste tracking and requests for menu input, high local power over quantity, re-cooking and reuse, and no rule-making power. (This revises the first placement of vendors at moderate power and low interest.) The three differ sharply under that shared placement. Yuktāhār (ABC) is a flat team with the fullest records and an explicit cost-and-waste goal. Kadamba's Prism holds the contract and runs a layered chain in which the supervisor sets each top-up. Bakul's manager, at the Hafizpet-supplied hall, calibrates the batch weekly by judgement above a no-shortage floor, and on the menu he sits at high interest and real expertise with low formal power. The **CDS team** is the visible institutional counterpart for complaints and scan data, placed provisionally while its relation to the Mess Office is unconfirmed. **Students** have very high interest and very low formal power, the textbook mismanaged ”Keep Informed” quadrant. **Mess Cell and Ping** have low formal power by design and moderate interest, with real influence running through informal channels.

Several stakeholders from the Stakeholder Map are placed here in text. **Event and exam organisers** have no power over the mess and little interest in it, yet move breakfast turnout, and their calendar reaches the kitchen only if mentioned. **Suppliers** have power over the quality of raw stock only, which operators answer by weighing, returning and switching. **The garbage contractor** has no power and no stake beyond collection, and is the single point where Kadamba's waste leaves. **The Facilities and Estate team** has power over repairs such as the broken toaster, with its identity unconfirmed. **The Portal and IT owner** has high latent power over what the platform records and permits and is named by no source. **The Student Parliament's mess seats** have high interest and power unconfirmed beyond one reported negotiation in December 2024. **CFS** is at the top of Kadamba's chain with high formal power and no observed action on quantity, as is **the CDS head**, whose identity is open between two names. **The facility team and QC** have high power over safety and taste every meal and none exercised over quantity. **Outside diners and Kadamba's own eating staff** have no power or voice yet absorb part of the first-batch surplus. **Serving and kitchen staff** have little power over rules but carry out every top-up and re-cook, and at Yuktāhār hold part of the planning through the daily meeting and peer checks. **Vindhya, the juice canteens and delivery platforms** have no power in the mess and an interest only in the demand it loses.

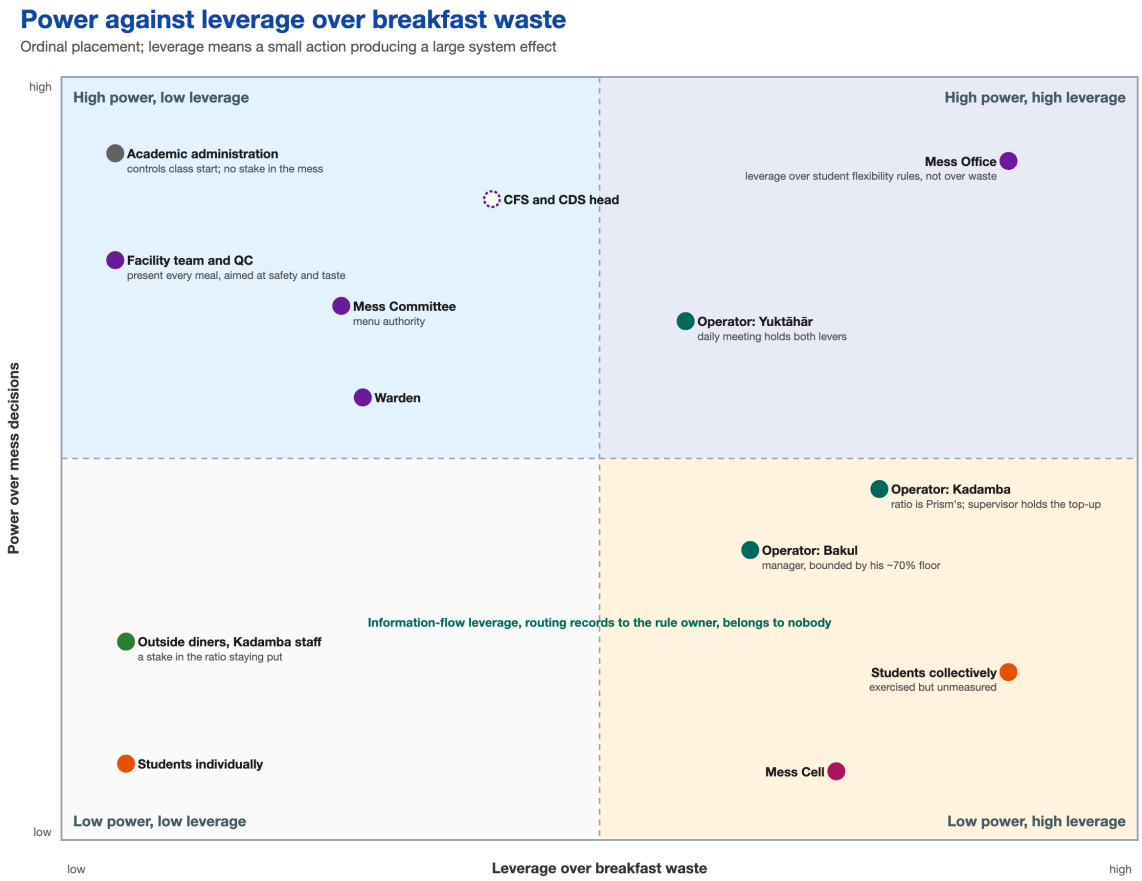


Figure 4. *Power against leverage. Placement is ordinal.*

Operators sit at high leverage over waste with local, not rule-making, power. Within Kadamba that leverage is exercised live by the supervisor, and at Bakul by the manager. The Mess Office's leverage is over student flexibility rules, not waste.

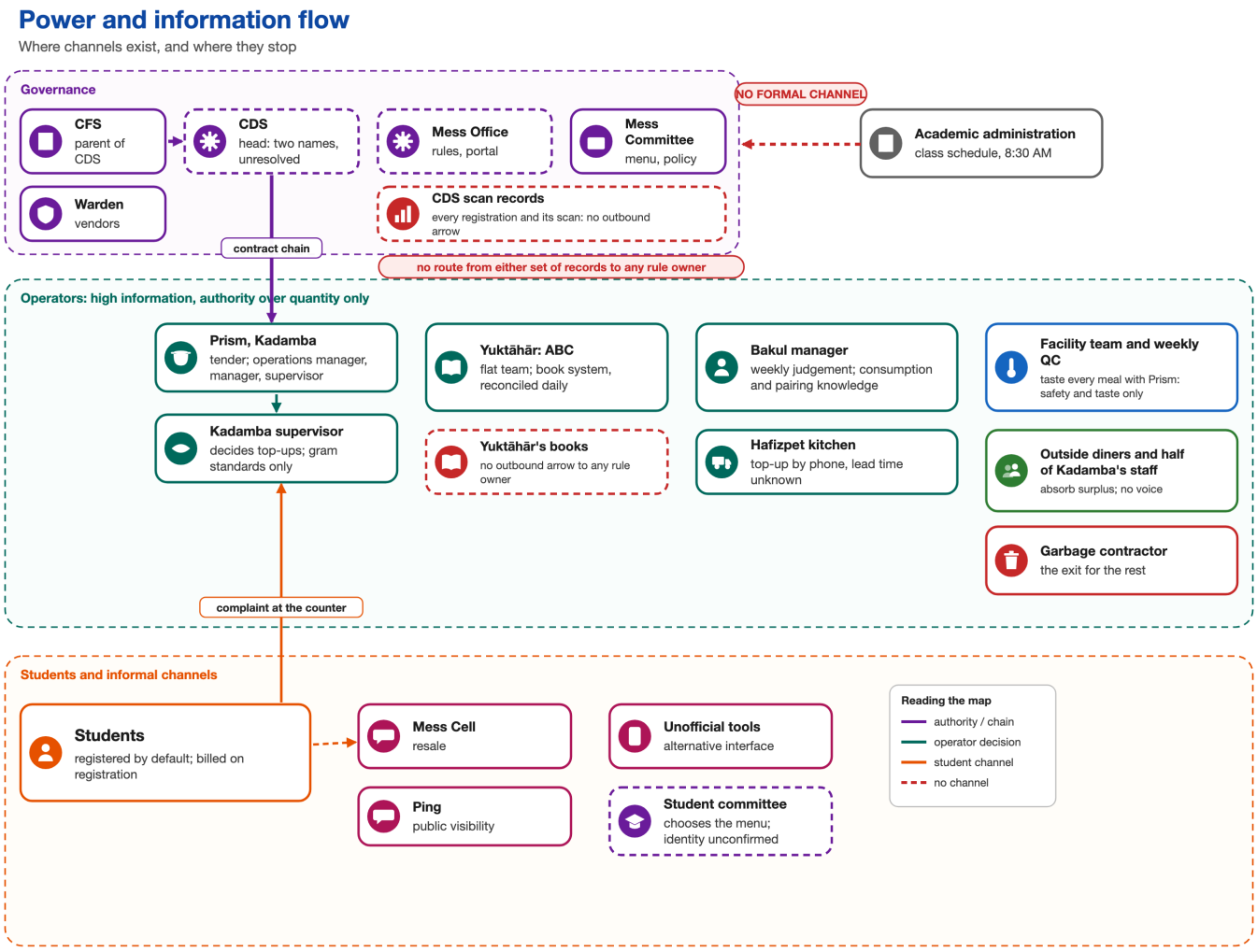


Figure 5. Power and information flow: where channels exist, and where they stop.

Dotted lines mark informal or unconfirmed channels, and a red ”no channel” marks a confirmed absence of any relationship, most consequentially between Academic administration and every mess-governance actor. The same absence runs from the operators' waste and attendance records to every actor that sets registration, cancellation or billing rules, and from the institute calendar to the kitchen. Kadamba's formal chain runs from CFS through CDS and Prism to the supervisor, and the facility team's per-meal channel into the kitchen carries safety and taste, never quantity.

Implications for System Behaviour

This distribution of power plausibly produces and sustains the Breakfast Paradox. Billing without an attendance incentive removes pressure on the one actor, the student, whose behaviour most directly decides the outcome. The one governance action on record served an internal operational goal, not the paradox, and may have tightened the underlying pattern (candidate). The actor best placed to resolve the second driver, the schedule overlap, is disconnected from every governance actor who might raise it.

The most concrete stabiliser sits in the kitchen. Each operator runs its own balancing loop: cook a fraction of registrations, watch the counter, top up when it runs low. It is confirmed at Yuktāhār and single-sourced at Kadamba and Bakul. It keeps the counter from running empty and keeps the gap from surfacing as a crisis. The Bakul manager states its goal outright, ”there should not be a shortage of food, even if some wastage occurs,” and keeps food for about 70% of registrations even on very low weekends. A run-out is visible and routes a complaint from the

CDS team through the manager towards the managing director, while a surplus is silent. The floor therefore reads as the kitchen's answer to a system that reacts only to what is loud (candidate). At Kadamba the surplus is further absorbed by outside diners, staff and reheating within service, which may explain why the waste the operator reports is small, though that is not established. Because its records never travel upward, the rule-setters receive no signal that anything needs changing. The system holds itself in place not because nobody measures the gap but because the one actor who measures it daily cannot change the rules that create it, and no route carries the measurement to those who can.

The channel that could generate collective pressure, the students, has been aimed at governance at least once, in an effort whose subject and outcome remain to be confirmed. The paradox may be sustained less by missing channels than by channels whose effectiveness nobody has measured, alongside one channel, from kitchen records to policy, that does not exist.

Key Findings

1. **Registration without an attendance incentive.** Billing follows registration, so nothing pushes students to register only when they mean to attend. This is the best-evidenced structure here, now sized from records: 37.1% breakfast turnout at Kadamba in April 2026, with registrations flat across every weekday.
2. **Unilateral operational authority in the Mess Office.** It made a real policy change in December 2024 with no evidenced Committee or student involvement beforehand (candidate; press-derived). Whether this is a standing pattern needs governance-source confirmation.
3. **Total independence of Academic administration from mess governance.** No mandate connects the two in either direction, through inattention rather than refusal. The gap reaches the kitchen, which learns of exams and holidays by word of mouth.
4. **An input nothing requires to be accurate.** Students control registration and bear no cost for its inaccuracy. Operators correct it privately: Yuktāhār with its own records, Kadamba with a fixed 70% as an obligation, and Bakul with a weekly 70 to 80% share and a floor near 70% on low weekends (single-sourced). Scan records measure the inaccuracy, no rule responds, and the institution never sees the corrections.
5. **Skip Meal as a dominated strategy.** It returns nothing while resale returns partial or full value, so resale wins whenever there is advance notice (confirmed, interviews).
6. **Exercised-but-unmeasured collective leverage.** Mess Cell shows real self-organising capacity, and an organised effort has been aimed at governance at least once. The December 2024 Ping-survey-and-Parliament episode is a strong but unconfirmed candidate match.
7. **A confirmed feedback input with an unconfirmed output.** Three student channels (in-app rating, email, Ping) carry input, and complaints demonstrably reach the operator's manager. None has a confirmed instance of producing a rule change.
8. **A data-rich, authority-poor operator.** Yuktāhār measures the gap daily, per item and weekday, in books it reconciles, and cannot change the cut-off, cancellation, billing or menu rules that create it. No route carries its measurement to those who can.
9. **A flexibility asymmetry against the student.** The cut-off binds students four days out while operators top up within the meal, and beyond five cancellations a month the booking cannot be released at all (confirmed).
10. **Oversight pointed at the wrong variable.** The institution is in the Kadamba kitchen every meal through temperature checks, a retained sample, the display plate and tasting, and weekly through QC, and every check covers safety and taste. None covers how much is cooked or wasted.
11. **Two operators, one billing rule, opposite stances.** Yuktāhār, with a profit and waste goal, a flat team and blame-free measurement, learns from its books. Kadamba, run by Prism through a layered chain, cooks a fixed ratio because it is paid for registrations. That the difference follows culture rather than payment rule is candidate, and it makes operator culture a candidate lever. Bakul adds shortage first with waste accepted, and its manager reports payment on plates served, in tension with Kadamba's remark. If confirmed, the operators do not share one billing rule (unresolved).
12. **Menu expertise without menu authority.** The Bakul manager holds consumption knowledge by item and about 20 years in North Indian food, disputes the pairings students choose and has no formal say (single-sourced). His

view of what residents would eat is untested. The finding is the mismatch between who knows what is eaten and who decides what is served.

Outstanding Data Collection

The September 2026 operator interviews and the Kadamba scan records answered most of what this section first listed. Kadamba's operator is Prism, which holds the contract by tender and runs all operations. The command chain from CFS to service staff is known. The cancellation cap is confirmed. The mechanism by which each kitchen decides quantities is known and the gap is measured from records. Vendor compensation is partly answered: Bakul's operator, Vijayalakshmi Caterers, won a tender and is paid monthly as a lump sum with deductions, on what its manager calls plates served (single-sourced), and it runs Bakul. Operators bargain with their own suppliers, but no bargaining with the institution is evidenced. The following remain open and **will be collected and submitted later**.

- Whether operator pay is per registered or per served plate, operator by operator, and the deductions. This decides whether the gap costs or earns the operator and is the most consequential open item for any later intervention. Bakul's ”served” needs confirming from the contract or CDS, including whether it means plates eaten, and whether Kadamba's basis differs. The reported plate rate of about Rs. 80, with bids near Rs. 60, comes from noisy audio and needs checking against the tender.
- Whether the December 2024 Ping-survey-and-Parliament episode is the ”organised email effort” otherwise heard only secondhand, and whether ”opposition registered, policy implemented anyway” is the full account. This is the highest-priority open governance question.
- Whether any institutional actor receives or asks for an operator's Wastage Book or attendance register, and whether operators know the institution lacks them.
- Who owns the four-day cut-off, whether an operator has ever asked to shorten it, and why the cap was set at five.
- The identity of the ”student committee” that operators say selects the menu (Parliament-internal committee, institute-level Mess Committee or CDS Committee), and whether Committee approval is formally needed for a Mess Office change; whether ”CDS” is the Mess Office; whether the CDS head named in the site account and the CDS Chair named earlier are two roles or one person; and who sees and acts on per-meal ratings once submitted.
- Whether QC visits can penalise or only inspect, and whether the facility team that tastes with Prism is the CDS site or facility team.
- Whether a Skip Meal declaration changes the count Kadamba cooks from.
- The base of Kadamba's 5 to 10% daily waste and how it squares with a 70% batch against 37% turnout; whether meals for the 30 to 50 outside diners and for Kadamba's staff are a written entitlement or a norm; and who pays the garbage contractor, whether waste carries any cost to the operator or institution, and whether the contractor composts Kadamba's waste off site.
- Who the Yuktāhār site head is relative to the mess in-charge and the two project managers.
- Whether Palash is served by the same operator and kitchen as Bakul (candidate).
- Whether anything in Bakul's weekly adjustment is written down, how long a Hafizpet top-up takes to arrive, whether the 9:20 reserve of about 50 rests on data or judgement, and which menu decisions students control and the manager can change.

Current-State Process Map / Process Trace

IIIT Hyderabad Breakfast Mess System

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Overview

The Stakeholder Map says who is involved. This trace says what actually happens, step by step, workarounds included. It follows one breakfast from the student's registration through the operator's planning and cooking to the counter, the close, and what happens to the food nobody eats.

Three timescales are kept apart. Registration is locked about four days ahead, and because every student is registered automatically it works as a standing month-long default, not a choice. Cooked quantity is set the day before: as a share of registrations at Kadamba, from the mess's own past records at Yuktāhār, and at Bakul as an item-dependent share the manager adjusts weekly from past attendance. The decision to eat is made on the morning itself, and the operator responds live at the counter. At Yuktāhār it also re-forecasts the rest of the meal from the first half hour, so the morning holds an operator forecast as well as a reaction.

(Revised: the first version covered only the student half and stopped at the kitchen door, because no source then showed how a registration became a cooked quantity.) Operator interviews at three messes, one mess's photographed planning and attendance registers, a month of timestamped Kadamba scans, a walkthrough of the Kadamba kitchen and the team's first-hand account of two visits now make the operator half visible. The two best-evidenced operators run on different logics, so the trace follows each end to end. Bakul, cooked off site, is traced on its manager's account alone, since no Bakul records exist. What stays unseen is narrow and listed at the end.

Conduct Interviews and Observations

Source	Represents	Reveals	Cannot establish
Original respondent (2026-09-03)	One habitual skipper, general system informant	System mechanics (registration, billing, QR, Mess Cell, governance) and one person's skip reasons	Whether the reasons generalise

Source	Represents	Reveals	Cannot establish
Portal screenshots and the printed CDS Student Handbook	The registration and billing platform and its published rules	Prices, capacities, registration counts, Skip Meal versus Cancel, the four-day registration and cancellation deadline, automatic allocation of a vegetarian meal	Attendance, since registration is not attendance
Five student interviews (2026-09-06)	Two daily eaters, three skippers	Decision sequences, queue and crowding, item run-outs, registration habits, Mess Cell use, timing preferences, self-estimated spend on unused meals	Whether six respondents represent the population
CFS Chair interview and Kadamba kitchen walkthrough (2026-09-13)	The dining-services side and one kitchen's layout	Governance and feedback triage, the entry cut-off and refill window, equipment, one person running all inventory on first-in-first-out and first-expiry-first-out	Day-to-day quantity decisions
Yuktāhār walkthrough with 26 photographed registers (2026-09-26)	The site head of ABC Hospitality Services walking through every register	The planning system: twice-daily meeting, Wastage Book, weekday attendance register, grocery and stock books, reuse, staff routines	Other messes; how typical one well-documented operator is
Kadamba manager interview (September 2026)	The mess manager of Prism	Waste handling, exceptions (scanner failure, gas shortage, broken equipment), complaint routing	Figures not covered by the team account; the audio is noisy
Team field account of the Kadamba and Yuktāhār visits (2026-09-27)	A team member present at both operator conversations, and a Kadamba student	The Kadamba command chain, quantity rule, gram standards, food-safety sequence, extra diners, timing and procurement; the Yuktāhār books, forecasting steps, organisation and culture; auto-registration from the student side	A verbatim operator transcript; the base of Kadamba's waste figure
Bakul manager and staff (September 2026), synthesised by a team member who understood the conversation	The operator of Bakul, cooked at Hafizpet	Tender and monthly payment, the weekly-adjusted first batch and its floor, early preparation, transport and hot holding, 30-minute checks and phone top-up, the late reserve, pairing views, the complaint route	Any Bakul record; the top-up lead time; whether Palash is served the same way
Kadamba breakfast records, April 2026	32,162 registrations and 11,941 timestamped scans from dining services	Attendance by day, weekday and veg or non-veg; the arrival curve minute by minute; when service really ends	Which individuals skip; lunch and dinner; quantities cooked or wasted
Power-Interest/Leverage analysis	Governance structure	Formal and informal power, precedent, the Parliament's Mess Secretary role	Kitchen operations

System visibility. Everything from a student's own decision through billing, and now everything from registration reaching the kitchen through cooking, service and disposal, is visible for two messes. Confidence differs. The Yuktāhār account is confirmed where it rests on the registers and single-sourced where it rests on the operator's explanation. (Revised: the Kadamba rule, gram standards, safety sequence and timing were first held as a noisy recording with doubtful numbers. A team member heard the same conversation directly, so they are now the operator's single-sourced statements with the figures no longer in doubt. The Bakul account was likewise first held as noisy audio from an unnamed mess, and is now the manager's single-sourced statements, still his own estimates with no records.) The Kadamba attendance figures are confirmed from records.

Observation. A live breakfast service has not been watched from start to finish. The 11,941 Kadamba scans answer what that observation was meant to capture, the shape and timing of the crowd, better than one morning of counting could. The team saw the wrapped display plate at Kadamba's entrance and the Yuktāhār books themselves. Waste disposal after close still needs direct observation.

Trace Decisions and Interactions

Registration is locked about four days ahead. The CDS handbook requires registration and cancellation at least four days in advance, and Kadamba's manager calls it ”weekly compulsory.” Students say they register ”for a month,” three of six explicitly. Both hold: the portal locks on a four-day weekly rhythm and students set a standing booking once. A team member's own experience as a Kadamba student adds that registration is automatic and mandatory, off campus and over holidays, and that the only exit is five cancellations per meal type per month made four days

ahead: ”we don't have any opportunity to nullify it.” The booking is a default. Kadamba's April records confirm the result: registrations flat at about 1,070 a day on every weekday including Sunday (range 928 to 1,123) while turnout runs from 42.1% on Tuesdays to 27.3% on Sundays.

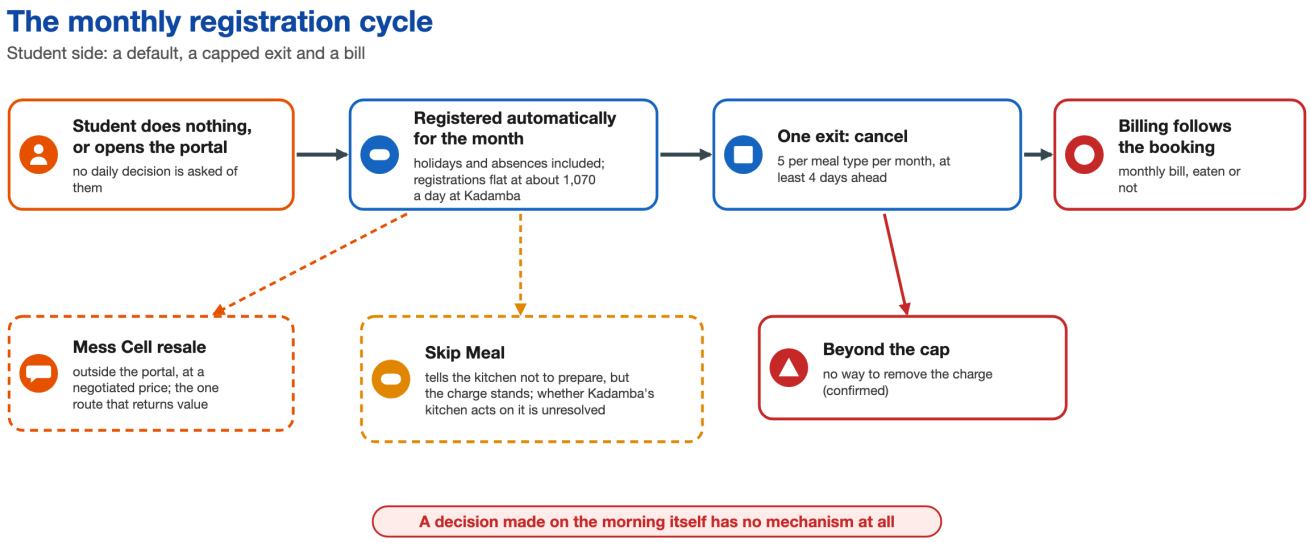


Figure 1. *The monthly registration cycle, from the automatic default to the bill.*

The daily cycle is where the go-or-skip decision happens. It is evidenced across all six student respondents and now measured at record level.

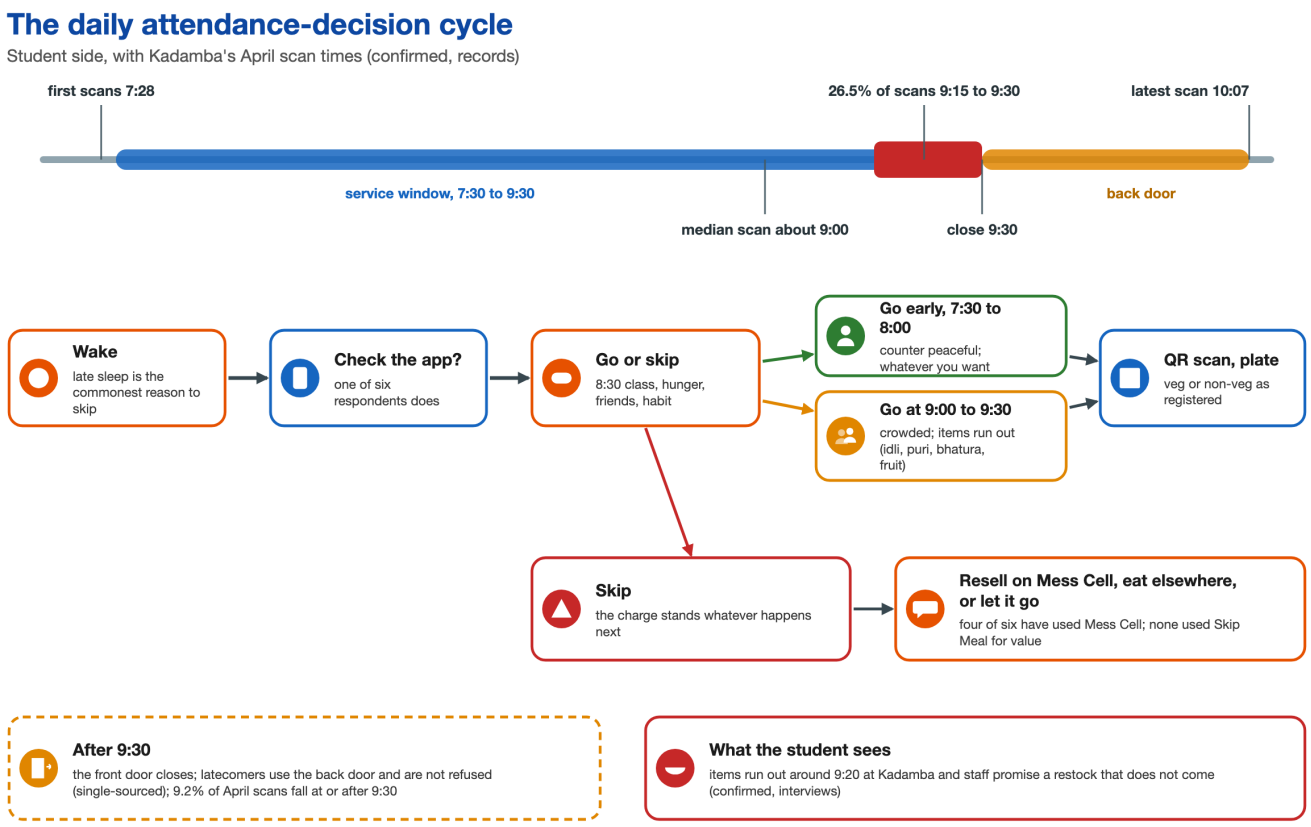


Figure 2. The daily attendance-decision cycle on the student side.

Decision points, supported by evidence only

Decision points: students

Decision	Who	When	Constraint	Options observed
Register or not	Student	At least four days ahead; in practice once for the month	Registration is automatic and mandatory, holidays included; popular messes fill; a student who does not register is allocated a vegetarian meal	All six register in advance, none day by day
Which mess	Student	At registration	Capacity, competition for popular messes	Kadamba chosen by four of six; Bakul mentioned once
Go or skip, that morning	Student	On waking	8:30 class and attendance policy (named by one respondent)	Go, skip, go later. The central real decision, evidenced across all six and confirmed by records: 37.1% of Kadamba breakfast registrations used in April, lowest on Sundays (27.3%); Yuktāhār September breakfast 17.7 to 40.2%, Sundays among the lowest. No Bakul records; its manager estimates at most about 60%, fewer at weekends (single-sourced)
Check the app first	Student	Before or instead of deciding	A crowd-check and menu view exist	One of six checks; the rest ignore it
Wait for or wake friends	Student	On waking	The friend's own wake state	Wakes them and goes regardless (two of six); not a factor (three); positive but non-determining (one)

Decision	Who	When	Constraint	Options observed
When to arrive	Student	That morning	9:30 entry cut-off	Skewed late. The median April scan fell between 8:51 and 9:11 by weekday, about half of scans came after 9:00 and 26.5% in the last quarter hour. Kadamba's rule of thumb is about 100 early, 100 in the middle and 150 in the last ten minutes: ”for any meal the highest flow is in the last 10 minutes.” Bakul's manager puts his main rush at 8:00 to 8:30, with 10 to 15 in the last ten minutes (single-sourced; no Bakul scans)
Cancel a registration	Student	At least four days ahead	Cap of five per meal type per month (confirmed)	Used by most at least occasionally; one does not, because they forget
Use Skip Meal	Student	Same day or ahead	Returns no money	Never used by any of the six who addressed it
Resell or buy via Mess Cell	Student	When a registration will go unused	Informal, no guaranteed buyer	Used by four of six; a bystander for one; not for breakfast for another
Eat as a walk-in	Student	Same day, unregistered	Price markup	Snacks only (one); breakfast (one); never (three). Yuktāhār's register counts Jain, unregistered, paying and faculty or staff diners separately; at breakfast the extras are at most about a dozen a day
Enter after the close	Student	After 9:30	The front door closes at 9:30	At Kadamba latecomers use the back door and staff do not refuse them

A hypothesis this data disconfirms. Early drafts treated social company as a meaningful driver of attendance. Six respondents tell a weaker story: peer presence is at most a motivational add-on, never the deciding factor. Nobody describes skipping because a friend skipped, or attending only because one did.

Decision points: operators

The three operators forecast in three ways. Kadamba cooks a fixed share of registrations and tops up reactively, with no learning record beyond per-dish gram standards. Yuktāhār learns from its own books, looking up the matching past week before service and re-projecting from the first half hour during it. Bakul sits between. (Revised: an earlier reading, from a noisy recording that gave only the ratio, had Bakul resembling Kadamba.) Its manager varies the share by item and adjusts it weekly from past attendance by judgement, never below a floor he keeps against shortage. Whether any history is written down is unknown.

Kadamba (run by Prism)

Decision	Who	When	Inputs and constraint	Confidence
Set the first batch	Prism kitchen management	The day before, from the weekly menu	About 70% of registrations, veg and non-veg planned separately; a per-person gram standard for every dish (idli 4 a head from about 120 g of batter, sambar 100 g, poha and pulao their own). Biryani lunch is the exception, at 100%	Single-sourced (operator, heard first-hand)
Cook for people outside the registration	Prism kitchen	Same planning step	About 30 extra portions for 30 to 50 outside people, such as night-shift security; about half of Kadamba's staff also eat from the same food	Single-sourced
Order raw material	Prism	In three tiers	Dry goods (rice, groundnut, masala, ghee, podi) a week ahead; perishables (milk, fruit, meat, eggs, most vegetables) the day before; some vegetables two to three days ahead	Single-sourced
Hold raw stock for a top-up	Kadamba kitchen	Morning	The uncooked remainder can be cooked in 10 to 15 minutes on request	Single-sourced
Watch the counter and call a top-up	The supervisor	Throughout service	Item levels at each time, registrations and turnout so far; the supervisor tells the kitchen how much more to cook. Items that ”float” above the gram standard are bonda, puri, vada and idli	Single-sourced

Decision	Who	When	Inputs and constraint	Confidence
Serve latecomers	Floor staff	After the 9:30 close	Latecomers enter by the back door and the mess ”can't say no to them”; on one biryani day the door was reopened for 5 to 10 minutes	Back door single-sourced; reopening single-sourced from noisy audio; breakfast application supported by scans after 9:30 on every April day
Dispose of leftovers	Kitchen	After close	Part is eaten by outside diners and staff; the rest goes to a bin collected by a garbage contractor	Single-sourced
Record the outcome	Kadamba kitchen	Daily and weekly	About 5 to 10% daily wastage on an unstated base, hard to square with a 70% batch against 37.1% turnout (see Key Findings); QC weekly. Per-item weighing is not established	Single-sourced

Yuktāhār (run by ABC Hospitality Services)

Decision	Who	When	Inputs and constraint	Confidence
Set tomorrow's breakfast and lunch quantities	Storekeeper, coordinator and kitchen staff in the first-half daily meeting; a second meeting sets snacks and dinner	The day before	Portal registrations as regular plus Jain plus cash meals; the Wastage Book page for the matching past week (the menu repeats on a two-week cycle: weeks 1 and 3 identical, weeks 2 and 4 identical) with its run-out times and quantities; the month's pattern; the weekday attendance register; holidays, exams and events, only if someone raises them (”We don't come to know until we sit in the meeting”). No fixed ratio; the site head believes breakfast runs at about 50%	Registers confirmed; forecasting steps single-sourced
Raise exceptions and discrepancies	Everyone, in the same meetings	Twice a day	The day's positives, negatives and discrepancies; impurities found in cleaning; store and cooked-quantity mismatches; leave requests (there is no formal policy and colleagues absorb the load)	Single-sourced
Issue raw material per meal	Storekeeper, cross-checked by the cross-verifier	Written at the planning meeting; issued the evening before or that morning	The per-dish ingredient list in kilograms from the store book; the Grocery Book with breakfast, lunch, snacks and dinner columns and a running balance	Confirmed from photographed registers
Clean and sort grain; judge the supplier	Store function	After issue	Raw material comes from many local shops and godowns, so stones are expected. An expected grams-of-impurity norm per item applies; exceeding it means a complaint or a new supplier, raised in the meeting	Practice confirmed; norm single-sourced
Cook in one batch or two	Kitchen	5:30 to 7:30	Most items in one batch; only items that dry out, such as poha, in two. The site head argues against general batching because ”student fluctuation is very high”	Confirmed
Re-project the rest of the meal	Coordinator and kitchen	First half hour of service	Turnout in the first half hour is a sample for the rest, driving top-up cooking	Single-sourced; breakfast use candidate
Monitor the counter	Kitchen	Every 30 minutes from about 45 minutes before close (about 8:45)	Food left against expected arrivals; a second roti counter opens at peak (handmade, no oil)	Checks confirmed; roti counters confirmed, meal not specified
Re-cook or not	Kitchen	When an item runs out	Stated for lunch: ”If it runs out at 1:30 we re-cook; if at 2:20 we don't.” The Wastage Book has run-out and re-cook columns. The same cut-off at breakfast is a candidate	Rule confirmed; breakfast candidate

Decision	Who	When	Inputs and constraint	Confidence
Reuse leftovers	Kitchen	After close	Idli becomes idli upma or idli poha for lunch; dosa batter goes back to the fridge; surplus milk is set as curd; paneer is made in-house	Confirmed
Record the outcome	The mess in-charge, cross-checked by peers	After each meal	The Wastage Book, where daily meal wastage is student plate waste in the bins plus leftover food	Confirmed

Bakul (Vijayalakshmi Caterers, cooked at Hafizpet). Every row is the manager's own account. No Bakul record of any kind has been seen, so every number is his statement or estimate.

Decision	Who	When	Inputs and constraint	Confidence
Set the first batch	The Bakul manager	Reviewed week by week	About 70 to 80% of registrations, by item, from attendance history by day (his example: if Mondays have run near 70%, cook about 80% of Monday's registrations; that 70% sits above his own breakfast estimate of at most about 60%, so it may not describe breakfast). (Revised: an earlier reading had a standing 70% raised to 80%. The two numbers describe variation by item and day, not change over time.)	Single-sourced; whether history is written down is open
Keep a shortage floor	The Bakul manager	Every day, visibly at weekends	His principle: ”there should not be a shortage of food, even if some wastage occurs.” Even on weekends, when breakfast attendance is ”very low,” he keeps food for about 70% of registrations	Single-sourced
Start labour-heavy items	Hafizpet kitchen	From about 3 AM	Items such as aloo paratha start around 3 AM to reach Bakul about 7, so some food is held for hours before it is eaten	Single-sourced
Transport and hold	Hafizpet, then the Bakul mess	Early morning	About 8 km, about 20 minutes early and 40 later; insulated boxes at about 60 to 72 °C, set in a hot-water holding setup. The heat is kept; steam softens hot parathas	Single-sourced
Check the counter	The Bakul manager	About every 30 minutes	Food remaining against registrations and demand still expected	Single-sourced
Call a top-up	The manager, by phone to Hafizpet	When food looks short	He gives an approximate quantity; the kitchen cooks and sends it. Lead time unstated. An earlier noisy reading said ”within 30 minutes,” possibly the check interval, and the drive alone is 20 to 40 minutes, so it is flagged, not relied on	Single-sourced; lead time open
Hold a late reserve	The manager	Around 9:20	He aims to still have food for about 50 at about 9:20, when 10 to 15 arrive in the last ten minutes. That it is held because a late top-up could not arrive is a candidate reading	Single-sourced; reason candidate
Dispose of leftovers	The Bakul mess	After close	Some leftovers go back to Hafizpet and the rest is thrown	Single-sourced (noisy audio)
Record the outcome	Not known	Not known	No attendance, consumption or waste record seen; his turnout figure (at most about 60%) is an estimate	Open

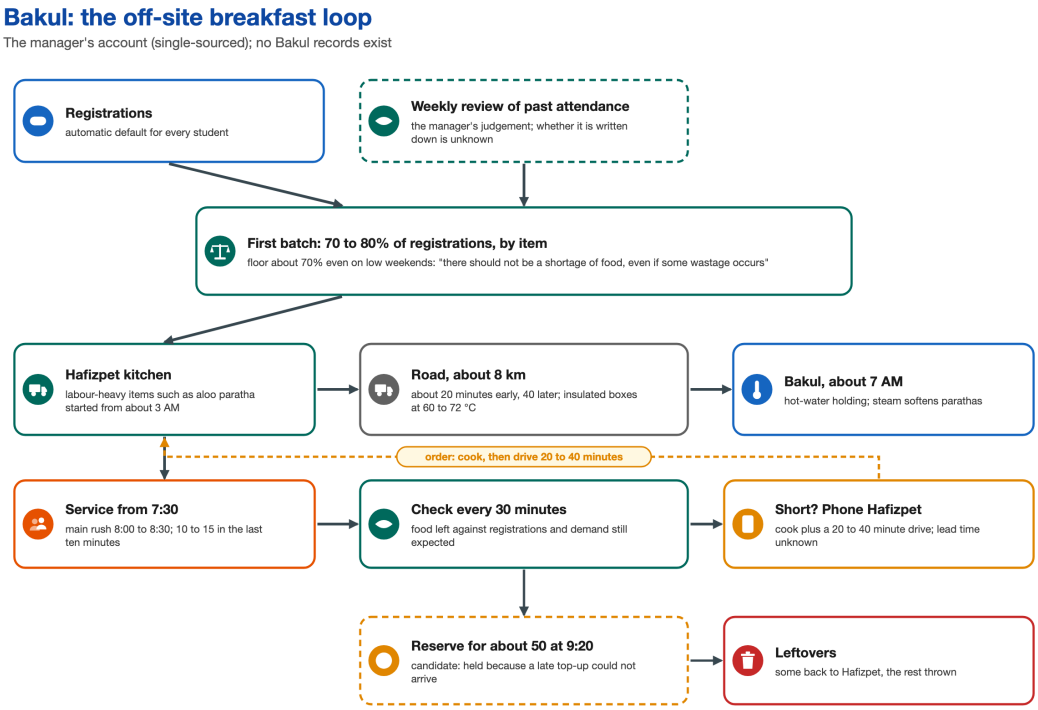


Figure 3. Bakul's breakfast loop from the weekly review to the late reserve.

Two readings follow from the manager's own numbers, both candidates. His first batch (70 to 80%) and weekend floor (about 70%) both exceed the highest turnout he expects (at most about 60%), so the batch exceeds turnout by design; if Bakul resembles the recorded halls at 28 to 37%, the gap is larger, and without records it cannot be sized. And the late reserve of about 50 is several times the 10 to 15 late arrivals he describes, the difference being insurance against a shortage a top-up from 8 km away could not fix in time.

Menu choice, all messes. The student committee chooses from an operator-supplied menu and the operator follows it. Operators see per-item waste (at Kadamba uttapam, set dosa and upma waste while puri, bonda, vada and idli move) and shift quantities between items, as Kadamba did in cutting ragi idli and increasing plain idli, but cannot change the menu. The Bakul manager asks for a say, and his objection is to pairings, not dishes: kachori belongs with chole, and he would serve Raj Kachori with aloo curry where the menu lists chana. He describes students choosing from balance, protein, carbohydrate and preference, and himself from traditional regional combinations, taste and what is eaten, with about 20 years in North Indian food. He also mentioned a protein target of about 30 to 40 g a meal, set by an unknown party, and cites pav bhaji as drawing more students and muesli and Chocos as eaten readily (single-sourced). This is the operator's view; whether Bakul residents share it is untested.

Yuktāhār: the book system as a process

Yuktāhār runs on paper books, each photographed by the team and explained by the operator. Everyone does every job, so ”storekeeper” is a function people rotate through, and each book is cross-checked by a peer.

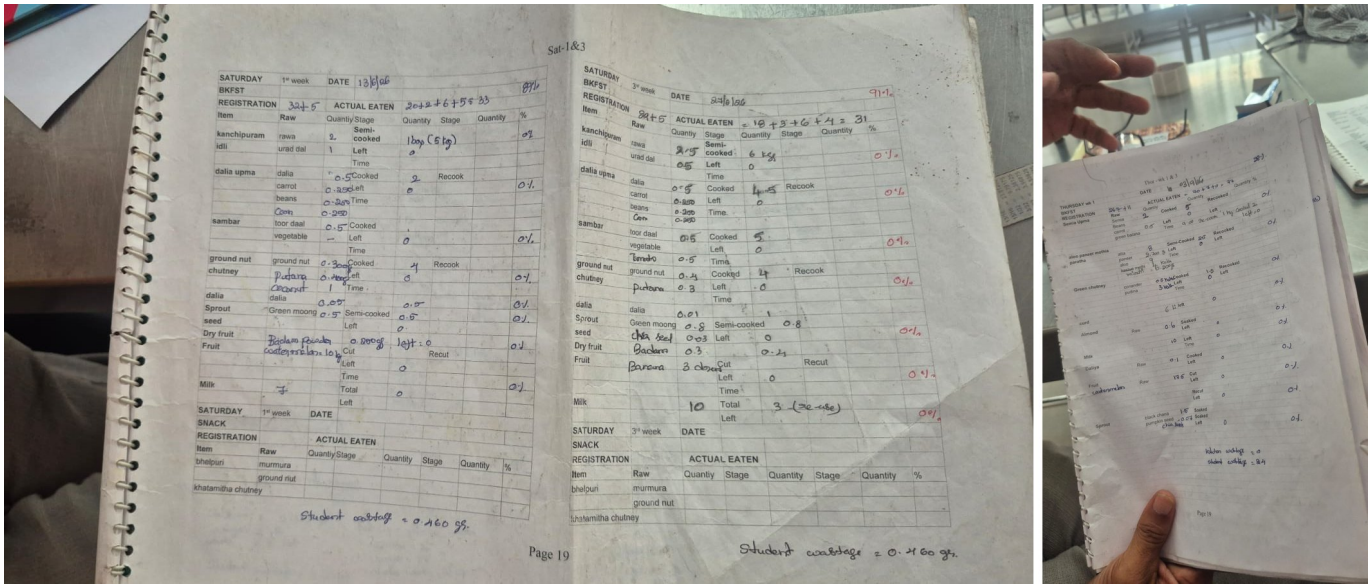


Figure 4. Yuktāhār's production and waste sheets for breakfast (a Saturday in June, a Thursday in September): raw issued, cooked, left, run-out time, re-cook and plate waste.

Book	Written by	When	Decision it feeds
Daily Meal Wastage Book: per meal and dish, raw issued, cooked, left, percentage, run-out time, re-cook quantity, registered (regular plus Jain), ate by category, and student bin waste plus leftover as the day's meal wastage	The mess in-charge, who also keeps the accounts	Every meal	The next matching-week quantity; the re-cook cut-off
Attendance register by weekday: registered against ate, every Sunday together, every Monday together	Coordinator	Every meal	Weekday turnout expectation
Wastage board	Coordinator	Daily	A target to push the waste share down
Grocery Book: issue per item per meal, day total, running balance; the header carries the day's registrations (18 September: breakfast 265+14, lunch 338+15, snacks 7+0, dinner 237+14)	Storekeeper, cross-checked by the cross-verifier	At the planning meeting for the next day	What the store pulls and cleans
Store and ingredient book: per-dish ingredient lists in kilograms	Storekeeper	Before each issue	The issue list
Stock register: one page per article, receipts in red	Store	Daily	Purchasing and the physical stock check
Vegetable book and vegetable indent: opening, received, used, balance; ordered against received, with remarks such as ”3 spoiled” or ”only 2 boxes”	The mess in-charge or the cross-verifier	Vegetables arrive two to three times a week	Vegetable orders; spoilage
Milk, fruit, gas, electricity and water register	Project team	Daily	Curd and paneer planning; machine run times (a Wednesday electricity spike comes from grinding idli batter)
Fridge temperature log and cleaning checklist	Hygiene team; the cross-verifier signs	Temperature three times a day; cleaning daily	Hygiene compliance

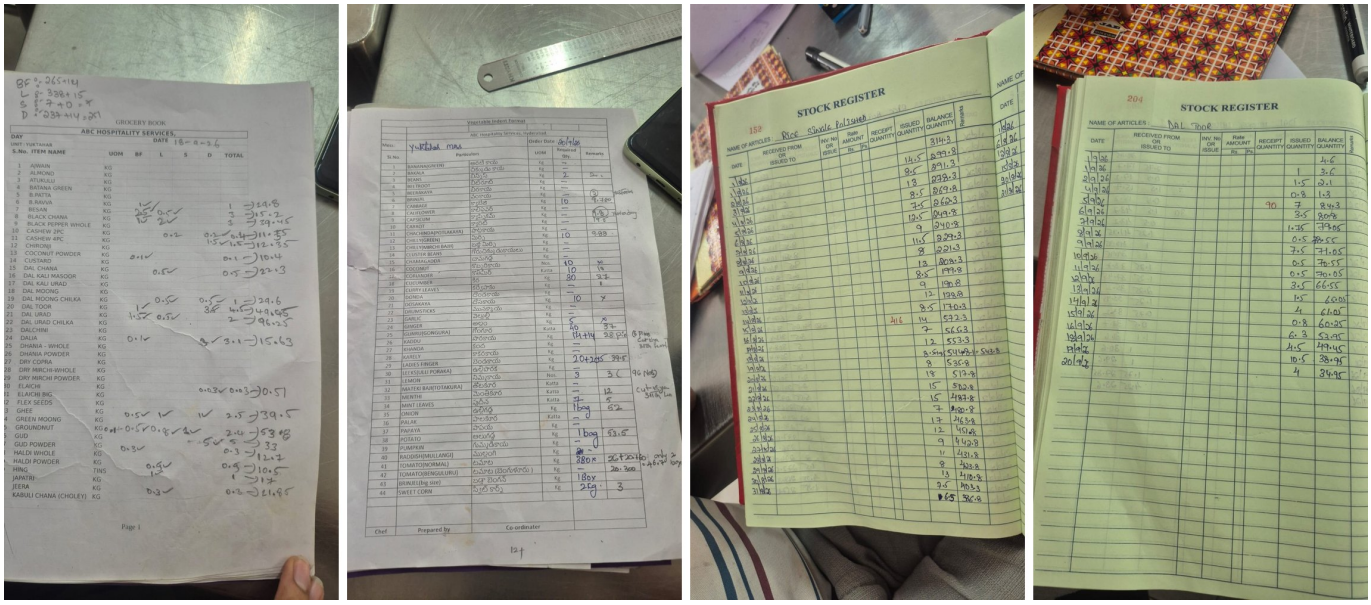


Figure 5. *The issue and stock books: Grocery Book page 1, the vegetable indent, and two stock register pages.*

One rule holds the books together: what the store book says was issued must match what the physical book says was cooked, and any mismatch is raised in the daily meeting. It held on the one day checked, 18 September, when 4.5 kg of toor dal issued and a 49.45 kg balance appear identically in the Grocery Book and the stock register (confirmed).

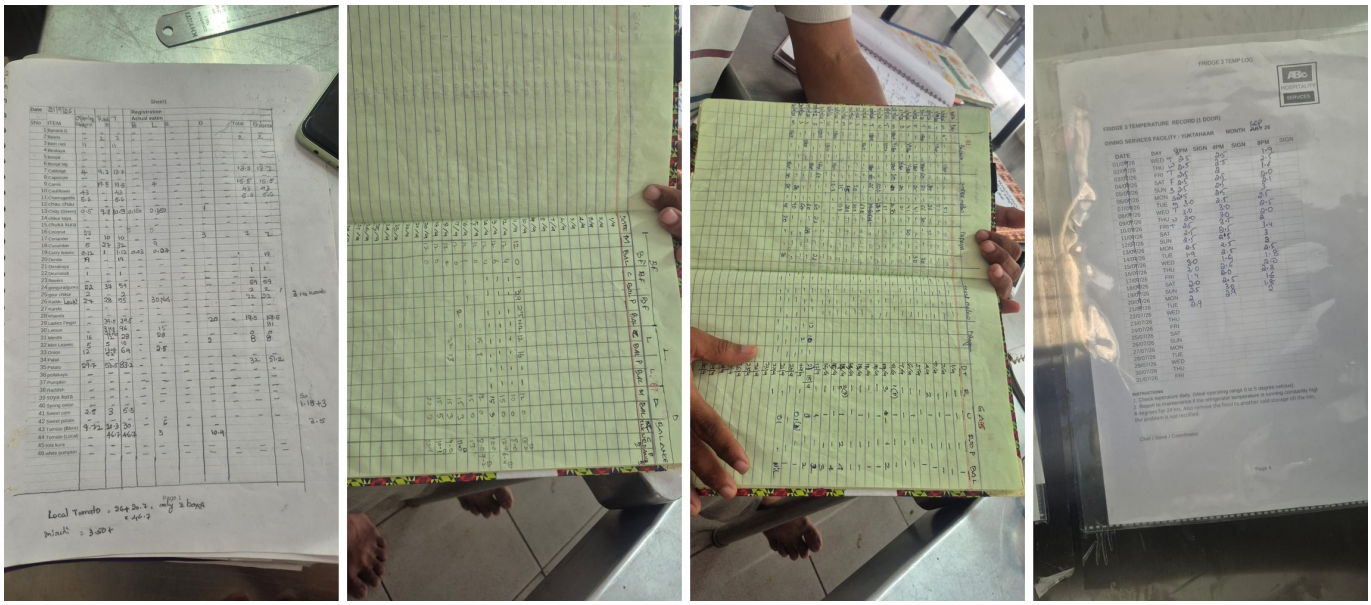


Figure 6. *The daily logs: vegetable stock, the milk and curd register, the fruit and gas register, and the Fridge 3 temperature record.*

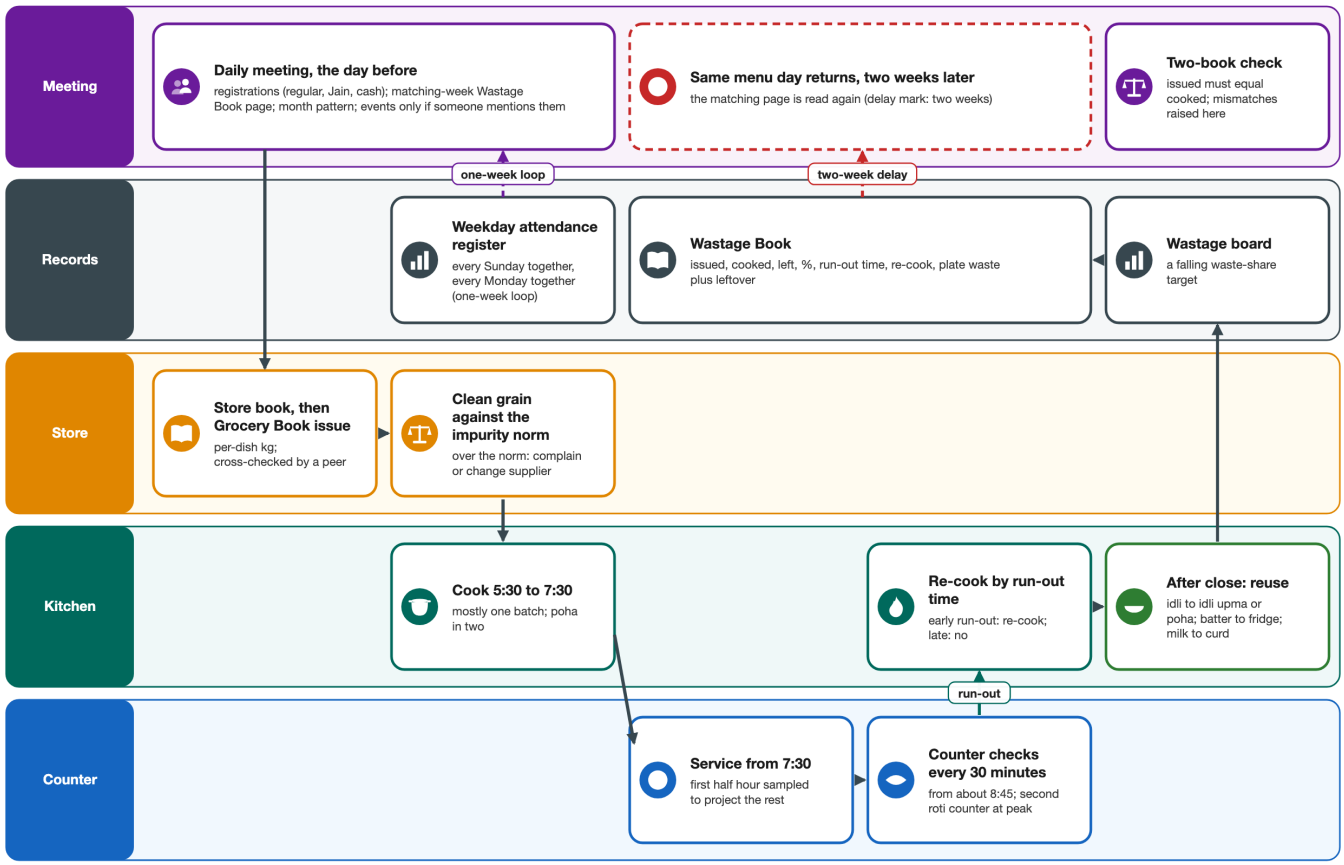


Figure 7. Yuktāhār's operator swimlane: from the daily meeting to the Wastage Book and back, with the two-week delay marked.

The breakfast day, operator side

The daily cycle diagram shows the student side. The operator side runs alongside it.

When	What happens	Confidence
One week ahead	Kadamba orders dry goods. Week by week, Bakul's manager resets each day's item-dependent share (70 to 80%)	Single-sourced
Four days ahead	Registration and cancellation close	Confirmed
Two to three days ahead	Kadamba orders some vegetables; Yuktāhār raises a vegetable indent two to three times a week and checks receipts against it	Kadamba single-sourced; Yuktāhār confirmed
Day before	Quantities set: Kadamba 70% of veg and non-veg registrations by gram standard plus about 30 outside portions; Yuktāhār from the matching-week Wastage Book page, weekday register, month's pattern and events. Kadamba orders perishables	Kadamba single-sourced; Yuktāhār confirmed
Evening before or early morning	Yuktāhār issues raw material and cleans grain against the impurity norm	Confirmed
From about 3 AM	Hafizpet starts labour-heavy Bakul items such as aloo paratha, aiming to reach Bakul about 7	Single-sourced
5:00 to 5:30	Kadamba's chef arrives at 5:30 and cooks batch items such as sambar and upma; the main veg chef leads. Yuktāhār cooking starts at 5:30 for a 7:30 finish. The early drive to Bakul takes about 20 minutes, against about 40 later (revised from an earlier 40-minute trip from a kitchen near the airport)	Single-sourced
Before the counter opens	Kadamba's food-safety sequence, in order: cooking temperature checked; a sample kept 72 hours so the food can be tested if anyone falls ill, including to tell whether the illness came from outside food; a wrapped display plate at the entrance so diners see the full menu; serving temperature per item; tasting by the facility team with two Prism staff; corrections. At Yuktāhār a tasting pair checks every shift	Display plate confirmed (observed); rest of Kadamba single-sourced; Yuktāhār confirmed

When	What happens	Confidence
7:00	Dosa and vada cooked in front of diners at Kadamba. Bakul's food arrives about 7 and its boxes go into hot-water holding, where parathas stay hot but soften in their steam	Single-sourced
About 7:30	First student scans	Confirmed
7:30 to 8:00	Yuktāhār samples this half hour to project the rest. At Kadamba it carried 15.3% of April scans	Kadamba share confirmed; Yuktāhār projection single-sourced
7:30 to 9:00	A steady, building flow; only a third of scans arrive before 8:30. At Kadamba each batch's surplus is held and reheated, and food is moved between vessels	Scans confirmed; holding and reheating is the team's judgement (candidate)
8:00 to 8:30	Bakul's manager's main rush, earlier than Kadamba's recorded curve (revised from an earlier 8:30 to 9:00 and later)	Single-sourced; no Bakul scans
Throughout service	Bakul's manager checks stock about every 30 minutes and phones Hafizpet when short	Single-sourced
About 8:45	30-minute counter checks begin at Yuktāhār; Kadamba top-ups take 10 to 15 minutes on site. A Bakul top-up's lead time is unknown: an earlier reading said up to 30 minutes, but the drive alone is 20 to 40	Confirmed at Yuktāhār; single-sourced at Kadamba; Bakul open
9:15 to 9:30	26.5% of Kadamba scans, rising to about 9 a minute by 9:29; a run-out this late likely falls past the re-cook cut-off. About 9:20 Bakul's manager aims to hold food for about 50 against 10 to 15 expected	Scans confirmed; re-cook link candidate; Bakul single-sourced
9:30	Formal cut-off. Scans continued past 9:30 on every April day, 9.2% of all scans came after it, and the last fell between 9:38 and 10:07. Latecomers use Kadamba's back door. The CFS Chair describes refills to 10:00 and lights off after	Scans confirmed; back door and refill window single-sourced
After close	Outside diners and staff eat the rest; leftovers are reused or binned; Yuktāhār records waste in the Wastage Book. At Bakul some leftovers return to Hafizpet and the rest is thrown, with no record seen	Confirmed at Yuktāhār; single-sourced at Kadamba and Bakul
Daily and weekly	Kadamba reads TDS on cooking and drinking water daily (a failure triggers a full check) and QC inspects weekly	Single-sourced
One and two weeks later	Yuktāhār's weekday register sets this day beside the same weekday a week on, and its Wastage Book page is read again when the same menu day returns	Confirmed

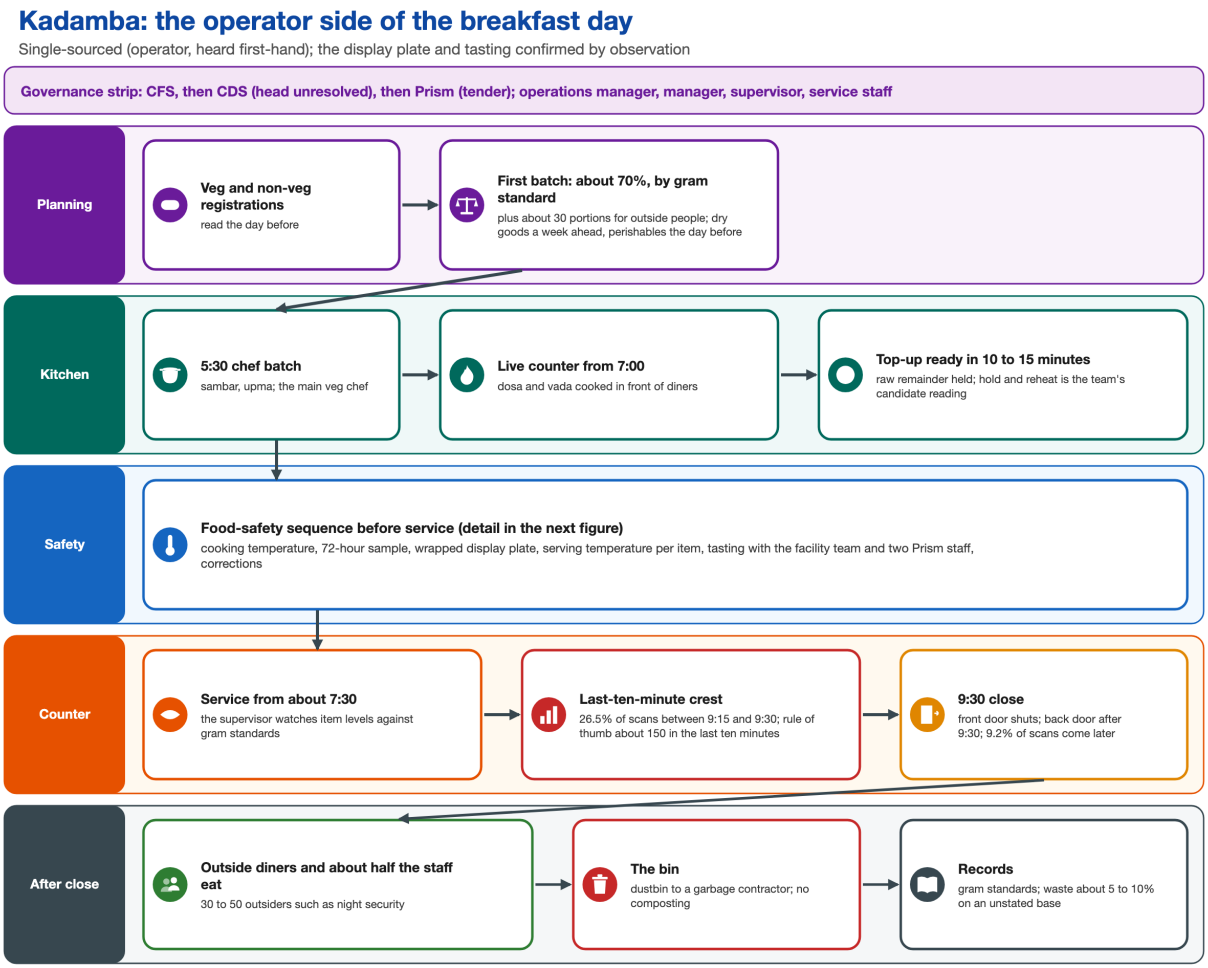


Figure 8. *Kadamba's operator swimlane: from registrations to the 9:30 close, the outside diners and the bin.*

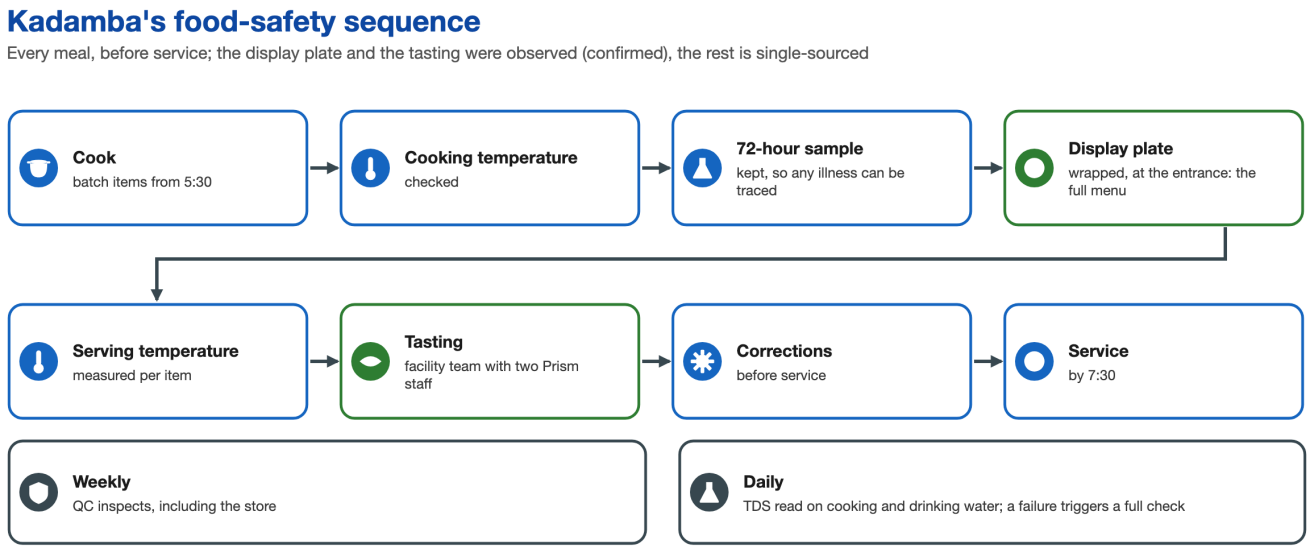


Figure 9. *The Kadamba food-safety sequence before 7:30.*

The last rows are a finding in their own right. (Revised: the first version said the best-documented correction acts only on a one-to-two-week delay.) Yuktāhār runs two corrections: a slow one, where a Thursday's breakfast shapes a later Thursday, and a fast one, where the first half hour of a meal projects the rest. The fast one samples the part of the morning least like the close. Using Kadamba's April curve as an illustration, the first half hour carries about 60 of about 398 daily scans, and projecting that rate flat across the two-hour window gives about 240, some 40% below the real total, because a quarter of diners arrive in the last quarter hour (candidate; Yuktāhār's own curve is unmeasured).

Document Formal and Informal Rules

Rules on the student side

Rule	Enforced by	Behavioural effect	Rationale known?
Registration required at least four days ahead, and automatic when a student does nothing, holidays included	CDS portal	All six respondents register in advance as a standing booking; registrations are flat across the week	Handbook: lets dining services estimate demand
Automatic vegetarian allocation for students who do not register	CDS portal	One respondent relies on it. Veg registrations at Kadamba were used at 32.4% in April against 48.5% for non-veg; whether default allocations explain part of that is a candidate	Stated in the handbook
Cancellation cap (5 per meal type per month), at least four days ahead	CDS portal	Used by most; one forgets; beyond the cap the charge cannot be avoided	Limits uncancelled no-shows
Skip Meal, no refund	CDS portal	Never used by any respondent who addressed it	Designed to reduce kitchen preparation, not billing; whether it lowers Kadamba's count is unconfirmed
Walk-in markup	CDS and operator	Used sparingly; two respondents avoid it for breakfast	Confirmed pricing incentive
Entry closes at 9:30	Mess	Four of six ask for it to be extended; at Kadamba latecomers already use the back door	Rationale unknown

Rules on the operator side

Rule	Formal or informal	Enforced by	Behavioural effect	Confidence
Paid monthly as a lump sum on plates, with deductions, under a tender	Formal (contract)	Dining services	Whether ”plates” means registered or served decides whether the operator gains or loses on every no-show; rate and deductions unverified. Kadamba describes registrations as the obligation it is paid for, Yuktāhār treats less waste as lower cost and higher profit, and Bakul's manager says payment is on plates served. That is one operator's word in tension with Kadamba's remark, and is not generalised	Single-sourced; basis unresolved
Cook a fixed share of registrations, top up live	Informal rule of thumb	Kadamba supervisor and kitchen	The stated 70% sits far above recorded turnout (37.1%). Kadamba treats cooking on registrations as a responsibility it is paid for	Ratio single-sourced; turnout confirmed; the gap candidate
Cook biryani at 100%	Informal	Kadamba	Full preparation when near-full turnout is expected; two pieces of 150 to 200 g with unlimited rice, and when some diners take three, later diners run short (lunch)	Single-sourced
Portion by per-dish gram standard	Informal but recorded	Kadamba	Standards are per person, yet the fast movers (bonda, puri, vada, idli) are the ones taken above standard	Single-sourced

Rule	Formal or informal	Enforced by	Behavioural effect	Confidence
Set an item share weekly from past attendance, never below a floor	Informal, the manager's judgement	Bakul manager	70 to 80% by item with about 70% kept on low weekends; corrects against surplus only down to the floor, and both share and floor sit above his own turnout estimate of at most about 60%	Single-sourced; written record open
Hold food for about 50 at 9:20	Informal	Bakul manager	Held rather than relying on a top-up from 8 km away; several times the 10 to 15 late arrivals he describes	Single-sourced; reason candidate
Look up the matching-week page and weekday trend, then re-project from the first half hour	Informal but documented	Yuktāhār staff in the daily meeting	The only calibrated correction seen in any mess; yet the site head believes about 50% where his register shows about 28%	Lookup confirmed; projection single-sourced
Issued must equal cooked	Informal but documented	Peers, in the daily meeting	Mismatches surface the same day; the rule covers materials, not diner counts	Single-sourced; held on the one record checked
Supplier impurity norm; re-cook only if an item runs out early; cook mostly in one batch	Informal	Yuktāhār store, daily meeting and kitchen	Impurity above the norm triggers a complaint or new supplier; late run-outs go unfilled; one batch front-loads cooked volume, and cooked food lasts only 3 to 4 hours, so surplus is recovered only by reuse	Confirmed except the norm (single-sourced)
Cook about 30 extra for outsiders; about half the staff eat at the mess	Informal, planned	Kadamba	30 to 50 outside people are fed on top of registrations and staff meals absorb surplus before it counts as waste	Single-sourced
Kadamba food-safety sequence (temperature, 72-hour sample, display plate, serving temperature, tasting, corrections); weekly QC; daily TDS	Formal (food safety)	Prism, facility team, QC	Shows the checking capacity operators already have	Display plate confirmed; rest single-sourced
Two-person tasting per shift; fridge temperature three times a day; first-in-first-out and first-expiry-first-out inventory	Formal (food safety)	Yuktāhār; Kadamba's inventory keeper; faculty visits to the store	Shows the documentation capacity operators already have	Confirmed at Yuktāhār; single-sourced at Kadamba
Blame-free waste measurement, peer cross-checking, no hierarchy	Informal norm set by management	Yuktāhār	Staff first resisted tracking (”they are catching us”) in August; managers made it blame-free and discussed it daily, and by December staff reported waste falling	Outcome confirmed; design single-sourced
Waste reduction as an explicit cost and profit goal; no formal leave policy	Informal (management)	Yuktāhār project managers and meeting	Goal: best quality, less waste, optimised workflow, lower cost, standardisation, higher profit; absences are agreed in the meeting	Single-sourced
Menu chosen by the student committee	Formal	Committee	Operators shift quantities between items but cannot drop one, and redirect menu complaints to the committee; Bakul's manager disputes pairings, not dishes	Single-sourced

Informal norms on the student side are unwritten and shared: registering for the whole month rather than day by day, which looks like the path of least resistance and which auto-registration makes the default, and one respondent's habit of checking the menu first. Going early to avoid queues is a minority behaviour, since only a third of Kadamba scans arrive before 8:30. At Yuktāhār the culture is described as family-like, and one project manager trained all staff at his farmhouse on what good quality looks like and what responsibility to take (single-sourced).

Workarounds compensate for a gap in the formal system. Mess Cell resale, used by four of six, compensates for Skip Meal's lack of refund and is mirrored on the operator side by Kadamba's manager describing students who register while away while a friend eats in their place. Eating snacks or a later meal compensates for skipping without a cost-recovery path. Early registration at Kadamba compensates for a popular mess filling up. Students enter by the back door after close. On the operator side the workarounds are a fixed ratio in place of a learned forecast at Kadamba, a weekly judgement with a shortage floor and a held reserve at Bakul, raw stock held ready for a quick top-up, holding and reheating each batch's surplus at Kadamba, leftover reuse into the next meal at Yuktāhār, and extra portions for people outside the registration.

Formal versus informal divergence

Where they diverge	Formal design	Actual behaviour
Registration as a demand signal	Registration lets dining services ”estimate demand accurately and prepare the right quantity”	Registrations are flat across the week while turnout varies. Kadamba discounts them by a fixed 70%, Bakul by an item-dependent 70 to 80% adjusted weekly but held near 70%, Yuktāhār through its records, believing about 50% where its register shows about 28%. Kadamba's ratio is not set against actual turnout, and Bakul's adjustment has no known record behind it
Registration cadence	A four-day weekly lock; the portal supports day-by-day choices	A standing monthly booking for all six respondents, made the default by automatic registration
Skip Meal	Designed to reduce kitchen preparation	Unused by any of the six. The institution's account says declarations reach the kitchen, while Kadamba says it cooks on registrations regardless; whether Skip Meal lowers the operator's count is unresolved between conflicting sources
Cancellation	The correct tool for planned non-attendance	Used inconsistently; one respondent forgets
Automatic allocation	A fallback for students who do not register	For one respondent it is their actual registration mechanism, and it applies over holidays
Closing time	Entry closes at 9:30	Scans continue past 9:30 every day, occasionally to about 10:00; refills run to 10:00; at Kadamba latecomers use the back door
Item availability at close	Designed to run until 9:30	Late run-outs are not re-cooked and staff make unkept promises to restock. At Bakul the manager holds food for about 50 at 9:20 because a Hafizpet top-up may not arrive in time (reason candidate)
Keeping off-site food hot	Insulated boxes and hot-water holding keep Bakul's food at about 60 to 72 °C	Temperature is kept, but items prepared from about 3 AM are held for hours and steam softens parathas: the chain protects heat, not texture (single-sourced)
Waste record	No waste record is part of registration and billing design	Yuktāhār keeps a detailed per-item record reconciled across its books and Kadamba reports a daily waste percentage with per-dish gram standards; no route carries either to anyone who sets registration or billing rules

Identify Negotiation and Escalation Mechanisms

(Revised: the first version said no escalation path was visible.) From the student side that still holds for rules and timing: asked whether they or anyone they know tried to change a mess rule or timing, all five new respondents said no. The operator interviews change the picture in one respect. An operational complaint path exists and works for same-incident fixes and supplier changes. The operators route it differently: Kadamba through a command chain, Yuktāhār through its daily meeting with no hierarchy.

Route	Path	Status
Operational complaints	Student, a director's email or the CDS site team, to the operator's manager, to a WhatsApp group, to the kitchen	Exists; works for same-incident fixes (single-sourced)
Kadamba command chain	CFS, CDS (head unresolved), Prism (won by tender), operations manager, manager, supervisor, service staff	Single-sourced
Yuktāhār (no hierarchy)	Staff, daily meeting and two project managers of ABC Hospitality Services, in both directions	Single-sourced
Bakul (Vijayalakshmi Caterers)	CDS team, manager, kitchen, managing director	Single-sourced, noisy audio
Upstream	Store to supplier: impurity above the norm leads to a complaint or new supplier; paneer brand switched after complaints	Single-sourced
Rule and timing changes	CDS Student Council to the CDS Committee (faculty), reached only by high-intensity feedback	Formal route on paper; unused from below
Student to rule owner	None: no respondent has used or seen a route	Absent
Operator to rule owner	None: no route for the waste record into registration or billing rules	Absent

(Revised: the first version's internal operator chain, staff to supervisor to manager to operations manager to director, came from a cut-off clip that could not be tied to one mess. The Kadamba chain above is the operator's own account, heard first-hand.) One identity in it is unresolved. The team account names one person as CDS head and the CFS Chair interview names another as CDS Chair, which may be two roles or one wrong name, and they are not merged. The ”facility team” that tastes Kadamba's food with two Prism staff is likewise unidentified and may or may not be the CDS site team.

A student complaint at Kadamba shows the boundary in action. Dinner sambar was dropped ”while they are taking more money from us,” and the operator answered that the committee decides the menu. The complaint reached the operator and stopped there, because the operator holds no menu authority.

Operators sit on the same side of the missing path as students. They have high power over quantities and none over the registration deadline, the 9:30 close, the cancellation cap or billing. At Kadamba perishables are ordered the day before and dry goods a week ahead against a lock four days out, and the operator has no channel through which to question that cut-off with anyone who sets it. The absence found in the power analysis, with feedback inputs but no outputs, is now corroborated from three sides: the governance structure, the student experience and the operators.

Track Exceptions, Workarounds and Dependencies

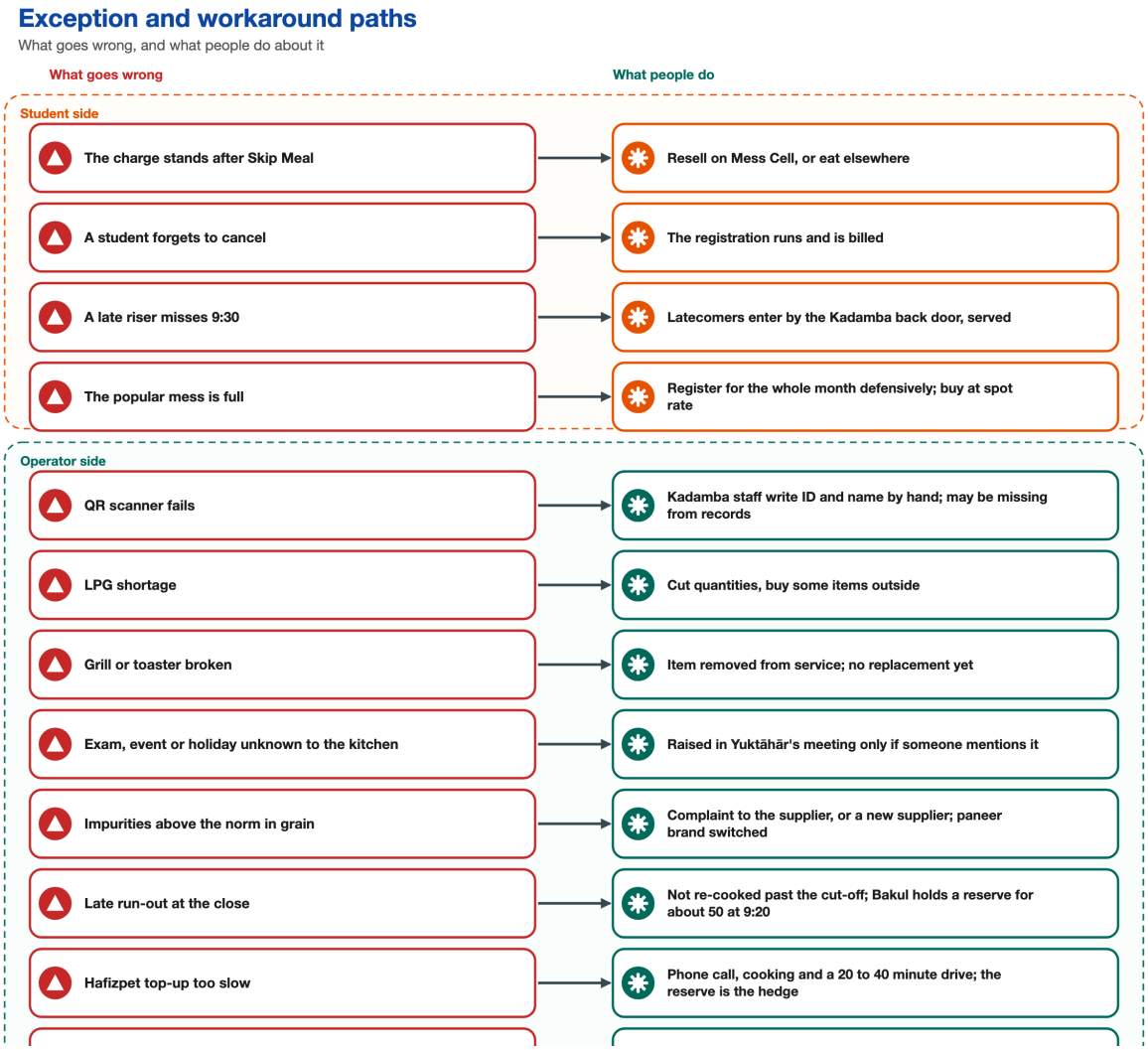


Figure 10. *Exception and workaround paths, student side and operator side.*

Operator exceptions

- **Scanner failure.** When the QR system fails, Kadamba staff write down ID and name by hand. The April export shows no sign of bulk entry (never more than two scans in one second), so hand-logged diners may be missing from the records, a candidate undercount.
- **Gas shortage.** During the LPG shortage Kadamba cut quantities a little and bought some items outside. This is the first operator account of the shortage, and its cause is still single-sourced.
- **Equipment failure.** Kadamba's bread grill is broken and unreplaced, which removes an item from service regardless of demand.
- **Events, exams and holidays.** Nothing formal tells the kitchen. They reach Yuktāhār's meeting only if someone mentions them, and hackathons and exams lower turnout in ways the site head calls unavoidable.
- **Supplier quality.** At Yuktāhār, impurities above the norm are raised in the meeting and lead to a complaint or a new supplier. Bakul's paneer has moved between three brands after complaints.
- **Biryani days.** Kadamba cooks biryani lunch at 100% rather than 70%, with a head chef brought in, so the ratio follows the operator's expectation of demand for a dish, not a fixed rule. Portion leakage (three pieces for two) still causes late run-outs at full preparation.
- **Off-site transport.** Every Bakul top-up is a phone call, a cook at Hafizpet and a 20 to 40 minute drive against a fixed close. The true lead time is unknown. The 9:20 reserve of about 50 is, as a candidate reading, the consequence: late in the window only food already held can meet late arrivals.
- **Latecomers.** When Kadamba's front door closes at 9:30, latecomers use the back door and demand food, and staff do not refuse. Last-minute registrations are called ”a pain.” The speaker's wording covers both the breakfast and dinner closes; scans after 9:30 on every April day support it at breakfast.
- **Records that disagree.** For breakfast on 3 September at Yuktāhār, the Wastage Book records 77 diners (28%) and the attendance register, on a partly obscured row, records 110 (40%). Which feeds planning is unknown. The two-book rule covers materials, not diner counts, and the two books have different keepers, which may be why the difference stands (candidate).

The operator workarounds alongside these (leftover reuse, outside diners and staff eating from the same food, holding and reheating surplus, late-door service, raw stock held for a quick top-up) are absorption mechanisms. Each soaks up part of the gap before it becomes visible, and each is locally beneficial. The outside diners and the half of Kadamba's staff who eat at the mess are real people fed by the surplus, so any change to the first batch has to plan for them. Holding and reheating is the exception among absorbers: the team judged Kadamba's food to be cooked ahead, reheated repeatedly and moved between vessels, so it degrades fast. That links over-preparation to quality as an open chain, and whether poorer quality feeds skipping is not evidenced. Bakul's hot holding carries a similar cost for a different reason: food is kept hot from an early start through transport, and steam softens parathas on the way. As a candidate reading, the food kept back for late arrivals is also the food held longest. Yuktāhār tried and dropped a commercial ERP for grocery management and rush prediction, because paying ”lakhs of rupees” for prediction made no sense when the data already sits in its books (”prediction will be based on our data only… I want to remain close to the data”), and because staff are not digitally inclined and too few people exist to digitise, so the office digitises later.

Dependencies

Eating breakfast at the mess depends on waking with enough time and alertness, the most cited constraint. It depends on the desired item still being available (run-outs confirmed by at least three of six, directly or by reputation), on no clash with class-attendance pressure (explicit for one respondent), and on the mess still being open, since the 9:30 cut-off is the most-requested change. The attendance-policy pressure depends on academic scheduling, which the governance analysis found disconnected from mess governance.

Item availability depends in turn on a chain this trace can now follow. (Revised: the first version said the Mess Office fixes each vendor's order four days out, so nothing on the day could change what is cooked.) The four-day lock applies to registration, not to cooking or buying. Yuktāhār buys groceries fortnightly by the stock register from many local shops and godowns, vegetables two to three times a week and milk daily. Kadamba buys in three tiers: dry goods a week ahead, perishables the day before, some vegetables two to three days ahead. Cooked quantity is set

the day before, and adjusted live through top-ups and re-cooks. At Bakul the first batch leaves Hafizpet before any sign of the morning's turnout, and a top-up can only add food, slowly. The binding constraint on waste is therefore cooked food prepared ahead of demand, not raw procurement: raw stock left over is issued again, while cooked food lasts only 3 to 4 hours.

The operator side has its own dependencies. They include supplier cadence and quality, and gas, electricity and water, which Yuktāhār logs daily and Kadamba checks for water quality daily. They include equipment (Kadamba's roti line, multi-item cooker and tilting curry machine, and the broken grill), and one person running all of Kadamba's inventory, a single point of failure. They include the matching-week Wastage Book page, which depends on the two-week menu rotation, and the daily meeting as the only channel for event information and exceptions at Yuktāhār. Transport from Hafizpet, about 8 km and about 20 minutes early, 40 later, bounds every Bakul top-up. Yuktāhār chooses inputs for quality (rock salt and cold-pressed oil for about six months, with visits to the mills), and staff numbers matter: Yuktāhār has 16 kitchen staff and two project managers with some site staff sharing the work, in separate tasting, cutting and cleaning teams without a hierarchy, and Kadamba about ten chefs under a main veg chef with a head chef for biryani days.

Where uneaten food finally goes is only partly settled. The CFS Chair described production and plate waste going via a garbage collector to an outside compost facility. Kadamba's manager, asked about composting, said there is none: the bin goes to the garbage contractor (single-sourced). (Revised: the first assumption that composting absorbs the physical surplus.) At Kadamba the surplus is absorbed by outside diners and staff, by holding and reheating, by latecomer top-ups and by the bin, and whether the contractor composts it off site is unverified. Yuktāhār's disposal route after reuse is not stated in any source. At Bakul some leftovers return to Hafizpet and the rest is thrown, with no onward route stated.

Key Findings

1. **The registration-attendance gap has several mechanisms, not one.** Habitual monthly over-registration made the default by automatic registration, scarcity-driven defensive registration at Kadamba, automatic allocation and simple forgetting all feed it, and the Kadamba records show their combined effect. Registration is an automatic standing booking, flat across the week while turnout varies by weekday, and not a daily demand signal.
2. **Peer influence is disconfirmed as a decision-determining factor.** Six respondents place it at most as a motivational add-on.
3. **There are three forecasting logics, not one or two.** (Revised: the first version gave every operator a private ratio, and a later one grouped Bakul with Kadamba.) Kadamba cooks a fixed 70% and tops up reactively, with no learning record beyond gram standards. Bakul varies an item-dependent 70 to 80% weekly from past attendance by the manager's judgement, never below about 70%, so its correction closes against surplus only down to that floor, and whether it rests on any written record is unknown. Yuktāhār has no fixed ratio: it learns from its books by matching week and weekday and re-projects from the first half hour, though its head believes about 50% where his register shows about 28%. None of the fixed ratios is set against actual turnout, confirmed at 37.1% for Kadamba in April. The gap is measured, locally and repeatedly, and what is missing is any route from that measurement into registration or billing rules. Even in the best-documented mess the fast correction samples the calm start of service, not the crest at the close.
4. **Broken restock promises.** Staff say an item will return before close and it does not, a trust and service finding first reported by students. The operator side offers a candidate mechanism: a re-cook cut-off that leaves late run-outs unfilled, top-ups of 10 to 15 minutes on site, an unmeasured 20 to 40 minute drive from an off-site kitchen against a close minutes away, equipment failures and items that float above their gram standard, together with a Kadamba chef's ”wait 5 minutes” that matches what students describe hearing.
5. **Under-provisioning is not the problem.** Two respondents independently said only about 30% of registrants attend and the mess still struggles to serve even that smaller group, which early drafts read as under-provisioning. The attendance half is now confirmed: about 28% at Yuktāhār in September and 37.1% at Kadamba in April, a gap of 63 to 72%. The under-provisioning reading does not hold. Every stated first-batch share sits well above that turnout, and at Bakul the share and weekend floor sit above even the manager's own estimate (candidate, no records), and Yuktāhār's book records leftovers. The shortage students feel is a timing and item-mix failure at the

close. 26.5% of Kadamba scans arrive in the last quarter hour, though the curve differs by hall, since the Bakul manager describes an earlier peak. The items students report running out (idli, puri, bhatura, fruit) include those Kadamba says float above gram standard (puri, idli), while the ones it says waste (uttapam, set dosa, upma) are left over, and late run-outs are not re-cooked. Over-preparation and run-outs happen on the same morning.

6. **The escalation-path absence is corroborated from the operator side.** An operational complaint path exists and works for same-incident fixes, but no path for changing rules or timing works for students or operators.

A tension the operator's own figures leave open. Kadamba's stated practice and stated waste do not yet add up. In April it averaged about 1,072 registrations and 398 scans a day (confirmed). A 70% first batch is about 750 portions against about 400 eaten, a surplus of about 350 a day before any top-up (candidate arithmetic from the ratio and the records). The operator reports about 5 to 10% daily wastage on no stated base (single-sourced). The two are compatible only if most of the surplus goes somewhere other than the bin: to the 30 to 50 outside diners and half the staff, to food held and reheated, to smaller top-ups than the ratio implies, or to a waste figure on a different base such as all meals or plate waste only. Which holds is not known, and this trace assumes none.

Outstanding Data Collection

Most of what this section first listed is answered. Kitchens prepare against a fixed share of registrations at Kadamba, an item-dependent weekly share with a floor at Bakul, and a matching-week lookup plus a first-half-hour projection at Yuktāhār. The Vijayalakshmi Caterers operator runs Bakul (single-sourced). Kadamba's 70% batch and the cancellation limit are heard first-hand, and so is its reporting chain. The ”30%” claim is confirmed for breakfast attendance. A candidate answer explains why particular items run out first: popularity against a fixed ratio, items floating above gram standard, and the closing surge. Unclaimed food goes to outside diners and staff, some reuse or reheating, and the bin. Operators record waste and no route carries it above them. What remains open:

- Whether a Skip Meal lowers the count Kadamba cooks against, given the earlier finding that it reaches the kitchen and the operator's statement that it cooks on registrations. One direct question settles it.
- Whether the CDS head named in the team account and the CDS Chair named earlier are two roles or one person, and who the facility team that tastes Kadamba's food is.
- What Kadamba's 5 to 10% waste figure is a percentage of (per meal or per day, cooked weight or plate waste), whether a waste record could be seen, and how the figure squares with a surplus of about 350 portions a day.
- Whether the 30 extra portions sit inside or on top of the 70% batch, how many staff ”half the staff” is, and whether staff meals are a written entitlement or a norm.
- Whether Yuktāhār uses its first-half-hour projection at breakfast and how it scales it given the closing rush; whether its two roti counters open at breakfast or only at lunch and dinner.
- Whether the mess in-charge is the person the site head's recording called by an honorific, and whether Yuktāhār's groceries are bought fortnightly (stock register), the week before use from many shops, or both.
- Confirmation that Kadamba's back-door entry happens at breakfast as well as dinner.
- The contract terms: the plate rate, how deductions work and whether costs pass through to the operator; whether Bakul is paid on plates served, as its manager says, and whether Kadamba's basis differs, confirmed from the contract or dining services.
- Where the garbage contractor takes the waste, compost or landfill; where Yuktāhār's residual waste goes; and direct observation of disposal after close.
- Whether Palash is served from the same off-site kitchen as Bakul (candidate).
- Whether Bakul's weekly adjustment rests on any written attendance, consumption or waste record by item; the lead time of a Hafizpet top-up and what happens when it is still not enough; whether the reserve of about 50 at 9:20 comes from data or judgement, and how much is left at close.
- Lunch and dinner records for Kadamba, and any records for Bakul and Palash.
- Why the Yuktāhār site head believes breakfast runs at about 50% when his register shows about 28%, and which of his two disagreeing diner records feeds planning.
- How hand-logged diners during scanner failures are reconciled into attendance records and billing.

System Timeline / Behaviour-Over-Time Map

IIIT Hyderabad Breakfast Mess System

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Overview

This map asks what pattern emerges when the process repeats, across people and across time. It rests on four kinds of evidence. The first is six student respondents describing their own breakfast habits. The second is two sets of records: thirty days of Kadamba breakfast registrations and timestamped scans from April 2026, and twenty days of Yuktāhār breakfast and lunch attendance registers from September 2026, photographed on site with the rest of Yuktāhār's daily books. The third is interviews with the operators of three messes (Yuktāhār, Kadamba, and Bakul, run by Vijayalakshmi Caterers). The fourth is a first-hand account by a team member present at both the Kadamba and Yuktāhār operator conversations. Numbers read off records are confirmed, and so is practice the team saw on site. Kadamba operator numbers, heard directly, are the operator's single-sourced statements about his practice. Bakul numbers come from a team member's structured synthesis of the recordings and are the manager's own statements and estimates, since no Bakul records of any kind exist. No Bakul figure, and not Kadamba's waste percentage, whose base is unstated, is used as a headline figure.

The timescales the evidence supports are these.

- **Within one morning:** measured. Every April scan at Kadamba carries a timestamp, the operator's rule of thumb has the same shape as the records, and both kitchens described their morning from the 5:30 start to the close.
- **Across a week:** measured. Kadamba registrations and turnout are recorded for every weekday and the Yuktāhār register covers nearly three weeks. Kadamba's weekly quality inspection and one-week dry-goods purchasing run on the same cycle, and the Bakul manager reviews attendance weekly to set the next first batch (both single-sourced).
- **Across a fortnight:** confirmed. The menu repeats on a two-week cycle and Yuktāhār's waste records are organised around it.
- **Across a month:** measured for Kadamba breakfast in April. Students are registered automatically and can cancel

only five meals per meal type per month (confirmed). Yuktāhār reads the month's turnout pattern when it forecasts (single-sourced).

- **Across a semester:** recall only. Six respondents each give a semester-level trend for themselves, and no multi-month records exist yet.
- **Across an operator year:** single-sourced for Yuktāhār in outline. The current operator took over the previous year, began per-item waste weighing in August on deliberately blame-free terms, and by December heard staff report lower waste. The switch to rock salt and cold-pressed oil dates from about six months before the visit, and software was tried and dropped at an unknown date.

No student records span multiple semesters, and no trend at that scale is claimed.

Collect Historical Versions, Records or Observations

Student respondents

Respondent	Behaviour	Frequency	Trigger	Time horizon
Original respondent (2026-09-03)	Habitual skipper	Most mornings	Intermittent fasting, poor hostel mess quality historically, late sleep	Longstanding habit
Daily eater 1	Near-daily attender	5 to 6 of 7 mornings	Habit; late-night work causes the rare skip	More regular this semester than before
Daily eater 2	Most-days attender	Most days	8:30 class plus late sleep causes skips; class-attendance policy is put ahead of breakfast	No explicit before and after comparison
Skip-1	Light eater	About once a week	Appetite-driven, menu-checked	Changed since just before classes started this semester; direction implied
Skip-2	Non-eater	Zero this semester	Not hungry that early; eats after 10 to 10:30	Zero for the whole semester
Skip-3	Declining attender	Rare, once more regular	Late sleep leading to no hunger	Declining over the last three to six weeks

The six do not agree on a direction. One reports attendance rising this semester, one a decline over three to six weeks, one zero all semester, and the rest no clear trend. Six people are moving in different directions at once, and the records show what those movements add up to.

Records and operator sources

Source	Mess	Period	What it measures	What it cannot show
Registration and scan export	Kadamba	Breakfast, 1 to 30 April 2026	32,162 registrations and 11,941 timestamped scans, split veg and non-veg	Student identity, lunch or dinner, quantities cooked, waste
Attendance register (photographed)	Yuktāhār	Breakfast and lunch, 1 to 20 September 2026	Registered (regular and Jain) and who ate, by category, with a daily percentage, grouped by weekday	Seven of twenty breakfast rows are partly hidden; the percentage is legible on thirteen days
Wastage Book pages (photographed)	Yuktāhār	June 2026 and 3 September 2026	Per item: raw issued, cooked, left, percentage wasted, run-out time, re-cook quantity; plate waste plus leftover gives the day's wastage	Only a few pages photographed; the archive runs back at least to June
Grocery Book and stock register (photographed)	Yuktāhār	18 September 2026	Issue list and running balance; the two agree, and a mismatch goes to the daily meeting	A single day
Daily registers (photographed)	Yuktāhār	September 2026	Vegetable book and indent with spoilage remarks; milk, curd and paneer; fruit; gas; two electricity meters; water; fridge temperature three times a day; daily cleaning checklist	One to three weeks each; one meter stops after 14 September

Source	Mess	Period	What it measures	What it cannot show
Operator interviews	Yuktāhār, Kadamba, Bakul	September 2026 visits	Quantity rules and their adjustment, planning routine, arrival pattern as the operator sees it, extra diners, procurement tiers, waste disposal, changes	Self-reported. Kadamba's numbers were heard first-hand and the Bakul account is a synthesis, not a transcript; no Bakul records
Team field account (first-hand)	Kadamba and Yuktāhār	September 2026 visits	Command chain, morning timings, pre-service food-safety sequence, procurement tiers, change history, staff organisation	A team member's synthesis, not a verbatim record

The two record sets come from different messes and months and cannot be joined. Direct observation of a live service was not needed for the arrival curve, since the Kadamba scans measure it. The team observed pre-service handling at Kadamba, the wrapped display plate and the tasting (confirmed). The team member judges that Kadamba food is cooked ahead, reheated and moved between vessels repeatedly, which degrades it, an observer's assessment and not a measurement (candidate). What happens to leftover food after service has still not been observed.

Develop Timelines

Daily service timeline: two kitchens

The students' morning and the kitchens' morning are different timelines. The kitchens start two hours before the first student arrives.

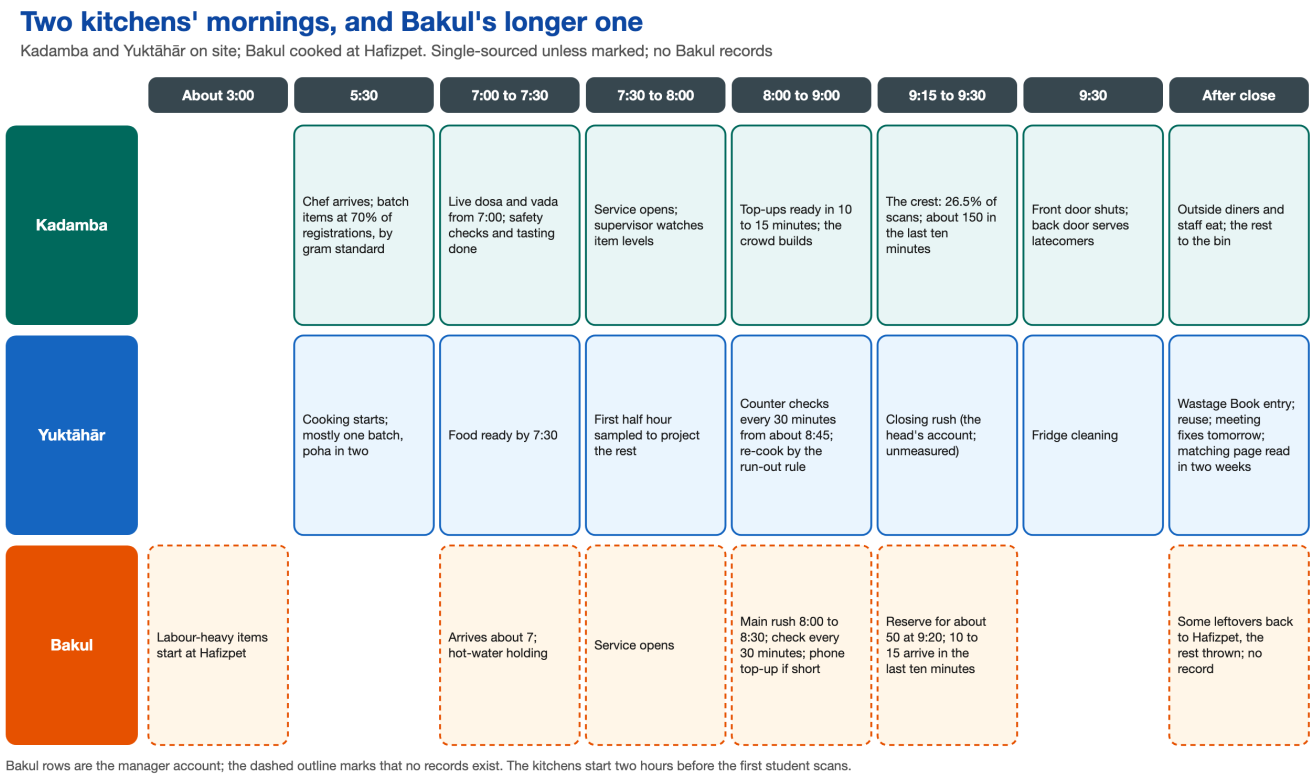


Figure 1. The two kitchens' mornings side by side, from 5:30 to the next day's meeting.

Clock time	Kadamba	Yuktāhār
5:30	Chef arrives and cooks batch items such as sambar and upma, a first batch at 70% of registrations, veg and non-veg planned separately and portioned by a gram standard per dish (single-sourced)	Breakfast cooking starts, mostly in one batch, poha in two (single-sourced)

Clock time	Kadamba	Yuktāhār
Before service	Cooking temperature checked, a sample kept 72 hours, a wrapped display plate at the entrance, serving temperature per item, then tasting by the facility team with two vendor staff and corrections (display plate and tasting confirmed; rest single-sourced)	Food ready by 7:30 (single-sourced)
7:00	Live items such as dosa cooked in front of diners; about eleven scans a day already come before 7:30 (confirmed)	
7:30 to 8:00	Service opens; the supervisor watches how much of each dish is left	The first half hour is used as a sample to project the rest of the meal, which drives top-up cooking (single-sourced)
During service	Supervisor orders top-ups, ready in ten to fifteen minutes (single-sourced)	Handmade oil-free rotis, a second roti counter at peak (confirmed; meal not specified); items re-cooked or not by the run-out-time rule (confirmed)
From about 8:45	Arrivals climb toward the closing crest (confirmed)	Counter checked every thirty minutes from about forty-five minutes before close (single-sourced)
9:30	Front door closes; latecomers still enter by the back door and are served (single-sourced)	Fridge cleaning (confirmed, records)
After service	Outside diners and about half the staff eat from the same food; waste goes to a dustbin collected by a garbage contractor (single-sourced)	Wastage Book entry with run-out time and re-cook quantity (confirmed); leftover idli turned into idli upma or poha for lunch, batter back to the fridge (single-sourced)
Same day	Water TDS read daily for cooking and drinking water (single-sourced)	A planning meeting fixes the next day's breakfast and lunch issue (confirmed)

Bakul runs a longer morning, because its breakfast is cooked at Hafizpet, about 8 km away, and driven in. Every step is the manager's description (single-sourced), and no Bakul record exists to time any of it. Labour-heavy items such as aloo paratha start about 3 AM and are dispatched in insulated boxes at about 60 to 72 °C, in about 20 minutes early in the morning (40 normally). They arrive about 7 AM and wait in a hot-water setup. Service peaks at about 8:00 to 8:30. About every 30 minutes the manager compares food left with registrations and expected demand and, if short, phones Hafizpet, which cooks and sends more (cooking plus 20 to 40 minutes on the road; lead time not stated). Around 9:20 he keeps food for about 50 people, with about 10 to 15 arriving in the last ten minutes. For labour-heavy items the timeline starts about four hours before the food reaches Bakul, and a top-up crosses the same road as the first batch. Hot parathas steam in the box and soften, so the food keeps its temperature and loses texture over the hours it is held (single-sourced; other items unknown).

Within-morning pattern, reconciled

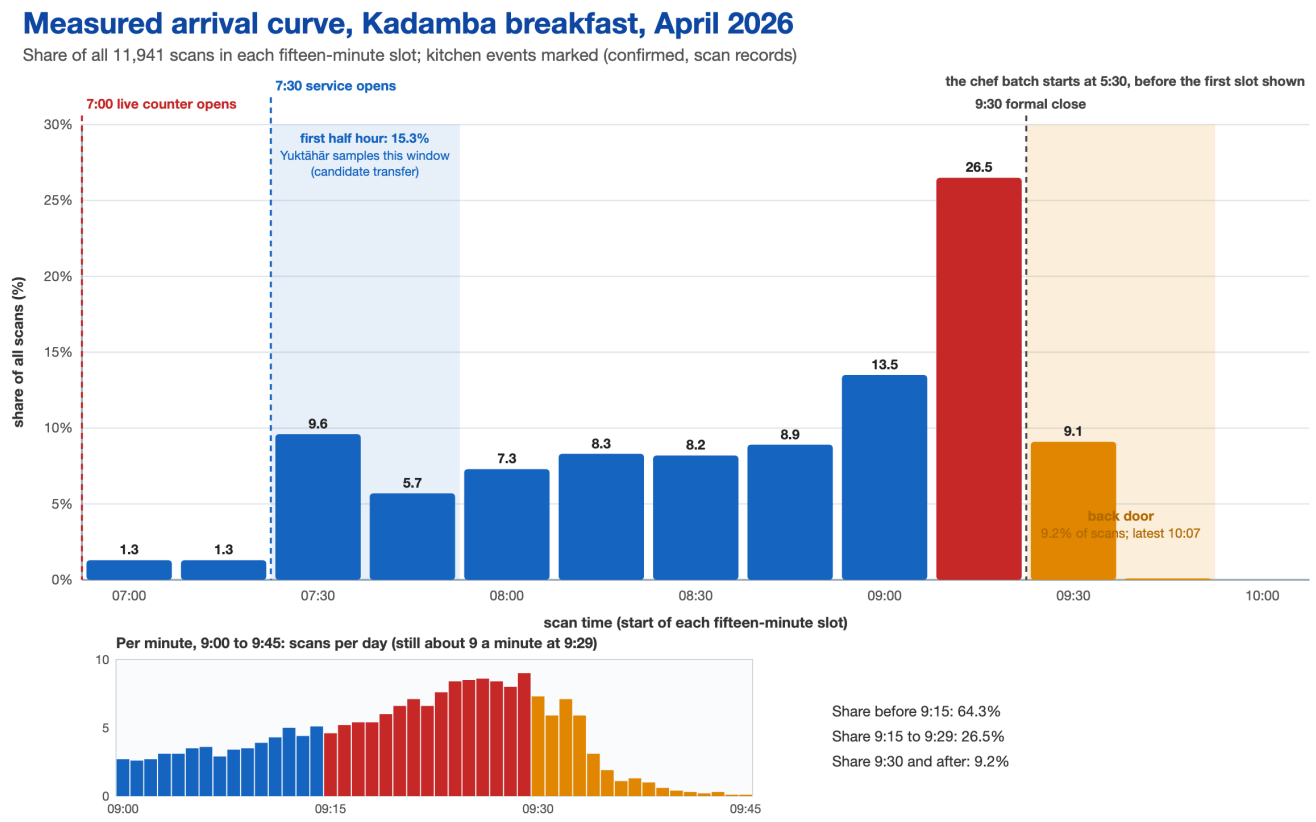


Figure 2. *Measured arrival curve, Kadamba breakfast, April 2026: share of scans per fifteen minutes, with kitchen markers.*

The crowd curve was first drawn from student recall. Two apparently competing peak claims, one respondent naming 8:00 to 9:00 and two naming 9:00 to 9:30, were reconciled into one continuously building crowd that crests just before the 9:30 close, on the reasoning that the respondent naming the earlier window arrives early to avoid the crowd. The Kadamba scan records now confirm that reading, which is no longer a working hypothesis.

Arrival slot (Kadamba, April)	Scans per day, average	Share of all scans	Cumulative share
Before 7:30	11	2.7%	2.7%
7:30 to 7:44	38	9.6%	12.3%
7:45 to 7:59	23	5.7%	18.0%
8:00 to 8:14	29	7.3%	25.3%
8:15 to 8:29	33	8.3%	33.6%
8:30 to 8:44	33	8.2%	41.9%
8:45 to 8:59	36	8.9%	50.8%
9:00 to 9:14	54	13.5%	64.3%
9:15 to 9:29	105	26.5%	90.8%
9:30 and after	37	9.2%	100%

The records add two features recall lacked. The curve opens with a burst: the first fifteen minutes after 7:30 carry more diners than any later slot until 9:00. It then runs as a flat plateau from 7:45 to 9:00 before climbing steeply, to about nine scans a minute in the final minute before 9:30. And roughly one scan in eleven falls after the nominal close, 1,100 across the month, most before 9:35, the latest at 10:07. The operator says latecomers come in by the back door once the front door closes and the mess cannot refuse them (single-sourced), which the records match. The effective close is 9:35 to 9:45, set informally at the door. No bulk marking appears: never more than two scans share a second.

Kadamba's rule of thumb is about 100 diners in the opening phase, about 100 through the middle and about 150 in the last ten minutes, ”for any meal the highest flow is in the last 10 minutes” (single-sourced). That has the shape of the records. The records put the last ten minutes before 9:30 at about 79 scans a day, about 115 counting the minutes just after close, of about 398 a day in total. The rule slightly overstates the final surge and understates the total. The records remain the measurement.

The closing crest is recorded at Kadamba and described at Yuktāhār, where the site head says the rush always comes at closing and staff check the counter every thirty minutes from about forty-five minutes before (single-sourced). It is not described at every hall. Bakul's manager puts the main breakfast rush at about 8:00 to 8:30, with a small late tail of about 10 to 15 in the last ten minutes, and keeps food for about 50 around 9:20 to cover it (single-sourced; his description, not a measured curve). (Revised: a noisy-audio reading placed the Bakul peak at 8:30 to 9:00 and later, arriving in waves, and that is withdrawn, as is the earlier claim that the closing crest is not particular to Kadamba.) Arrival curves differ by hall, and the Kadamba profile cannot be carried to another hall without that hall's own scans (candidate).

The two timelines meet badly at one point. Yuktāhār projects the rest of the meal from its first half hour. On the Kadamba curve, the first half hour carries about 15% of the morning's diners and the last half hour about 40% (confirmed for Kadamba). If Yuktāhār's curve has the same shape, a straight projection from the first half hour reads low and pushes top-ups late, toward the crest (candidate; Yuktāhār keeps no arrival timestamps).

Daily and weekday pattern: measured

The central behaviour-over-time pair is registrations against actual diners, day by day.

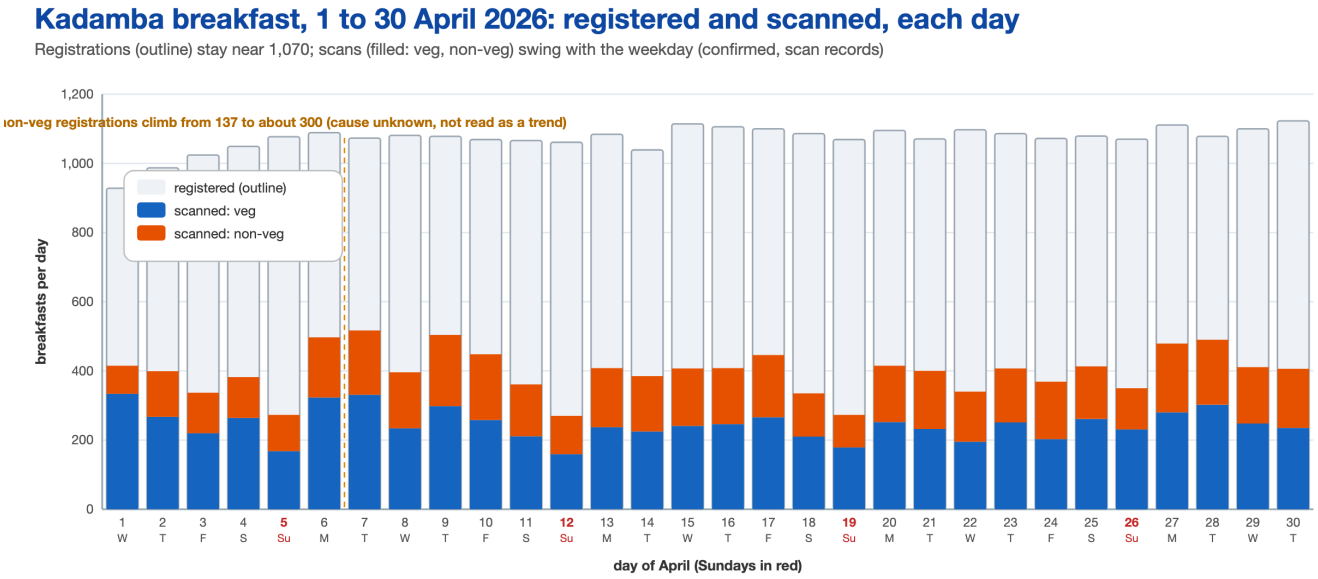


Figure 3. *Kadamba, April 2026: registered and scanned breakfasts each day. Registrations are flat; turnout is not.*

Kadamba breakfast, April 2026	Registered per day	Ate per day	Share who ate
Whole month	928 to 1,123	270 to 517	37.1% (11,941 of 32,162)
Monday	1,095	450	41.1%
Tuesday	1,065	448	42.1%
Wednesday	1,064	394	37.0%
Thursday	1,076	425	39.5%
Friday	1,066	400	37.5%
Saturday	1,070	373	34.8%

Kadamba breakfast, April 2026	Registered per day	Ate per day	Share who ate
Sunday	1,069	292	27.3%

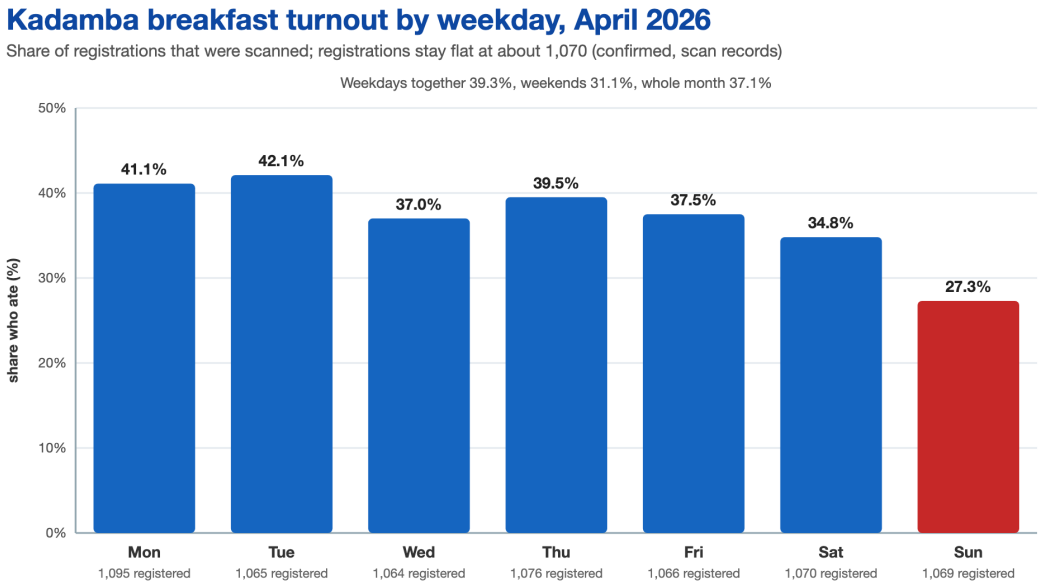


Figure 4. Share who ate, by weekday, against flat registrations.

Registrations are flat. From 6 April they sit between 1,039 and 1,123 every day, Sunday included, while the number who eat swings with the weekday. The three lowest days of the month were all Sundays (25.3%, 25.4% and 25.5%), the fourth Sunday came in at 32.7%, and Sunday is also the latest-arriving day, its median scan at 9:08 to 9:15 against 8:36 to 9:03 on most weekdays. Weekdays together run at 39.3% and weekends at 31.1%. Non-vegetarian registrants turn up far more often than vegetarian ones (48.5% against 32.4%). Registration does not follow the daily decision to eat. It is set once and left standing (confirmed).

Kadamba plans its first batch separately for veg and non-veg (single-sourced), but the two turn up at very different rates. If the same 70% applies to both, veg is prepared at about 2.2 times what veg registrants eat and non-veg at about 1.4 times (candidate; the operator did not say whether the ratio differs by type). The operator's worked example of 400 non-veg and 300 veg registrations does not match April's mix of roughly 750 veg and 320 non-veg a day, and reads as illustrative or a lunch figure (candidate). The first week of April shows registrations rising from 928 to about 1,070, almost entirely because non-veg registrations climb from 137 on 1 April to about 300 by 6 April. The cause is unknown and it is not read as a trend.

The Yuktāhār register shows the same shape at a second mess, and a comparison with lunch on the same days.

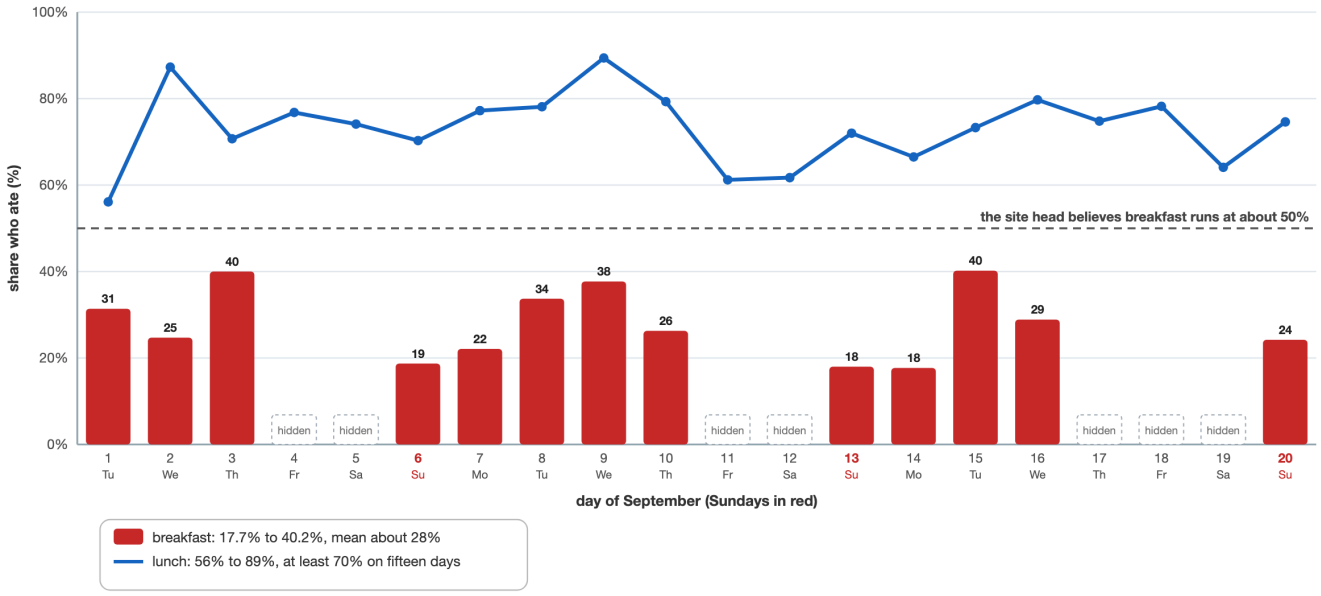


Figure 5. Yuktāhār, 1 to 20 September 2026: breakfast against lunch attendance each day.

Yuktāhār, 1 to 20 September 2026	Share who ate	Registered per day
Breakfast (thirteen legible days)	17.7% to 40.2%, averaging about 28%	228 to 404
Breakfast, Sundays	18.0% to 24.2%	228 to 269
Lunch (all twenty days)	56% to 89%, at least 70% on fifteen days	161 to 376

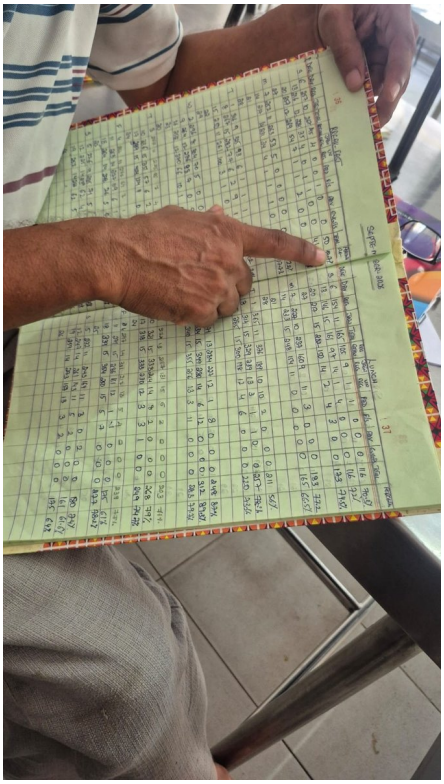


Figure 6. *The attendance register, with the breakfast page on the left and lunch on the right; the hand covers part of the breakfast rows.*

Breakfast is the structurally worst meal. On the same days in the same mess about seven in ten lunch registrants eat and fewer than three in ten breakfast registrants do (confirmed). The register also shows a pattern the Kadamba export cannot. On Sundays lunch registrations fall (165, 161 and 232 against 237 to 376 on other days) while Sunday breakfast registrations do not (267, 228 and 269). Students adjust their Sunday lunch booking and leave the breakfast booking in place, then do not come. That breakfast registration behaves as a standing default in a way lunch does not is a candidate pattern on three Sundays of evidence. The two messes differ on Mondays, among Kadamba's best breakfast days in April and Yuktāhār's worst in September (22.1% and 17.7%), which different months and populations could each explain and which is unresolved.

The kitchens keep a weekly rhythm of their own. Kadamba buys dry goods a week ahead and inspects weekly (single-sourced). Yuktāhār's operator attributes a Wednesday rise in electricity use to grinding idli batter and sets machine run times by it. The meter only partly supports this: Wednesday 2 September (134.4 units) and Wednesday 9 September (118.4) were high, but Saturday 12 September (136.2) was higher, so the pattern is a candidate.

Aggregate stability despite individual divergence

The aggregate does not move the way individuals do. Each seven-day block of Kadamba's April sits between 35% and 39% with no drift across the month, and the Sunday trough recurs week after week. Individual respondents are rising, falling or at zero, and the records cannot see individuals because they carry no student identity. Both hold together: heterogeneous individual trajectories sum to a stable aggregate with a strong weekly rhythm (confirmed at the aggregate). The three-to-six-week decline one respondent reports cannot be tested against April data from another month with no identities, and April itself shows no within-month decline.

Planning-horizon timeline: when each decision is fixed

The gap between registered and actual diners is produced partly by when each quantity is fixed relative to the meal.

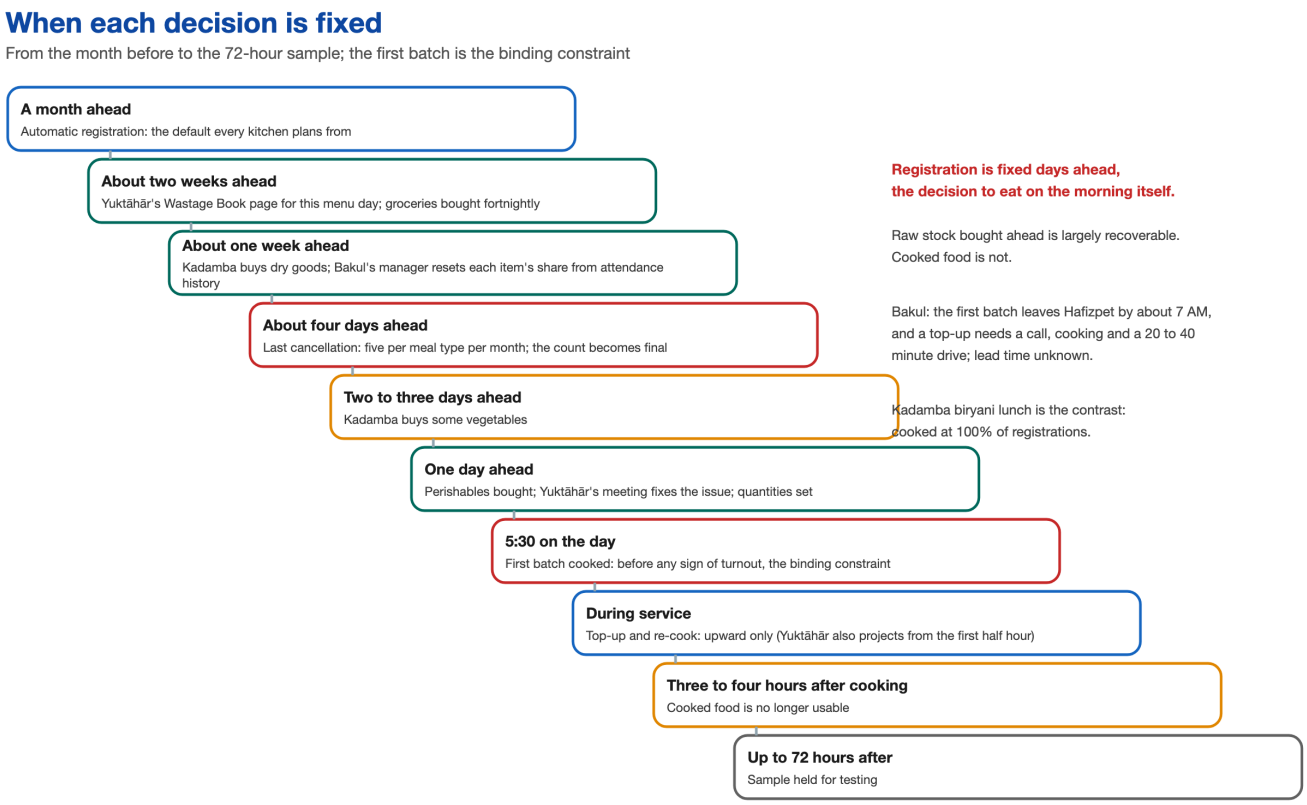


Figure 7. *When each decision is fixed, from the month before to the 72-hour sample.*

When	What is fixed	Where it is described
About two weeks ahead	The Wastage Book records waste and run-out times for each item on the matching day of the two-week menu cycle, and that page becomes the reference for sizing the same day two weeks later. Groceries are bought fortnightly and vegetables arrive two to three times a week; the operator also describes raw materials bought the week before from many local shops, in mild tension with the fortnightly register	Yuktāhār, confirmed (weekly description single-sourced)
About one week ahead	Dry goods (rice, groundnut, masala, ghee, podi) are bought at Kadamba. Bakul's manager reviews historical attendance by day and adjusts the share each item is cooked for in the coming week (his example: a Monday that historically sees about 70% attendance is cooked at about 80%); whether the history is written down is unknown	Kadamba and Bakul, single-sourced
About four days ahead	The last point at which a student can cancel. Students are registered automatically and may cancel five meals per meal type per month, each at least four days ahead; beyond that the charge cannot be removed. The automatic booking, less early cancellations, becomes the number every kitchen plans from	Published rule and student experience, confirmed
Two to three days ahead	Some vegetables are bought	Kadamba, single-sourced
One day ahead	A planning meeting fixes the next day's issue from registrations (regular, Jain and cash), the matching Wastage Book page, the weekday register, the month's pattern and any holidays or events; grain with debris above a grams-per-item norm leads to a complaint or new supplier. Kadamba buys perishables (milk, fruit, meat, eggs, most vegetables)	Yuktāhār confirmed for meeting and books, inputs and norm single-sourced; Kadamba single-sourced
Before service	A first batch is cooked at 70% of registrations at Kadamba, veg and non-veg separately, by gram standard (idli four per person, sambar 100 g), then passes the safety checks and tasting. At Bakul the first batch is 70 to 80%, varying by item and day, with food for about 70% kept on low weekends, cooked at Hafizpet (from about 3 AM for items such as aloo paratha) to reach Bakul about 7 AM	Single-sourced

When	What is fixed	Where it is described
During service	More is made as the counters empty: ten to fifteen minutes at Kadamba on the supervisor's call; at Bakul by phone to Hafizpet after a check about every thirty minutes; at Yuktāhār by projection from the first half hour plus the run-out-time rule. An earlier noisy-audio reading put the Bakul top-up at about thirty minutes, but the drive alone takes 20 to 40 minutes and cooking time is unstated, so the lead time is open	Kadamba and Bakul single-sourced; Yuktāhār rule confirmed, projection single-sourced
Three to four hours after cooking	Cooked food is no longer usable. At Kadamba, surplus is held, reheated and moved between vessels until then, degrading quickly (the team member's judgement)	Operator single-sourced; reheating candidate
Up to 72 hours after the meal	A sample of each item is kept for testing if anyone falls ill	Kadamba, single-sourced

Kadamba's biryani lunch is the contrast case: cooked at 100% of registrations, two pieces of 150 to 200 g each with unlimited rice, with later diners running short when some take three (single-sourced).

The real-time demand signals, the scan and the counter staff's view of each dish, arrive after the first batch is cooked. Yuktāhār alone turns the first half hour into a forecast. The student's decision to eat is made on the morning itself, against a booking made automatically and removable only through five early cancellations a month. Raw stock bought ahead is largely recoverable and cooked food is not, so the binding time constraint on waste is the ratio at which cooked food is prepared against registrations (fixed at Kadamba, adjusted weekly by judgement at Bakul), not the procurement lead time. At Bakul it is tighter still: the first batch leaves Hafizpet before any turnout signal exists, and a top-up needs a call, cooking and a drive (single-sourced).

Provisional system-level timeline

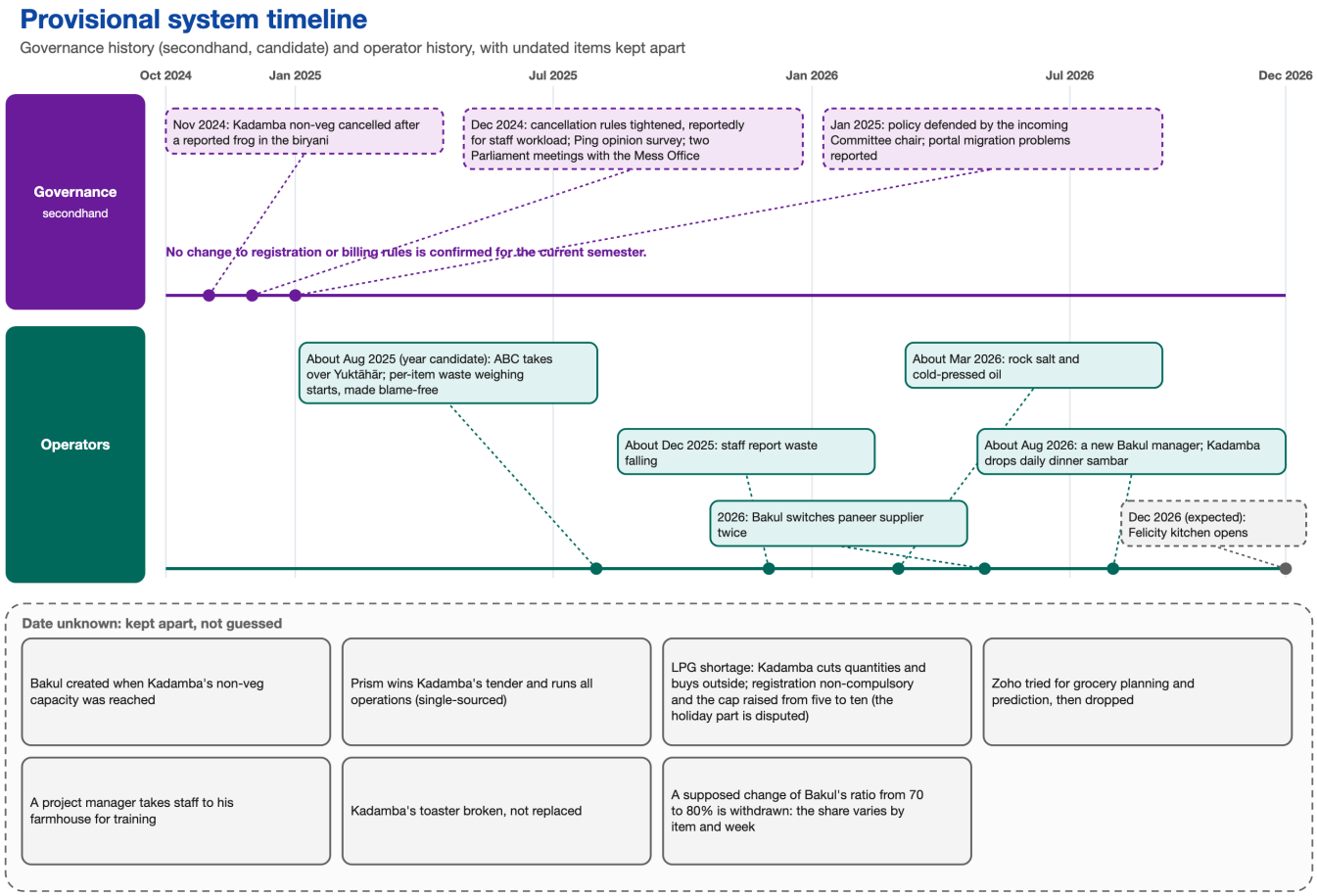


Figure 8. Provisional system timeline: governance history above, operator history below, undated items in a separate gutter.

This timeline shows institutional and governance history drawn from precedent research, much of it secondhand. No student respondent mentions a recent rule change, and no change to registration or billing rules has been confirmed for the current semester. The operator side, however, has changed repeatedly over 2025 and 2026, as set out below.

Identify Significant Changes and Their Triggers

Date or period	Change	Trigger
Date unknown	Bakul created	Kadamba's non-vegetarian capacity ceiling was reached
Date unknown	Prism, an outsourced vendor, wins the Kadamba tender and runs all operations, reporting through operations manager, manager and supervisor to the service staff, under CDS and CFS (single-sourced)	Tender
November 2024	”Let Them Eat Frogs”: Kadamba non-veg meals reportedly cancelled unilaterally (secondhand; candidate)	A frog reportedly found in Kadamba's chicken biryani
December 2024	Cancellation rules reportedly tightened, weekly caps, portal migration (secondhand; candidate)	Reported Mess Office workload concerns, compounded by a Kadamba renovation
December 2024, same window	Ping student opinion survey; two Parliament and Mess Office meetings, reportedly (candidate)	Organised pushback against the tightening
January 2025	”Cleaning up the Mess?”: policy defended by the incoming Committee chair; portal migration issues reported (secondhand; candidate)	Continuation of the December episode
The previous year (candidate: 2025)	ABC Hospitality Services takes over Yuktāhār and grinds whole spices in-house (single-sourced)	Tender (not stated)
August, the same year	Yuktāhār begins per-item waste weighing: issued, cooked, left, percentage wasted, run-out time, re-cook quantity (confirmed)	Operator initiative tied to an explicit goal of lower cost and higher profit alongside quality. Staff first resisted (”they are catching us”); managers made the weighing blame-free and discussed it daily (single-sourced)
About December, the same year	Kitchen staff themselves report wastage has fallen (single-sourced)	About four months of blame-free weighing and daily discussion
Date unknown	A project manager takes all staff to his farmhouse for training in what good quality looks like and what responsibilities to take (single-sourced)	Team building; the operator links it to making weighing work
About March 2026 (six months before the visit; candidate date)	Yuktāhār replaces white salt with rock salt and moves to cold-pressed oils; the manager visits the mills (single-sourced). Later than the takeover, which revises the first dating	The operator's quality policy, adopted despite the higher cost
Date unknown	Zoho software tried for grocery planning and rush prediction, then dropped (single-sourced)	Paying ”lakhs of rupees” for prediction made no sense with the data already in its books, and staff are not digitally inclined
Date unknown	During an LPG shortage, registration made non-compulsory and the cancellation cap raised from five to ten; the same was reported for holidays and fests (single-sourced). The holiday part is disputed: the team's account says students stay automatically registered over holidays, and Kadamba's manager said holidays follow registration as usual	LPG supply shock; holidays and fests disputed
Date unknown	Kadamba cuts quantities and buys some items outside during an LPG shortage (single-sourced), the first operator account of the shortage, which had rested on one student's statement	LPG shortage
2026	Bakul switches paneer supplier twice, Amul to Heritage to Milky Mist (single-sourced)	Supply and quality problems
About August 2026	A new manager takes over Bakul, about a month in the role at the visit (single-sourced)	Staff turnover
August or September 2026	Kadamba stops serving daily sambar at dinner (single-sourced)	Not stated; a student links it to rising charges, and the operator says the committee sets the menu
Date unknown	Kadamba's toaster is broken and not yet replaced (single-sourced)	Equipment damage

Date or period	Change	Trigger
Since at least June 2026	Yuktāhār's Wastage Book is kept by the two-week menu cycle, sizing each day from the matching day two weeks earlier (confirmed)	Menu rotation
2026-09, current semester	No confirmed change to standing registration or billing rules; several operator-side changes in quantity rules, suppliers, menu items, inputs and staff	None identified

(Revised: the first version read the system as not in the middle of changing.) The standing rules students face have been stable since early 2025, but the kitchens behind them have not. The only documented change to the registration model since then is temporary: non-compulsory registration and a raised cap during the LPG shortage, after which the standing model returned. Whether the relaxation ever applied over holidays and fests is open, because the team's own account is that students stay registered through holidays. Every other recent change sits on the operator side, and none has touched how registration works or how students are billed. An earlier reading of the Bakul recordings listed a dated change in Bakul's first-batch rule from 70% to 80%. That entry is withdrawn: the clearer account describes a share of 70 to 80% that varies by item and day and is adjusted weekly, which the noisy reading had taken for a change over time (single-sourced). Whether any standing Bakul ratio was ever changed is open.

Examine Feedback Loops and Recurring Patterns

R1: ”Shifting the Burden” via Mess Cell (open chain, held as a candidate reinforcing loop that sustains the dysfunction rather than growing it). Its central mechanism is confirmed. Zero of six respondents who addressed Skip Meal have used it, and the respondent who explained why did so in exactly the terms the incentive analysis predicts: it does not return the money. Four of six have used Mess Cell in some form.

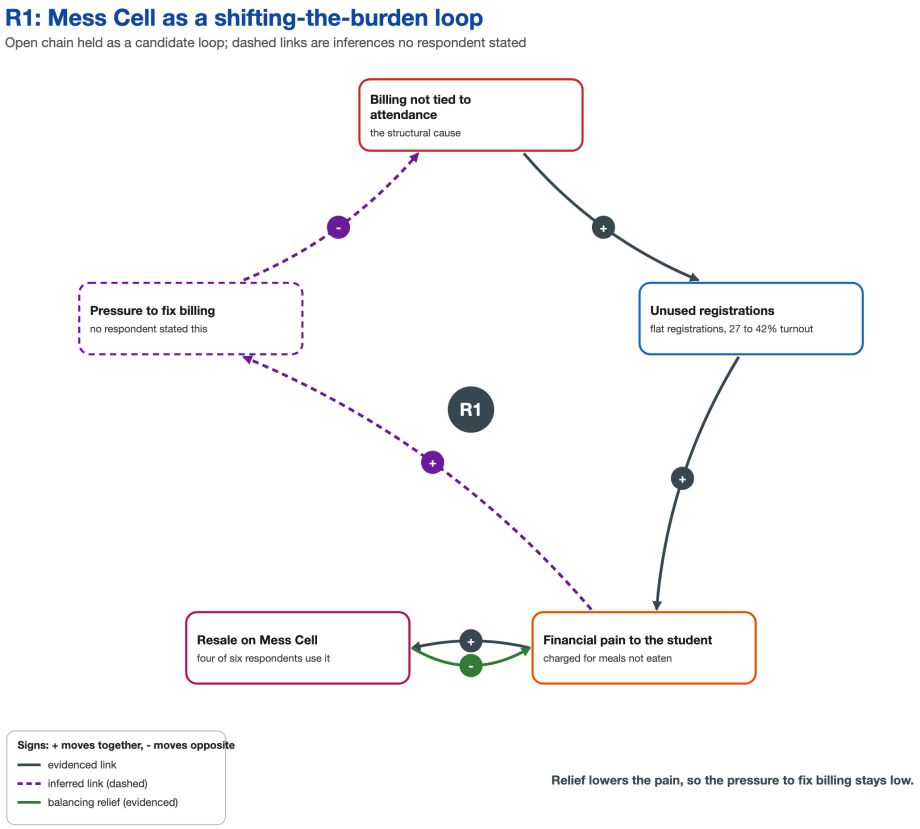


Figure 9. R1: Mess Cell as a shifting-the-burden loop. Dashed links are inferences no respondent stated.

The informal resale market is a symptomatic relief valve. It makes the underlying problem, billing that does not track attendance, tolerable enough that nobody is forced to fix the cause. Every link is directly evidenced except the last, that relief reduces pressure to fix billing, which is an inference no respondent stated. Until it is evidenced, R1 is an open chain with a loop-shaped hypothesis.

B1: Kitchen waste control (balancing; revised). The first reading was that because no respondent uses Skip Meal, the kitchen-side balancing loop might not close at all. Kitchen waste is balanced by the operator, not by Skip Meal, and each mess runs its own balancing loop with a different delay.

- **Yuktāhār** runs two balancing channels. The first sizes each item from registrations, the Wastage Book page for the matching day two weeks earlier, the weekday register and the month's turnout, in a meeting held twice a day. Having seen 18.7% breakfast attendance on Sunday 6 September, the site head said the quantity would be cut "not next month, next week" (confirmed). Its delay is two weeks and it acts on surplus as well as shortage. The second works within the meal: the first half hour projects the rest (single-sourced), and items are re-cooked or not by a written run-out-time rule (confirmed). Its delay is minutes and it acts upward only.
- **Kadamba** cooks 70% of registrations, veg and non-veg separately, by gram standard. The supervisor watches the counters and orders top-ups in ten to fifteen minutes (single-sourced). The delay is minutes, and nothing recorded shrinks the next first batch.
- **Bakul** also runs two channels. The manager's weekly review of attendance history sets 70 to 80% by item and day, but on his account not below about 70%, which he keeps even on low weekends under his rule that "there should not be a shortage of food, even if some wastage occurs" (single-sourced). Its delay is about a week and it acts on surplus only down to that floor. The second channel is the 30-minute check and phone top-up, upward only, with cooking plus a 20 to 40 minute drive in its delay and the lead time unstated.

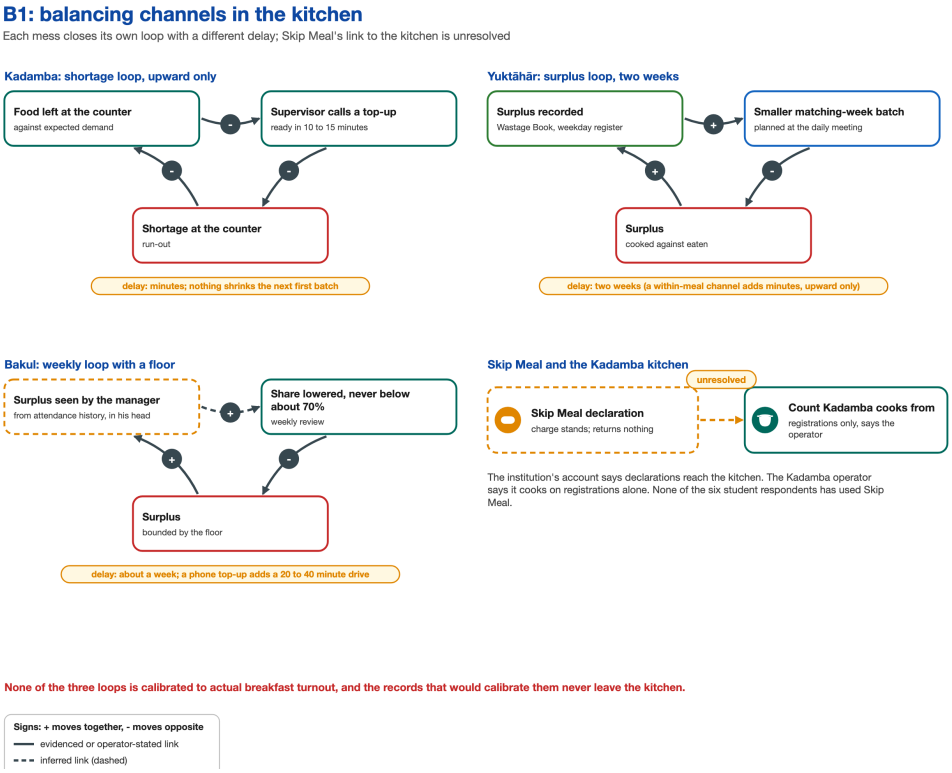


Figure 10. *B1: balancing channels in the kitchen. Skip Meal's link to the kitchen is unresolved (dashed amber).*

Kadamba's loop closes against shortage only. It adds food when the counter runs low and has no route that shrinks the first batch when it proves too large. Yuktāhār's first channel closes against surplus through the Wastage Book and weekday register, with a two-week delay. Bakul's weekly review also closes against surplus, by one person's judgement with no known record, and only down to its floor, so it sits between Kadamba's fixed ratio and Yuktāhār's books (revised from a first reading of Bakul as a fixed ratio). None of the three is calibrated to actual breakfast turnout.

Kadamba's ratio sits well above it, the Yuktāhār site head's working figure for breakfast is about 50% against about 28% in his register, and at Bakul the first batch and weekend floor sit above even the highest turnout the manager expects, about 60% (candidate; no Bakul records). Late in the window a Hafizpet top-up is unlikely to arrive before close, so Bakul's only hedge against late arrivals is food already held, and its reserve of about 50 at 9:20 is several times the 10 to 15 late arrivals (candidate reading). Skip Meal's zero student-side use is a separate fact.

Whether Skip Meal reaches Kadamba's kitchen at all is now an open conflict. The earlier finding was that Skip Meal declarations reach the kitchen for same-week preparation. The operator's account is that the kitchen cooks for the number registered, ”because it is a responsibility and they are getting paid for it” (single-sourced). The two fit together only if Skip Meal lowers the registration count the kitchen sees, which is unconfirmed.

B2: Capacity redistribution (weak or broken). One respondent registers for the whole month specifically because ”Kadamba is hard to get and it gets full,” direct evidence of scarcity-driven demand concentration on the popular mess, not organic redistribution to Bakul or Palash.

Same meal, two failures (open chain; revised). The first candidate was that preparation might be calibrated to the low attendance number, leaving real attendees under-served. That does not hold. At the aggregate, preparation runs well above turnout: Kadamba's first batch is 70% of registrations, about 750 portions a day in April, while 37.1% of registrants ate, about 398 a day, so the first batch alone is roughly double what is eaten before any top-up (the 70% is single-sourced and the doubling a candidate inference against the confirmed 37.1%). Yuktāhār's head believes about 50% where his register averages about 28%. What survives is that real attendees do meet run-outs, driven by timing, not volume. A quarter of Kadamba's diners arrive in the last fifteen minutes, Yuktāhār cooks mostly in one batch, specific items draw longer queues, and the kitchen responds by re-cooking or serving latecomers at the back door. At Bakul the first batch exceeds turnout by design on the manager's own numbers, and the gap is larger if Bakul resembles the two recorded halls (candidate).

Kadamba's operator puts waste at about 5 to 10% a day, without saying whether per meal or per day, by weight, or plate waste or kitchen leftover (single-sourced). That sits in tension with a first batch about twice what is eaten. Either the surplus is absorbed before it is counted, by 30 to 50 outside diners, about half the staff, reheating into later service and smaller top-ups, or the percentage uses a different base. The tension is recorded, not resolved. Over-preparation in total and shortage at the crest both come from fixed-ratio batch cooking against a curve that crests at close. No evidence links run-outs back to how students register, and none links reheated food to students skipping, so this is an open chain, not a loop.

Recurring pattern: the weekly rhythm. Turnout falls every Sunday at both messes with records (Kadamba 25 to 33%, Yuktāhār 18 to 24%) while registrations stay put. The system cycles weekly and registration does not cycle with it (confirmed). Bakul's manager describes weekend breakfast as very low too, a third hall showing the pattern on his description alone, and he still keeps food for about 70% of registrations on those days (single-sourced).

Recurring pattern: three kitchens, three forecasting regimes (revised from two). Kadamba prepares a fixed ratio and reacts at the counter, with no record carrying one morning's surplus into the next first batch. Yuktāhār learns from its books across a two-week cycle and projects within the meal from its first half hour. Facing the same automatic registration, they state opposite stances: Kadamba cooks for registrations because that is what it is paid for, and Yuktāhār names lower cost and higher profit as goals alongside quality (candidate reading of the difference). Both are exposed to the closing crest, Kadamba through top-ups under time pressure and Yuktāhār through a projection window that sees only about 15% of a morning's diners on the measured curve. Bakul sits between them: its manager adjusts the share cooked, by item and day, from weekly attendance history, a calibration by judgement with no known record and a floor of about 70% (single-sourced). Its stance is a third, stated outright: a shortage is the failure to avoid and some waste its price. He also says payment is on plates served, in tension with Kadamba's remark about being paid to cook for registrations. Contracts may differ, and the basis is flagged here, not resolved. Bakul's manager describes an earlier rush rather than a closing crest, so its late exposure is carried by the held reserve, not by top-ups (candidate).

Recurring pattern: operator beliefs drift from operator records. (Revised: Kadamba's ”300 to 400 by 8:30” came from a noisy recording and is superseded by his rule of about 100, 100 and 150 in the last ten minutes, which has the records' shape.) At Kadamba the belief about timing is close to the records, and the gap is in the level of turnout. Yuktāhār's site head believes breakfast turnout is at most about 50% beside his own register's 18 to 40%. The Bakul

operator says Kadamba sees 90 to 95% (single-sourced) against 37% in Kadamba's April records. Bakul's manager estimates his own turnout at most about 60% (single-sourced), and if Bakul resembles the two recorded halls at 28 to 37%, that is an upper bound and a further instance of the pattern (candidate). The numbers are confirmed from records. The narrower pattern, that operators misjudge the level of breakfast turnout more than its timing, is a candidate.

Identify Unintended Consequences

Rule or action	Intended function	Observed response	Unintended consequence
Cancellation cap	Limit uncancelled no-shows	Students use cancellation inconsistently; one forgets. Registration is automatic, so the booking is a default (confirmed)	The cap constrains a behaviour some students do not reliably perform for reasons unrelated to the cap
Walk-in markup	Discourage same-day uncertainty	Monthly advance registration, universal across all six respondents	Registration counts inflate independent of intent. Kadamba registrations stay flat at about 1,070 a day while turnout swings 27 to 42% by weekday, and 62.9% of April breakfast registrations were no-shows (confirmed)
Skip Meal, no refund	Reduce kitchen waste	Zero real usage across six respondents	Kitchen waste is managed by the operators' own rules instead, so the student-side tool carries almost none of its intended load. Whether a declaration changes what Kadamba cooks is disputed
Cancellation deadline four days ahead	Give the kitchen lead time	Kadamba buys dry goods a week ahead, some vegetables two to three days ahead and perishables the day before (single-sourced); registration becomes a standing booking (confirmed)	The lead protects dry goods bought a week ahead anyway, not the perishables bought the day before, and registration loses its value as a daily demand signal
First batch set at a share of registrations (70% at Kadamba; 70 to 80% at Bakul, adjusted weekly, about 70% on low weekends)	Avoid run-outs (at Bakul: no shortage, even if some wastage occurs)	Kadamba turnout is 27 to 42% by weekday; Bakul's manager estimates at most about 60% (no records)	Cooked breakfast is prepared at roughly double what is eaten at Kadamba and above the manager's own highest estimate at Bakul (both candidate); cooked surplus is unusable after three to four hours. Held and reheated surplus is a candidate route to quality complaints, and the link to skipping is not evidenced
Bakul's reserve for about 50 at 9:20	Serve latecomers when a top-up could not arrive (candidate reading)	About 10 to 15 arrive in the last ten minutes (single-sourced)	A shortage-insurance premium paid in food, several times the late arrivals, and the food held longest (candidate; whether it rests on data and how much is left at close is unknown)
Insulated hot-box transport from Hafizpet	Keep food at about 60 to 72 °C over 8 km	Temperature is kept; hot parathas steam and soften (single-sourced)	Food arrives hot but less pleasant, and the longer it is held the more texture it loses, so late diners at Bakul get the softest food (candidate; effect on turnout untested)
Nominal 9:30 close	Bound the service window	About 9% of scans come after 9:30; latecomers use the back door and the mess cannot refuse them (single-sourced)	The real close drifts to 9:35 to 9:45, enforced by nobody, and serving latecomers from top-ups keeps the late crest workable for students
Kadamba waste to a dustbin and a garbage contractor; outside diners and staff eat from the same food	Hygienic disposal	About 30 extra portions for 30 to 50 outside diners such as night security, and about half the staff eat at the mess (single-sourced); the rest leaves in the bin	The cost of over-preparation is absorbed out of sight by diners who are uncounted, so cutting the first batch would take food from them unless planned for. There is no composting at Kadamba (revised from the first assumption that composting absorbs surplus); whether the contractor composts off site is unverified
Per-item waste weighing at Yuktāhār	Reduce waste	Staff feared blame; managers made it blame-free and discussed it daily, and staff reported lower waste about four months later (single-sourced)	A positive side effect: staff took ownership of the numbers. The data never leaves the mess, so it informs cooking and never registration or billing

Rule or action	Intended function	Observed response	Unintended consequence
Buying raw material from many local shops	Keep input cost down	Debris is expected, cleaned out in the store and checked against a grams-per-item norm (single-sourced)	Cleaning labour in the store, and supplier changes when the norm is exceeded
Menu chosen by students or the committee	Give students a voice	Operators see per-item waste (uttapam, set dosa, upma) but have little say; daily dinner sambar was dropped at Kadamba; Bakul's manager says students choose pairings he considers wrong, such as kachori without chole (single-sourced; untested with Bakul residents)	The people who hold per-item waste data and the people who decide the menu are different groups with a weak link between them

Key Findings

1. **No single trend exists across individuals.** Respondents move in different directions at once. At the aggregate, Kadamba breakfast turnout was stable at 35 to 39% in every week of April, with a Sunday trough that recurs weekly. Any claim that the paradox is getting worse or better needs multi-month records, not recall.
2. **The within-morning crowd is measured, not inferred.** It opens with a burst at 7:30, runs flat until 9:00, then climbs steeply: 40% of diners arrive in the last half hour, a quarter in the last fifteen minutes, and about 9% after the nominal close, many by the back door. Kadamba's rule of thumb has the same shape. The curve is Kadamba's: Yuktāhār's head describes the same closing rush, but Bakul's manager describes a main rush at about 8:00 to 8:30 with a small late tail (single-sourced; no Bakul records), so arrival curves differ by hall.
3. **Skip Meal's student-side use is zero, but kitchen waste is balanced anyway,** by operator rules with delays from minutes to two weeks. Yuktāhār runs a two-week lookup against its books, which acts on surplus as well as shortage, and a within-meal projection that acts upward only and sees about 15% of a morning's diners if its curve matches Kadamba's (candidate). Kadamba acts within minutes against shortage only. Bakul adjusts weekly by judgement, acting on surplus only down to about 70%, and tops up by phone upward only (single-sourced). None is calibrated to actual breakfast turnout.
4. **The same breakfast fails in two directions at once:** too much food in total (candidate, from the single-sourced first-batch ratio against the confirmed 37.1%) and item-level run-outs at the closing crest. Both come from a fixed first batch meeting a curve that crests at close. Part of the surplus is eaten by 30 to 50 outside diners and about half the staff, and the operator's 5 to 10% waste figure has no stated base and sits in unresolved tension with a batch about twice what is eaten. The earlier candidate, that preparation is calibrated to the low attendance number and under-serves real diners, does not hold at the aggregate.
5. **Registration does not track the daily decision to eat.** Kadamba registrations are flat at about 1,070 a day every weekday, Sunday included, while turnout swings from 27 to 42%. Sunday is the lowest and latest-arriving day at both messes with records, and Bakul's manager describes weekend breakfast as very low (no records). Registration is automatic, with five early cancellations per meal type per month as the only way out.
6. **Breakfast is the structurally worst meal.** At Yuktāhār about 28% of breakfast registrants eat against 56 to 89% at lunch on the same days, and at Kadamba 62.9% of April breakfast registrations were no-shows. The ”about 30%” gap students cited is a lunch-sized figure, and the breakfast gap is 63 to 72%.
7. **Every recent change sits on the operator side,** apart from the temporary non-compulsory registration during the LPG shortage (whether it also covered holidays is disputed), and none has touched registration or billing. The one reported waste reduction (Yuktāhār, August to December, reported by staff themselves) came from a paper routine made blame-free and discussed daily over about four months, alongside team training at a project manager's farmhouse. Software was tried and dropped because it cost more than the data it would add, and staff are not digitally inclined. The switch to rock salt and cold-pressed oil came about six months before the visit, later than the takeover.
8. **The two best-evidenced kitchens run opposite forecasting philosophies:** a fixed ratio plus reactive top-up at Kadamba, and learning from its own books plus a within-meal projection at Yuktāhār. Both are exposed to the

closing crest, one through top-ups under time pressure, the other through a projection window that sees about a seventh of the morning's diners on the measured curve (candidate transfer to Yuktāhār). Bakul sits between, with a weekly judgement bounded by a shortage floor, and an operator who states outright that shortage is worse than waste (single-sourced).

Outstanding Data Collection

The new records, operator interviews and first-hand account answered several earlier items: the shape and timing of the crowd peak (measured for Kadamba); whether the gap sits near 30% (not for breakfast: about 63% at Kadamba, about 72% at Yuktāhār); how stock and registration numbers become per-item quantities (answered for all three messes); whether any records-based time series exists (two now do); the wording of Kadamba's 70% batch (heard first-hand, still the operator's own claim); the cancellation limit (five per meal type per month, at least four days ahead, confirmed); whether service after 9:30 is a formal extension (informal, by the back door); and which hall Vijayalakshmi Caterers serves (Bakul, single-sourced; whether Palash shares the kitchen remains a candidate). The earlier kilogram figures for Kadamba waste are withdrawn. Still open:

- Multi-month records: Kadamba breakfast from May onward, any lunch and dinner exports, and exports for Yuktāhār, Bakul and Palash, so semester-level and exam-period patterns can be tested. Yuktāhār's site head expects exams to push attendance down, but no record covers an exam window.
- A Bakul breakfast scan export, to test the manager's 8:00 to 8:30 rush, at most 60% turnout and very low weekends; whether anything in Bakul's weekly adjustment is written down; the Hafizpet top-up lead time; and whether the reserve for about 50 at 9:20 rests on data or judgement and how much is left at close.
- Records carrying student identity, to test individual trajectories such as the reported three-to-six-week decline, and the Yuktāhār Wastage Book archive from June 2026, which would give a per-item history of waste and run-out times.
- The base of Kadamba's 5 to 10% waste figure and how it reconciles with a 70% batch against 37% turnout; whether a Skip Meal declaration lowers the count Kadamba cooks against; and whether the first-batch ratio differs between veg and non-veg.
- Whether the non-compulsory registration relaxation ever applied over holidays and fests; whether the CDS head named in the team's account and the CDS Chair named earlier are one person or two roles; and whether the back-door service after 9:30 is known to or sanctioned by CDS.
- Kadamba's internal staff headcount, to size ”about half the staff eat at the mess”, and whether outside diners are counted anywhere; and whether breakfast service formally opens at 7:00 or 7:30, given the live counter from 7:00 and about eleven scans a day before 7:30.
- Whether Yuktāhār keeps any arrival timestamps, so its first-half-hour projection can be tested against its own closing crest, and what waste target its wastage board shows, which was not captured clearly.
- Dates for the Prism tender, the Kadamba LPG response, the non-compulsory registration precedent, the Kadamba dinner-sambar removal, any change to a standing Bakul ratio, each paneer switch, the Yuktāhār takeover year, the salt and oil switch and the farm training trip.
- Whether the Wednesday electricity rise at Yuktāhār holds over more weeks; why Kadamba non-veg registrations roughly doubled in the first week of April; whether Yuktāhār's Sunday pattern (lunch registrations fall, breakfast do not) holds across more weeks; and why Monday is a strong breakfast day at Kadamba and a weak one at Yuktāhār.
- Why the Yuktāhār production sheet and attendance register disagree for breakfast on 3 September (77 against 110), and whether Skip Meal use, even if rare in a larger sample, reduces item-level run-outs.
- When Skip Meal was introduced and why, why the cancellation cap is five, and whether breakfast hours have ever differed from 7:30 to 9:30. These are governance questions for CDS, whose head or Chair has not been interviewed, along with the vendor contract terms.

Not yet claimed: that breakfast attendance is declining or rising across the student population over a semester; that any one student's trajectory matches the aggregate; that the Kadamba April profile holds for other months or messes without records; that Skip Meal has no effect on kitchen behaviour, as opposed to near-zero observed student use (now also disputed from the kitchen side at Kadamba); and that reheated food at Kadamba causes students to skip.

Phase 1 Consolidated Report

The Breakfast Paradox: IIIT Hyderabad Breakfast Mess System

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Introduction

Phase 1 of the Systems Thinking and Design capstone asks for five deliverables, one per activity: the System Context Brief, the Stakeholder Map, the Stakeholder Power-Interest/Leverage Map and Power Analysis Brief, the Current-State Process Map / Process Trace, and the System Timeline / Behaviour-Over-Time Map. This report condenses the findings of all five.

Students at IIIT Hyderabad are registered for breakfast far more often than they eat it. The evidence covers both halves of the system. On the student side, six respondents, the published rules and a governance review describe how registration works and why mornings are skipped. On the kitchen side, interviews with the operators of three messes, the photographed books of Yuktāhār, a month of timestamped Kadamba scans and a team member's first-hand account of two visits show how registrations become cooked food and where it goes. Several first-version findings were overturned by this operations evidence and are listed in the corrections table below.

Task 1: System Context Brief

System boundary. Breakfast as it is registered, billed, cooked and served in four halls (Kadamba, Palash, Bakul, Yuktāhār), in three layers: the registration and billing platform with the QR scan; the kitchen layer, each operator's planning, cooking, food-safety checks, top-up, reuse and disposal (Prism at Kadamba, ABC Hospitality Services at Yuktāhār, Vijayalakshmi Caterers at Bakul, cooking at Hafizpet); and the governance layer that sets menu, rules and contracts. Suppliers, the garbage contractor, the institution's site and facility teams and the tender sit on the boundary. So do non-student diners: at Kadamba, 30 to 50 outsiders such as night-shift security, and about half the operator's staff (single-sourced). Lunch and dinner are out of scope and serve as the comparison. At Yuktāhār in September 2026 lunch turnout ran 56 to 89% against 18 to 40% for breakfast on the same days (confirmed, records).

Purpose and actual mechanism. The institution's method is ”minimizing food wastage through advance meal planning.” In practice registration is a default. Every student is registered automatically, and the only exit is five cancellations per meal type per month, at least four days ahead (confirmed). Kadamba's count stayed near 1,070 a day in April while turnout ran from 42.1% on Tuesdays to 27.3% on Sundays (confirmed, records). No kitchen cooks to the count. Yuktāhār plans from its Wastage Book for the matching week, a weekday register, the month's pattern and events, then re-projects from the first half hour of service. Kadamba cooks 70% of registrations to a gram

standard and a supervisor calls top-ups. Bakul cooks 70 to 80% by item, revised weekly by its manager, and under his rule of ”no shortage, even if some wastage” keeps food for about 70% even on low weekends (single-sourced; no records). Only Yuktāhār learns from recorded outcomes, and its head still believes breakfast turnout is about 50% where its register shows about 28%. The on-site operators also read the count with opposite aims: Kadamba as an obligation it is paid for, Yuktāhār as a cost to cut.

Context. Registered breakfast costs Rs. 48 (vegetarian) or Rs. 66 (non-vegetarian), Rs. 53 at Yuktāhār, and walk-in access about 70% more. Governance splits across the Mess Committee, the Mess Office and the Warden, and the Kadamba operator describes a line from CFS to CDS to Prism. How these map onto each other, and who heads CDS, is unconfirmed. Operators are paid monthly on plates, with deductions. Bakul's operator says payment is on plates served and Kadamba's remark points to registrations, so the basis stays unconfirmed for every operator.

Symptoms and underlying issues. About two thirds of breakfast registrations go unused: 62.9% at Kadamba in April and about 72% at Yuktāhār in September (confirmed). Arrivals pile up at the close, with 26.5% of Kadamba scans between 9:15 and 9:30, so shortage and waste share a morning. Billing follows registration, Skip Meal returns nothing and the Mess Cell resale market recovers value the formal tools do not. Items that always waste are seen only by the operator, with no route to the menu.

Initial problem framing. Breakfast registration does not reflect who will eat, by roughly two thirds, while the same system works at Yuktāhār lunch. Why not, and why must each kitchen guess its own conversion when the data to calibrate it is already recorded?

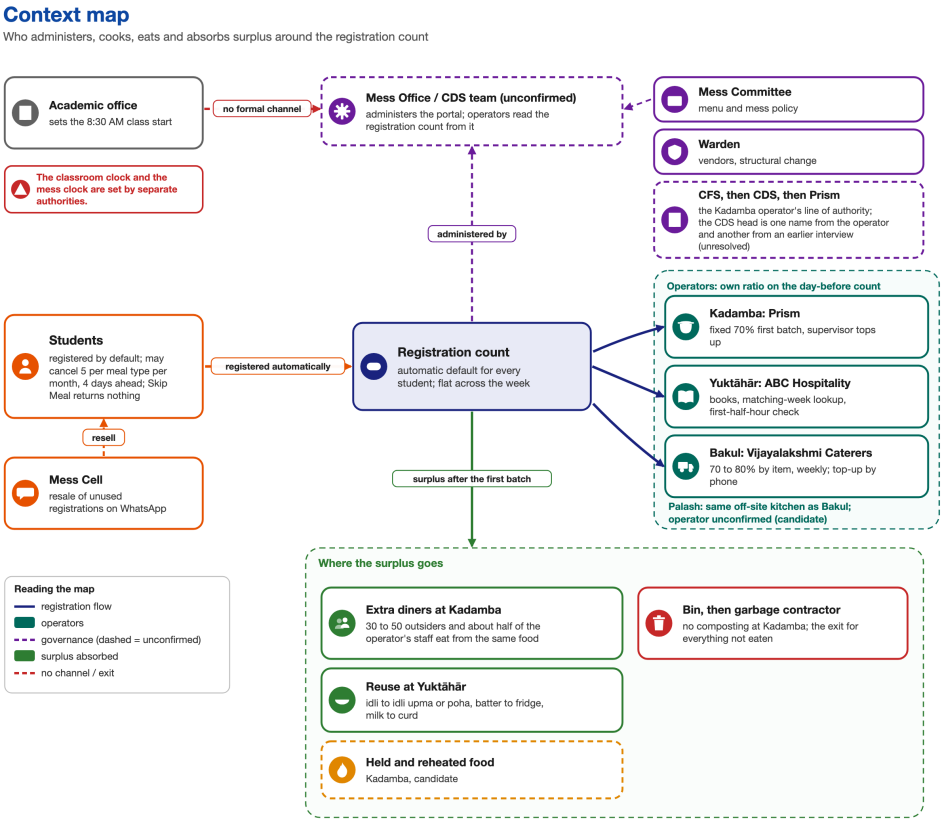


Figure 1. The registration count at the centre of the context map.

Task 2: Stakeholder Map

Thirty-two candidate stakeholders narrowed to twenty-four: seven primary (students, the three operators, operator site management, serving staff, kitchen staff), five secondary (Mess Committee, Mess Office, CDS site team, Warden, the Parliament's mess seats) and twelve indirect. The new stakeholder is the non-student diners at Kadamba.

The two on-site operators are organised as opposites. Kadamba is a ladder from the institution through Prism to a supervisor who decides each top-up. Yuktāhār has no hierarchy, two project managers, rotating roles, peer cross-checking and paper books reconciled daily. Bakul splits across two sites, with one manager deciding at the mess and a kitchen 8 km away.

Four findings matter most. Academic administration holds the lever in conflict with breakfast, the 8:30 class start, and has no relationship to mess governance. The Mess Office owns the four-day rule but does not place food orders. The gap between registration and turnout is measured twice, in the scan records and the operators' registers, and neither measurement reaches whoever sets the rules. And the outside diners and staff who eat Kadamba's surplus would lose food if the first batch were cut without planning for them.

Task 3: Stakeholder Power-Interest/Leverage Map and Power Analysis Brief

Power is split across governance bodies whose relations are only partly established, and part of it is exercised by actors with no standing: friends, Mess Cell, Ping, unofficial tools and the operators' own accommodations. Two absences define it. No relationship links Academic administration to any mess-governance actor, through inattention rather than refusal. And no route carries the kitchens' records of the gap to anyone who sets the rules that produce it.

The operator is the data-rich, authority-poor actor. It sets the parameter with the largest effect on waste, the share of each registration cooked, and has no authority over the cut-off, cancellation, billing or menu. The institution is in the Kadamba kitchen every meal through temperature checks, a 72-hour sample, a display plate and a tasting, and every check covers safety and taste, never quantity. Students are locked in four days ahead while operators adjust within minutes. Precedent shows how rules change: a reported December 2024 tightening of cancellation rules over organised pushback, a temporary relaxation during an LPG shortage, and the one deliberate waste reduction on record, which stayed inside Yuktāhār's kitchen.

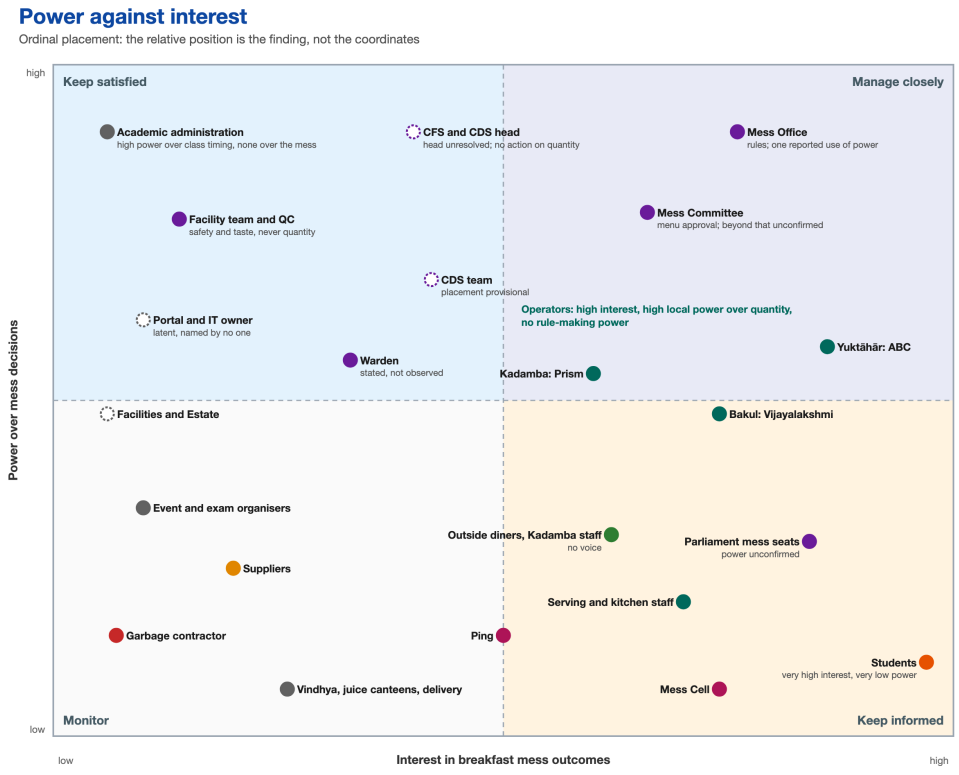


Figure 2. Power against interest. Placement is ordinal.

Task 4: Current-State Process Map / Process Trace

Three timescales are kept apart. Registration closes four days ahead and works as a standing default. Cooked quantity is set the day before, as a fixed share at Kadamba, a weekly-revised share at Bakul and from the mess's own books at Yuktāhār. The decision to eat is made each morning, driven by wake time, the 8:30 class and whether the wanted item is still there.

The kitchen half is now traced. The four-day lock applies to registration, not to buying. The binding constraint on waste is the cooked first batch, since cooked food lasts three to four hours while raw stock returns to the store. At Bakul that batch is fixed even earlier, when it leaves Hafizpet, and a top-up needs a call, cooking and a 20 to 40 minute drive. Yuktāhār runs two corrections, a matching-week lookup with a two-week delay and a first-half-hour projection that samples the calmest part of the morning. Total food runs well above turnout, so the shortages students meet are a timing and item-mix failure at the closing crest. Peer influence is disconfirmed as a deciding factor. An operational complaint path works for same-incident fixes, and no path for changing rules or timing works for students or operators. Surplus leaves through reuse at Yuktāhār, the outside diners and staff, held and reheated food at Kadamba (candidate), latecomers served at the back door, and the bin, with no composting at Kadamba.

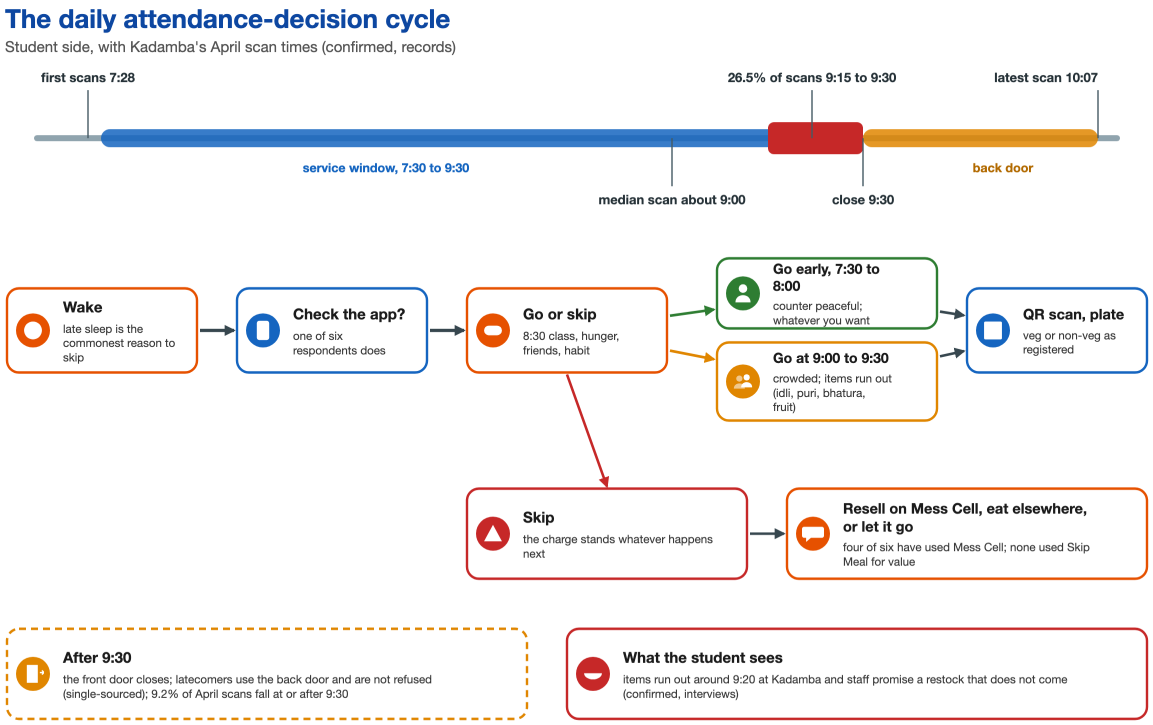


Figure 3. The daily attendance-decision cycle on the student side.

Task 5: System Timeline / Behaviour-Over-Time Map

The within-morning crowd is measured. A burst at 7:30 and a flat plateau until 9:00 give way to a steep climb, with 40% of Kadamba diners in the last half hour and about 9% after the nominal close, many by the back door (confirmed, records). The curve is Kadamba's. Yuktāhār's head describes the same closing rush, while Bakul's manager places his rush at 8:00 to 8:30, so curves differ by hall. Across the week, registrations stay flat while turnout falls every Sunday at both recorded messes (Kadamba 25 to 33%, Yuktāhār 18 to 24%), and each April week at Kadamba sits between 35 and 39% with no drift. Standing registration and billing rules have been stable since early 2025, while the kitchens behind them have changed repeatedly.

R1, Shifting the Burden via Mess Cell, is an open chain: its last link, that resale reduces pressure to fix billing, is not evidenced. B1, kitchen waste control, is balanced by operators, not by Skip Meal. Kadamba closes against shortage

only, within minutes. Bakul's weekly review closes against surplus only down to its floor. Yuktāhār's matching-week channel closes against surplus with a two-week delay. None is calibrated to breakfast turnout. The same-meal-two-failures chain, a fixed first batch meeting a curve that crests at close, gives surplus in total and run-outs at the crest.

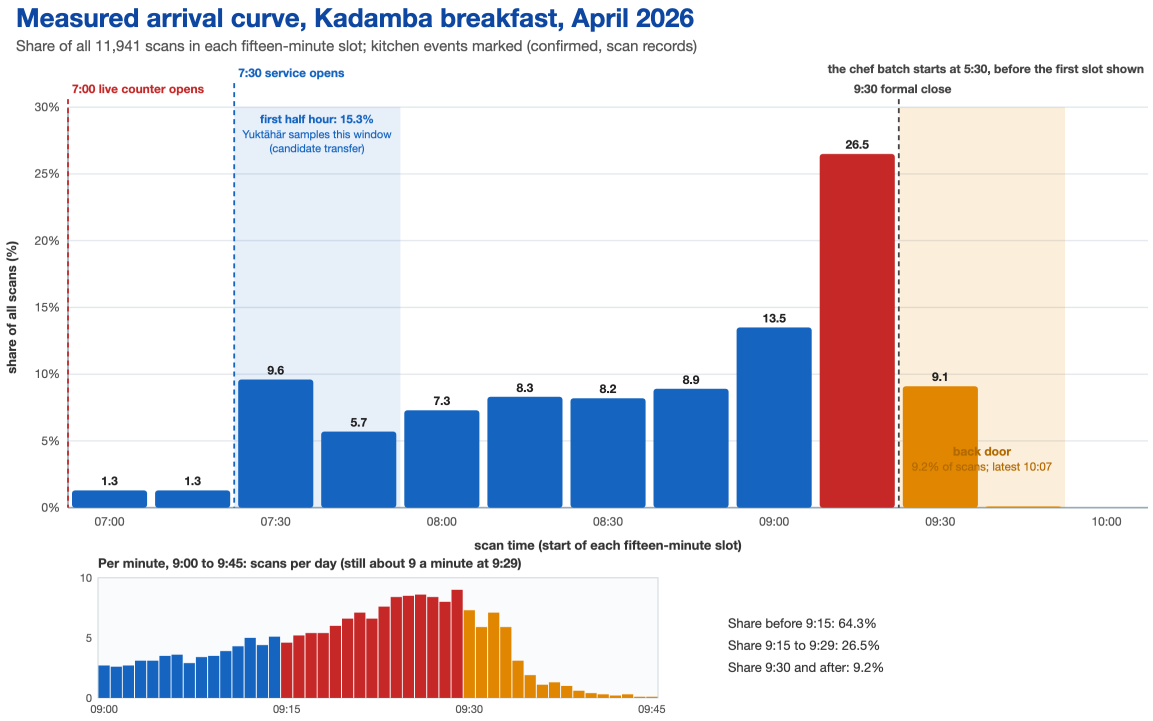


Figure 4. *Measured arrival curve, Kadamba breakfast, April 2026.*

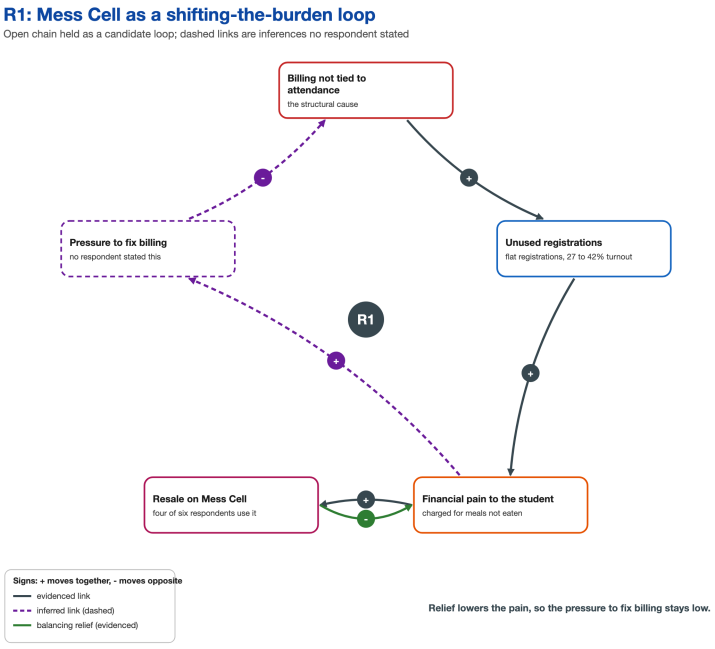


Figure 5. *R1: Mess Cell as a shifting-the-burden loop.*

Consolidated Findings Across All Five Deliverables

1. **Registration is a default, not a demand signal.** Automatic registration, five early cancellations as the only exit, billing on registration, a Skip Meal that refunds nothing and a walk-in premium of about 70% together reward staying booked and ignore attendance. About two thirds of breakfast registrations go unused, while lunch at the same mess mostly works.
2. **Every kitchen guesses its own conversion, and every guess is too high.** Kadamba cooks 70% and Bakul 70 to 80% (single-sourced), Yuktāhār's head believes at most about 50%, and recorded turnout is 37.1% at Kadamba and about 28% at Yuktāhār. Bakul's manager estimates at most about 60% (no records).
3. **The gap is measured and goes nowhere.** The scan records and the operators' books both measure it, and neither reaches the owner of registration or billing rules or each other.
4. **The binding constraint is the cooked first batch, not procurement.** Nothing shrinks a batch once it is cooked and approved.
5. **Shortage and waste happen on the same morning.** At Kadamba a late crest meets a fixed first batch.
6. **Academic administration is the clearest sleeping stakeholder.** It holds total authority over class start time, has no relationship to mess governance, and the calendar reaches the kitchen only by word of mouth.
7. **Oversight and feedback change kitchens, not rules.** Institutional checks cover safety and taste, never quantity, and no confirmed instance exists of student input changing registration, billing or hours. The one reported feedback-driven policy change is the fortnightly menu rotation.
8. **Surplus is absorbed out of sight.** Resale absorbs the student's loss, and reuse, outside diners and staff, reheating and the bin absorb the food. Kadamba's reported 5 to 10% waste, on an unstated base, sits in unresolved tension with a first batch near twice turnout. Each absorber is locally useful and some feed real people, so any cut to the first batch must plan for them.
9. **Respondent-proposed changes.** Extending the 9:30 close was the most requested fix, and a pay-as-you-go alternative to fixed registration was proposed as a structural option.

Corrections Since the First Submission

First version said	Now
The registration gap is about 30%	62.9% at Kadamba (April) and about 72% at Yuktāhār (September); 30% fits lunch
Weekly blocks of registration; the Mess Office places food orders four days ahead	Registration is automatic, with five cancellations per meal type per month at least four days ahead; operators buy their own stock and set quantities the day before
Vendors cook to a forecast and are passive	No operator cooks the full count; each applies its own ratio and tops up
Bakul cooks 70%, raised to 80%, and its food is cooked at Palash or near the airport	70 to 80% by item, adjusted weekly with a floor near 70%; Vijayalakshmi Caterers cooks at Hafizpet, about 8 km away
Composting absorbs the surplus	Kadamba has no composting; surplus goes to outside diners, staff, reheating and the bin
20 to 30 staff eat leftovers	About 30 portions for 30 to 50 outside diners, plus about half of Kadamba's staff
Kadamba waste of 16 to 23 kg	Withdrawn; the operator reports 5 to 10% daily on an unstated base
Yuktāhār has a five-person project team and a cross-checker called ”Adit Amma”	Two project managers; the mess in-charge keeps the Wastage Book (the ”Adit Amma” identity is unconfirmed)
Every operator distrusts registration	Only Yuktāhār does; Kadamba treats it as an obligation and Bakul adjusts by judgement
Cuisine is the stronger explanation of low uptake	Within Kadamba, veg registrants attend at 32.4% against 48.5% for non-veg; cuisine is one factor
All three operators describe a closing rush	Recorded at Kadamba and described at Yuktāhār; Bakul's manager places his rush at 8:00 to 8:30
Skip Meal declarations reach the kitchen	Conflicting: Kadamba says it cooks on registrations alone

Outstanding Data Collection

The operator visits, the first-hand account and the Kadamba records answered what the kitchens prepare against, Kadamba's first-batch rule and the cancellation limit, whether the ”about 30%” figure was real (it is a lunch-sized figure), why items run out, where unclaimed food goes, who runs Kadamba, which hall Vijayalakshmi Caterers serves and the shape of the crowd. What remains open **will be collected and submitted later**.

- Whether each operator is paid per registered or per served plate, and the deductions, confirming Bakul's reported served-plate basis from the contract or CDS.
- The base of Kadamba's 5 to 10% waste figure, and how it squares with a 70% first batch against 37% turnout.
- Whether a Skip Meal declaration lowers the count Kadamba cooks from.
- Who heads CDS, given two differing names; whether CDS is the Mess Office; how CFS relates to the governance bodies; and who the facility team that tastes Kadamba's food is.
- How the outside diners are registered, who pays for their portions, and whether staff meals are a written entitlement.
- Whether Palash is served by the same operator and kitchen as Bakul.
- A Bakul breakfast scan export; whether anything behind Bakul's weekly adjustment is written down; and the Hafizpet top-up lead time.
- Where the garbage contractor takes the waste, and direct observation of disposal after close.
- Multi-month records for all four halls and all meals, including an exam window, and any Yuktāhār arrival timestamps.
- Whether any institutional actor has ever received an operator's books, and whether the December 2024 pushback is the ”organised” student effort heard of secondhand.

Status Summary

#	Deliverable	Status
1	System Context Brief	Complete; revised on the operations evidence and the first-hand account
2	Stakeholder Map	Complete; twenty-four stakeholders, non-student diners added
3	Stakeholder Power-Interest/Leverage Map and Power Analysis Brief	Complete; payment basis, CDS leadership and facility-team identity open
4	Current-State Process Map / Process Trace	Complete for the student and kitchen halves; disposal after close unobserved
5	System Timeline / Behaviour-Over-Time Map	Complete; measured within the morning, week and month; semester trends unmeasured